



Commvault® Professional Advanced

Student Guide
Module 1 – Deployment

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Introduction

Course Overview

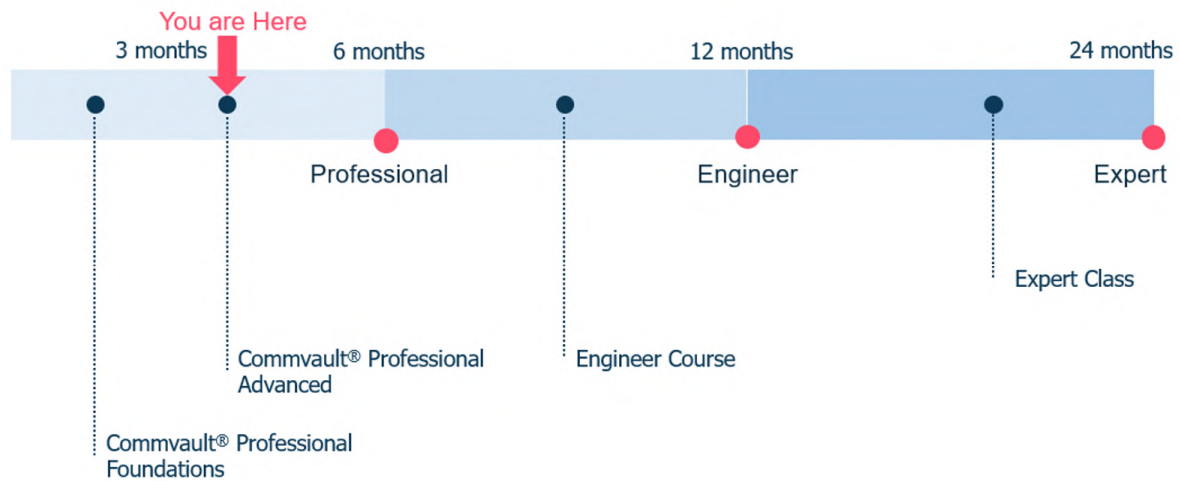
1. Deployment
2. Storage Advanced Configuration
3. Storage Policies
4. Monitoring

Course Objective

This presents advanced configurations of Commvault® Software. It focuses on the CommCell® Console and builds upon the Commvault® Professional Foundations course.

Commvault Education Career Path

Suggested learning path for administrators and engineers



The Commvault platform provides capabilities and functionality for the performance, security, compliance, and economic benefits of a holistic data management strategy. The key concepts covered in this first step learning module highlight the core features of the platform. To realize the full value of these features, Commvault provides multiple levels of education and certification for users at all levels.

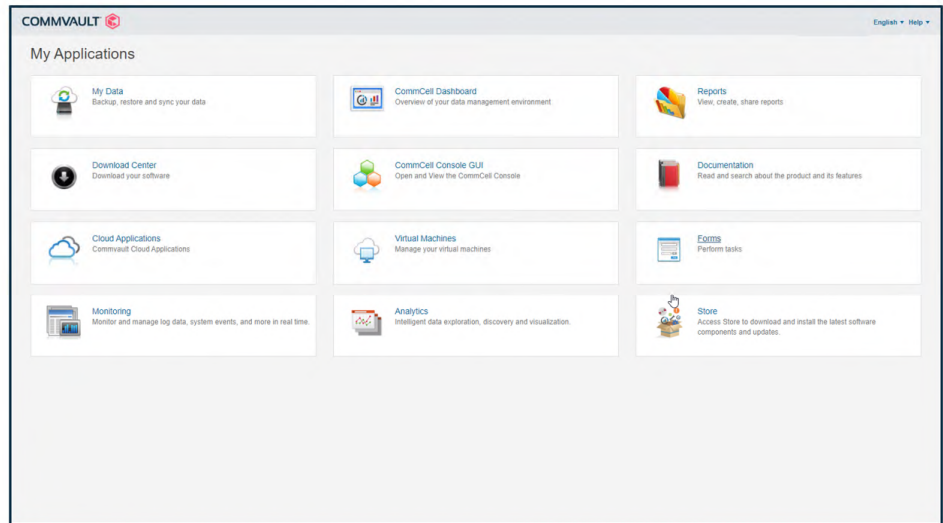
Commvault Certification

Badges	Certification Program
	Pass Commvault Professional Foundations and Advanced Exams
	Must have current Professional certification and pass the Engineer Exam
	Must have current Professional and Engineer certifications, and pass the Expert Exam

Commvault's Certification Program validates expertise and advanced knowledge in topics, including Commvault Professional, Engineer and Expert technologies. Certification is a valuable investment for both a company and the IT professional. This program provides certification based on a career path, and enables advancement based on an individual's previous experience and area of focus.

Cloud Resources

- Personal Data Management
- Reporting
- Reports to download (Store)
- Software to download
- Documentation



Go to <http://cloud.commvault.com> for resources to help you manage and maintain your Commvault environment.

Download Center – lets you download feature releases, hot fixes, and binaries for new installations.

Forms and Store – allows new features and components like workflow or custom reports to be downloaded and installed into an existing installation. Commvault regularly releases updates to product. In addition to new features and functionality, Commvault may retire some components or features.

Documentation – provides a link to Commvault's technical documentation site

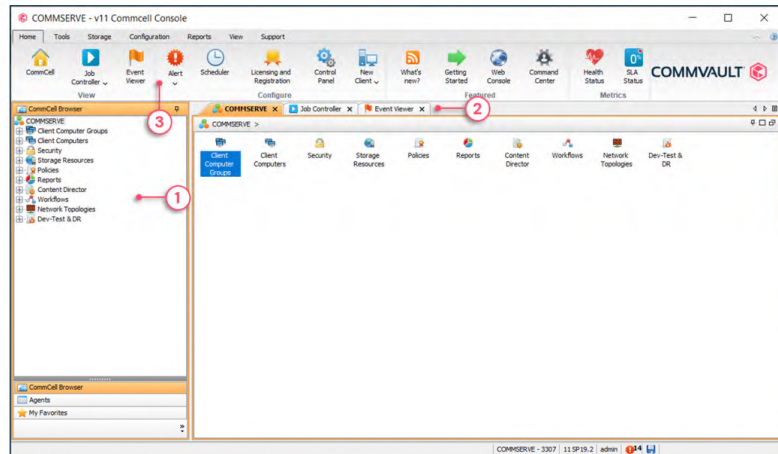
For the new customer, the Quick Start section of the documentation site guides you through steps to review prerequisites and steps to install and configure Commvault components.

The documentation site is divided into two sections: Essential and Expert. To view Expert information, you must log into with your Maintenance Advantage credentials.

CommCell® Console

CommCell® Console Overview

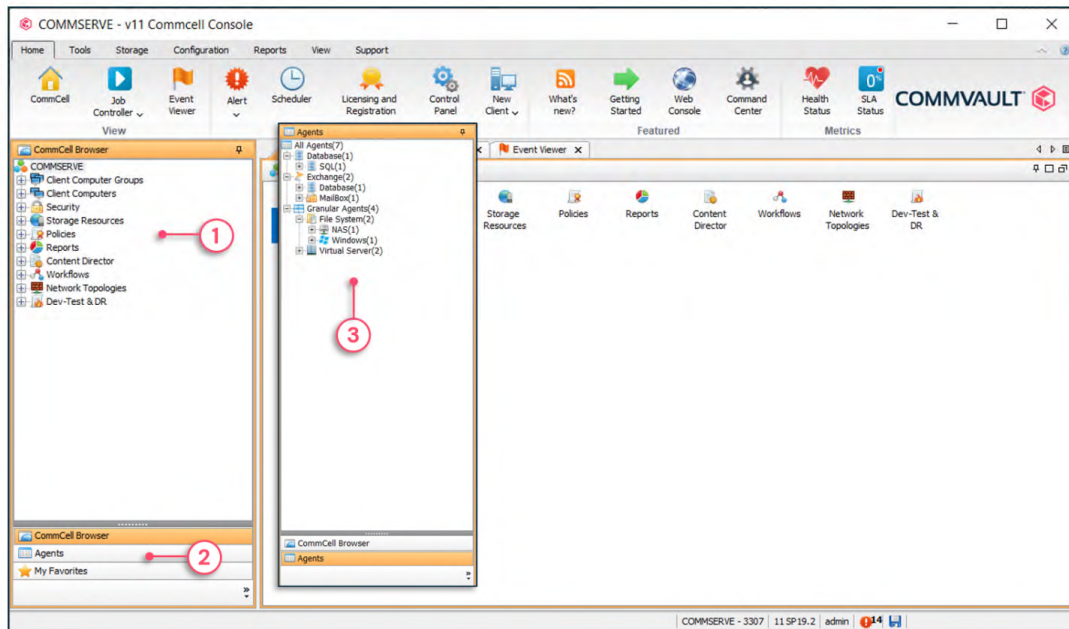
- Local install vs. web page
- Layout
 - CommCell® Browser
 - Content / Summary view
 - Job Controller
 - Event Viewer



The CommCell console is the interface used to configure and manage the CommCell environment, monitor, and control active jobs, and view events related to all activities. The console is accessed using local CommCell accounts or by using Active Directory integrated account authentication.

1. The CommCell browser displays all CommCell components in an easy-to-navigate tree structure
2. Tabbed windows include details of item selected in CommCell browser, Job Controller, and Event Viewer
3. The toolbar provides access to CommCell configuration settings

CommCell® Console Browser Window



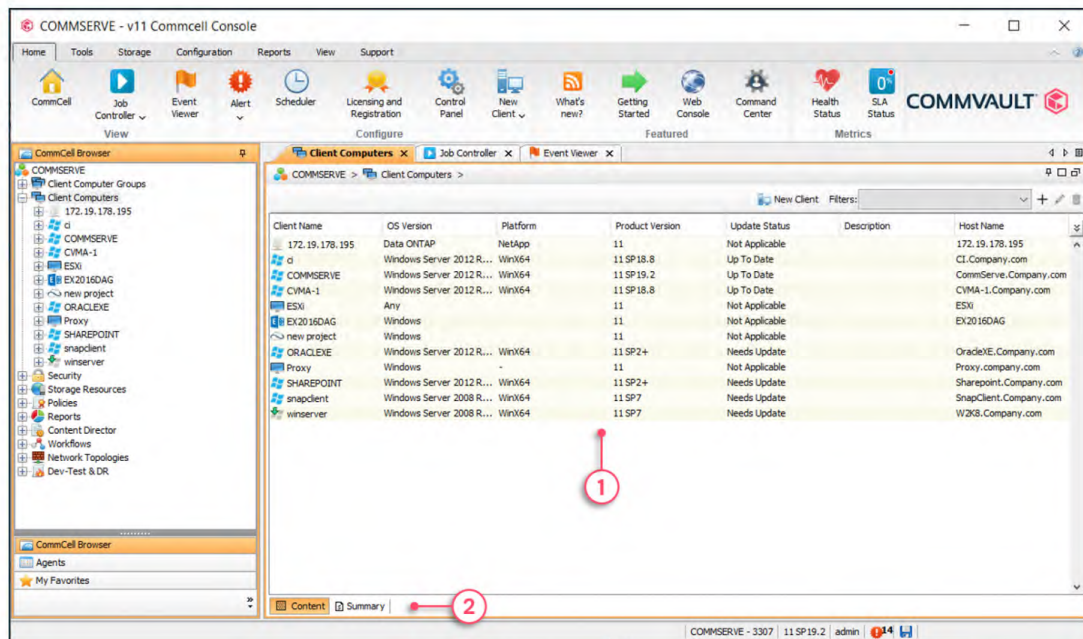
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The browser provides two views: Browser (default) view and Agents. The navigation is hierarchical, and most settings are customizable.

1. Hierarchical structure showing all CommCell components
2. Select the *Agents* view to displays a categorized view of agents in the CommCell environment
3. The browser displays all agents categorized by type

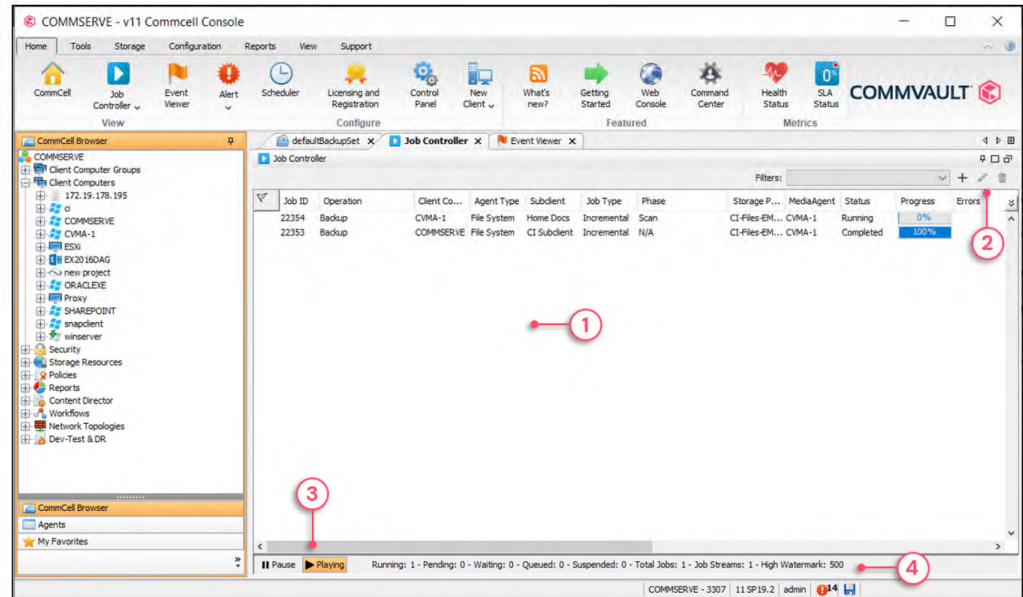
Content and Summary Window



1. **Content View** - Displays detailed information based on what is selected in the CommCell browser window
2. **Summary View** - Can be selected to display summarized information
 - o Extended Summary – When specific objects in the browser are selected, the extended summary view provides greater detail than the summary view
 - o Feature View – Displays feature related to the selected object

Job Controller

- Displays active jobs
- Job details
- Job control
- Job Controller filters
- Job retention time
- Job update frequency

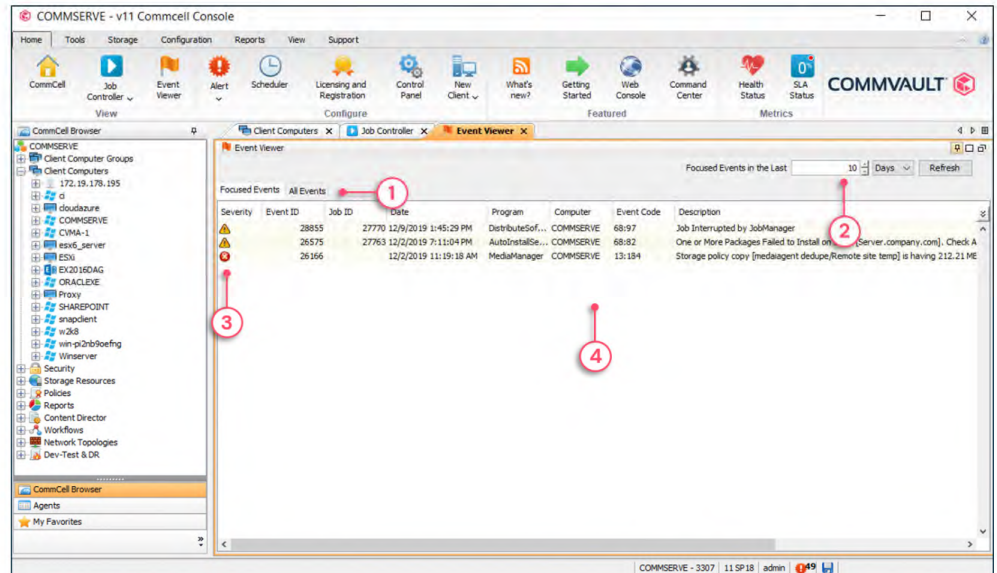


The Job Controller provides monitoring functionality for all active jobs within the CommCell environment. Jobs remain in the Job Controller for five minutes after the job completes, fails, or is killed. The default time jobs are maintained after they finish and can be modified in User Preferences applet in the Control Panel.

1. Currently active jobs and job status are displayed in the Job Controller
2. Add or apply Job Controller filters if needed
3. Pause and play buttons are used to freeze and unfreeze the Job Controller window refresh, but do not pause the actual jobs
4. CommCell summary data for all active jobs and streams is displayed here

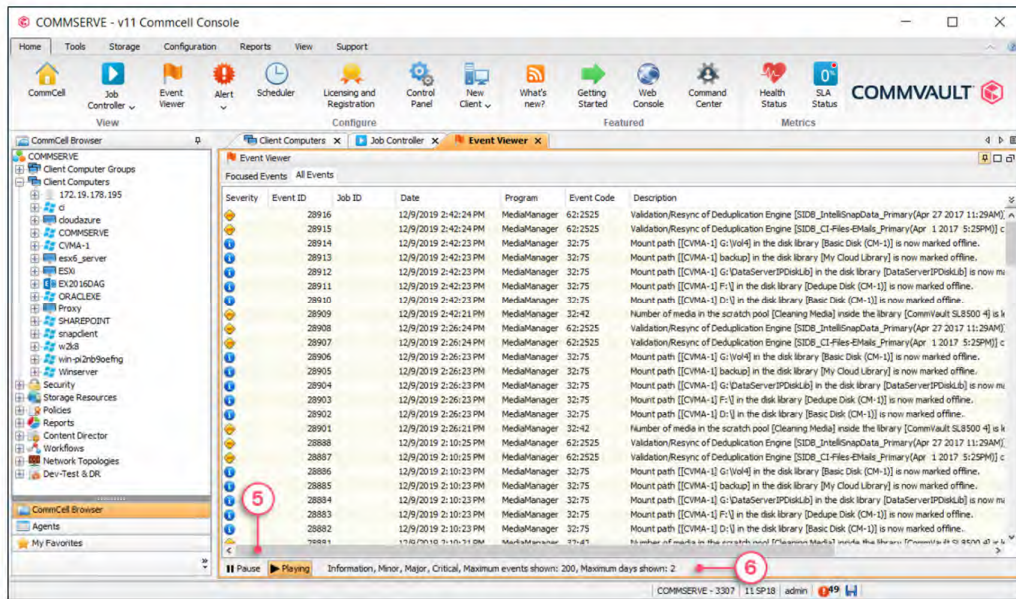
Event Viewer

- Centrally displays all CommCell® events
- Focused vs. All Events
- Defaults to 200 events, but can be increased up to 1000
- Filters
- Search events
- Events retention



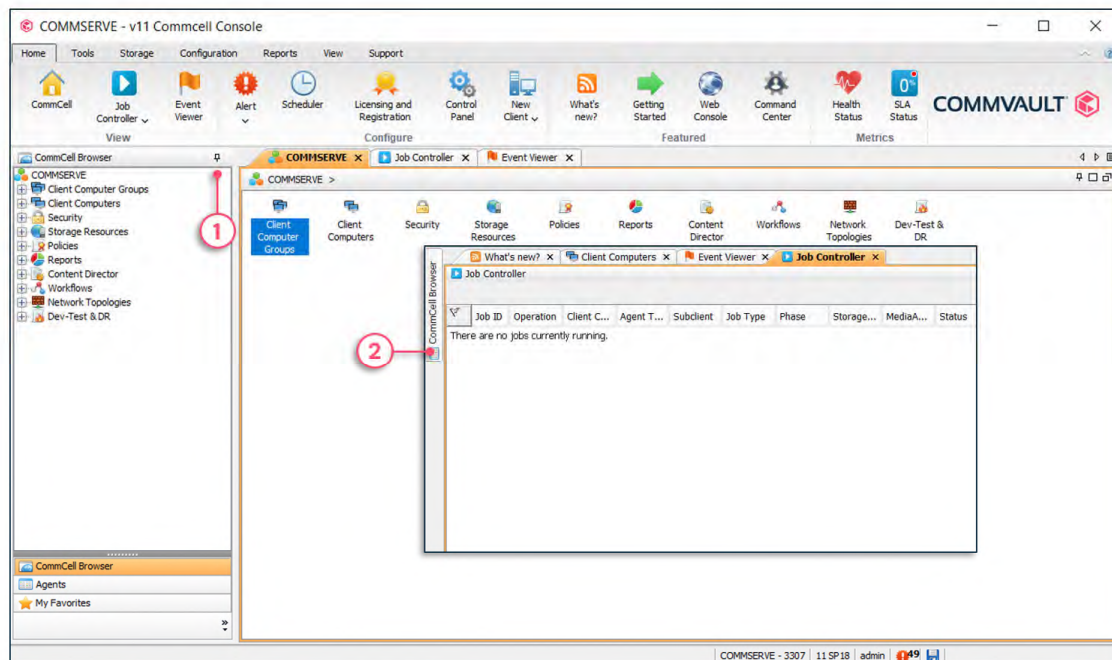
All Commvault software related events are reported in the Event Viewer. By default, 200 events are displayed in the event viewer, but the event log can maintain up to 10,000 events or 7 days of events. These options are configured in the System Settings applet in the Control Panel.

- Select the tab for *All Events* or *Focused Events*
- Set a time range for the Focused Events tab
- These icons indicate job severity
- The main window displays event details



5. Pause and play buttons to freeze and unfreeze Event Viewer window refresh
6. This is the maximum number of events that will be displayed

Customize the CommCell® Console

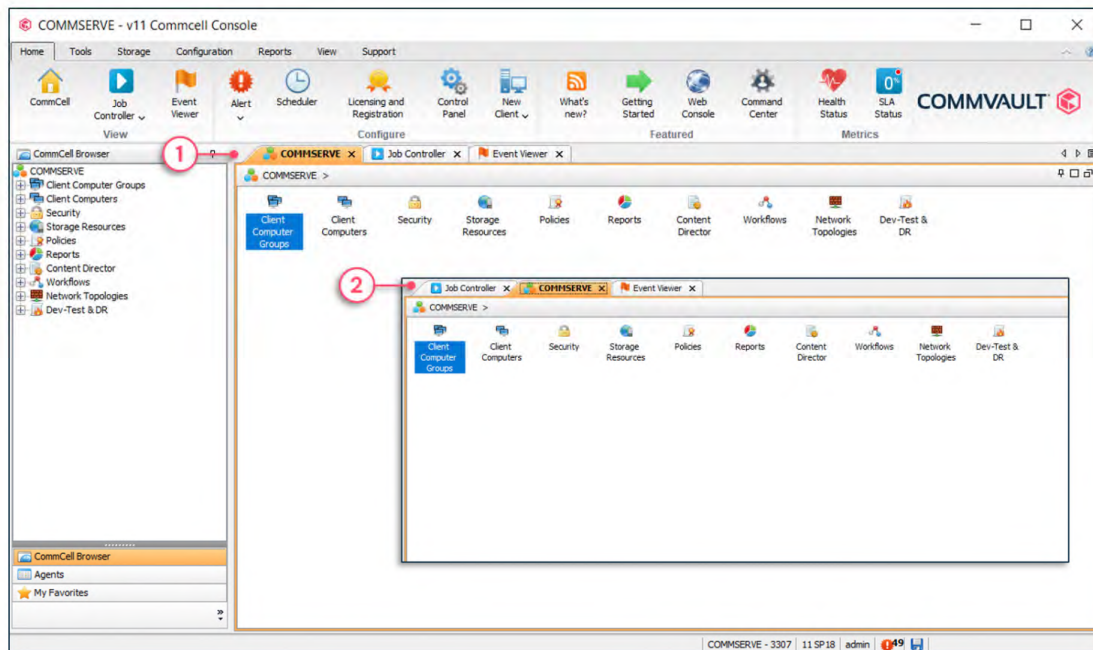


Hide the CommCell Browser

The CommCell Browser is hidden by unpinning the browser window. This can free up space in the console to view more information in the Job Controller and Event Viewer windows. Once hidden, the browser window can be temporarily displayed by hovering the mouse over the browser tab on the left side of the window. To lock the browser window back, double-click the pin button on the right side of the browser window.

1. Click CommCell Browser pin
2. Hover over the hidden browser title bar to view browser and click the pin to browser in the window

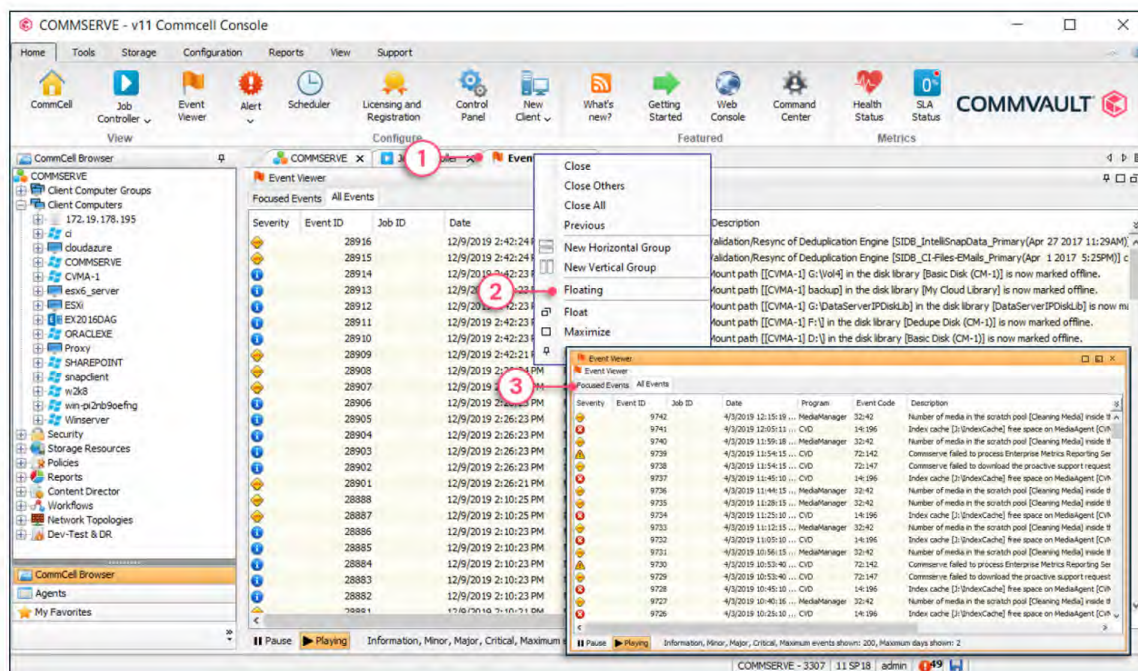
Move Tabbed Windows



Tabbed windows can be rearranged in any order by dragging and dropping the tab. This is beneficial when many windows are open at the same time.

1. Click and drag the tab to the tab location
2. Tabs are reordered in console

Float Windows



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Use float windows for improved visibility and access.

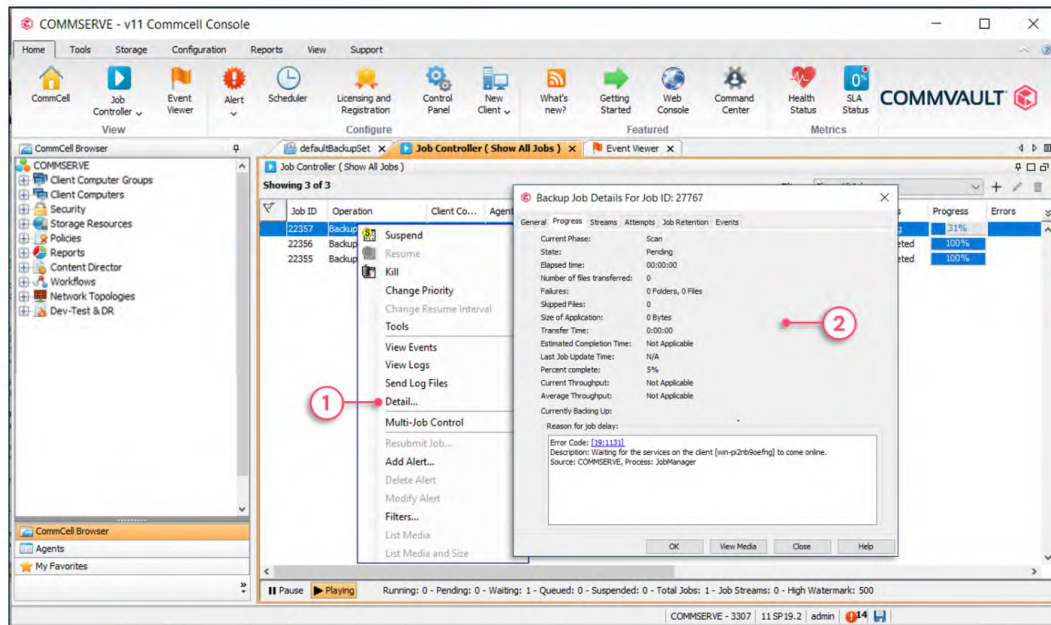
1. Click and drag the tab to an empty space in window
2. Alternatively, right-click the tab > **Floating**
3. The floating window is not bound to the CommCell Console

Customize the Job Controller

The Job Controller is the most effective tool for managing and troubleshooting all active jobs within the CommCell environment. Fields can be customized.

- **Move a field** – Click and drag the field to the location in the Job Controller
- **Resize the field** – Click the outer edge of the field and drag to the correct size
- **Sort the field** – Click on the field and an arrow on the right side appears. You can sort in order or reverse order for the field
- **Add/remove field** – Fields can be added and removed by using the menu arrow in the upper-right corner
 - **Number of readers in use** – displays the number of streams per job that are being used for the operation
Note that the 'readers in use' field only displays a number during the backup phase.
 - **Current throughput** – displays the current throughput in GB per hour
 - **Priority** – displays the job priority for the operation

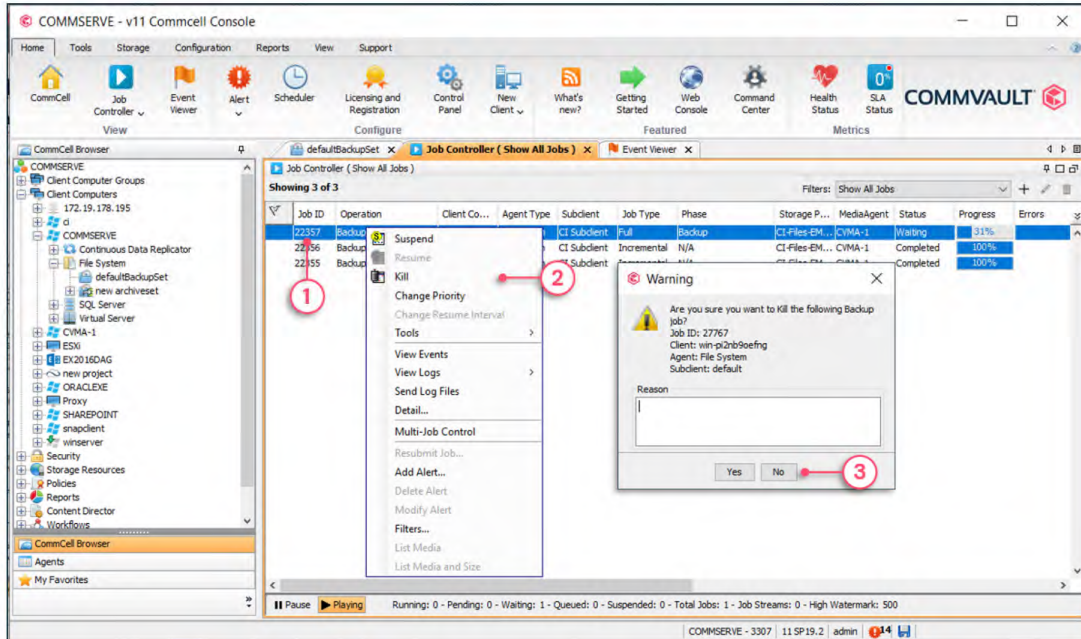
View Job Details



Each job includes information about the job status, data path, and media usage or job errors.

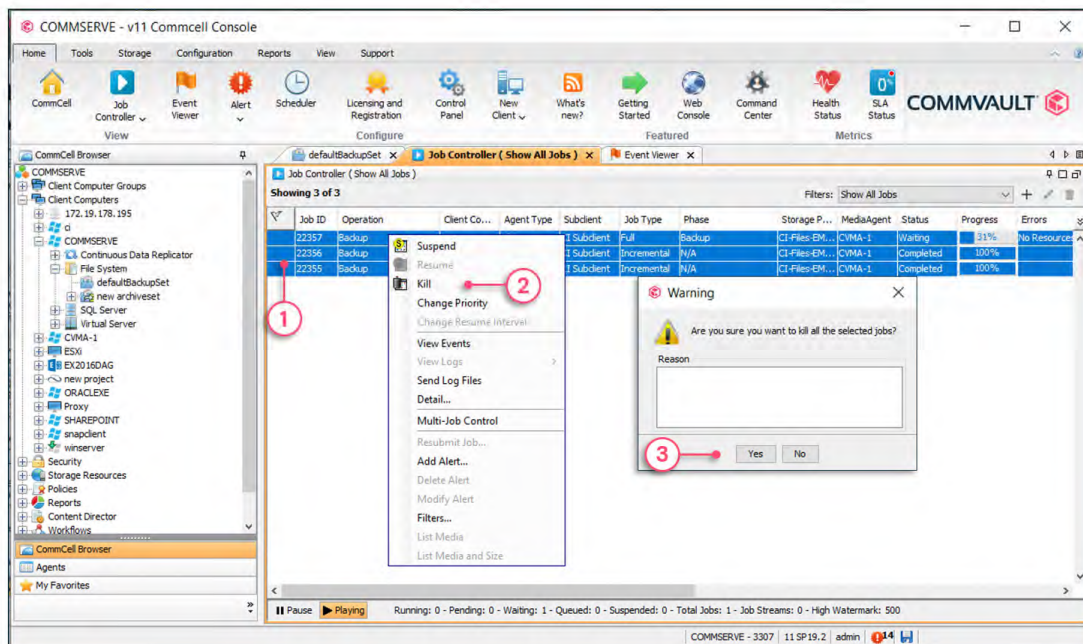
1. Right-click on the job and select Detail or double-click the job
2. Details for job are displayed here

Change Job Status – Single Job Control



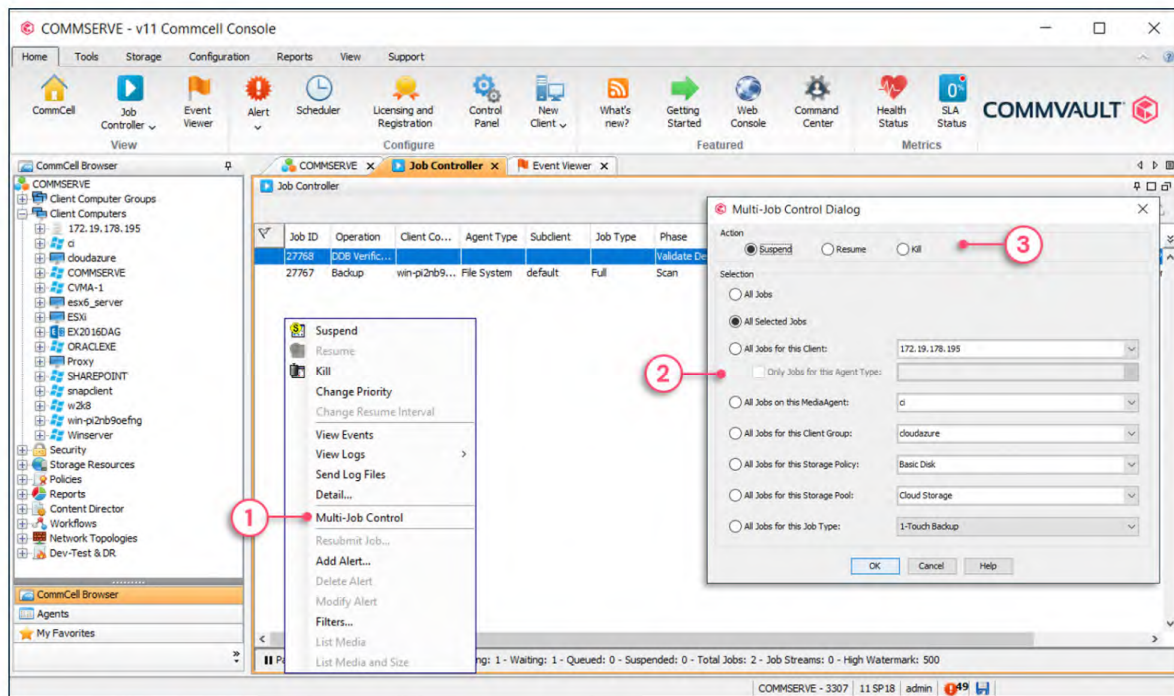
1. Right-click the job
2. Choose to suspend, resume, or kill the job
3. For a kill, enter an explanation click **Yes** to confirm

Change Status – Multi-Job Control



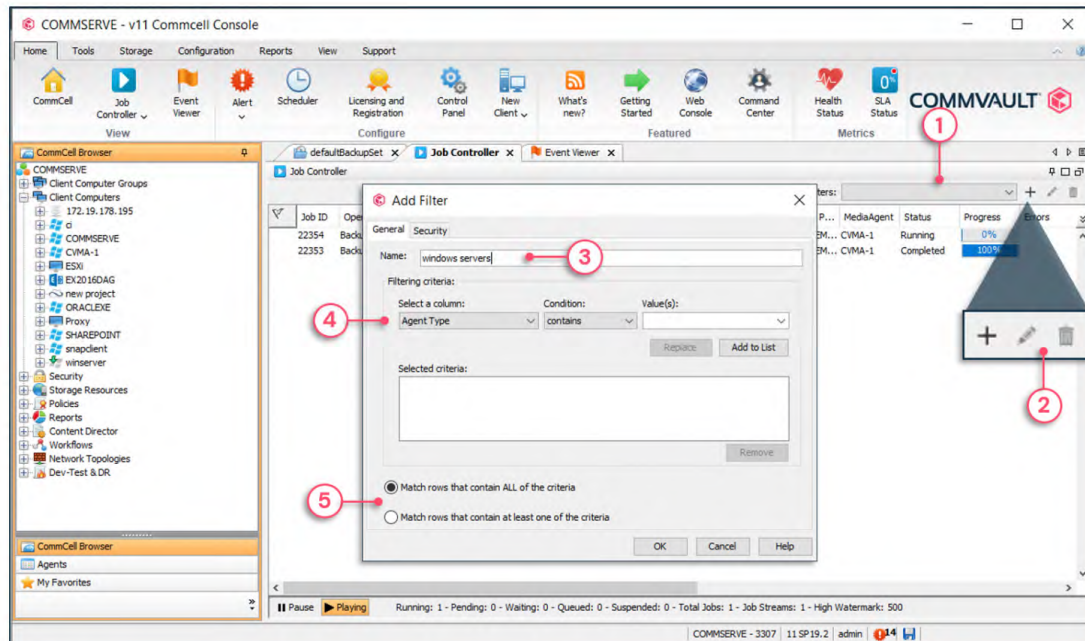
1. [SHIFT + click] or [CTRL + click] to select multiple job
2. Right-click and select suspend, resume, or kill
3. For a kill, enter an explanation click **Yes** to confirm

Change Job Status for Groups of Jobs



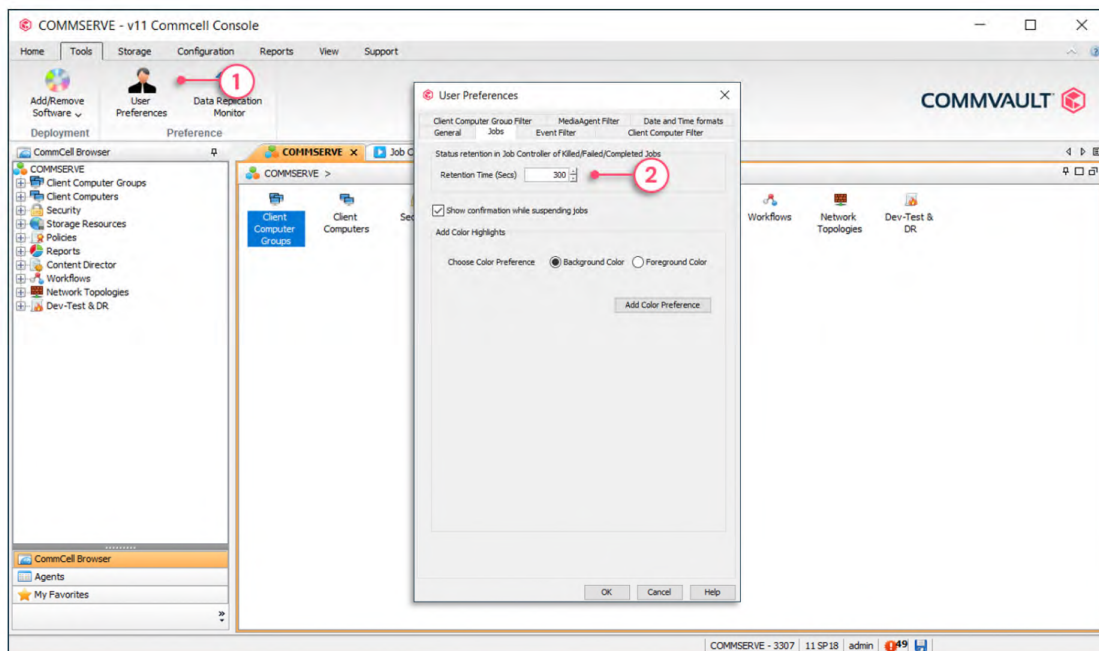
1. Right-click in the job controller > **Multi-Job Control**
2. Choose all jobs or specific jobs
3. Choose to suspend, resume, or kill

Apply Job Controller Filters



1. Click the drop-down to apply an existing filter
2. Click + and select the icon to add, delete, or edit a filter
3. Enter a name for the filter
4. Create one or many rules and add to the list
5. Specify if it is required to meet ALL or ANY of the rules created above

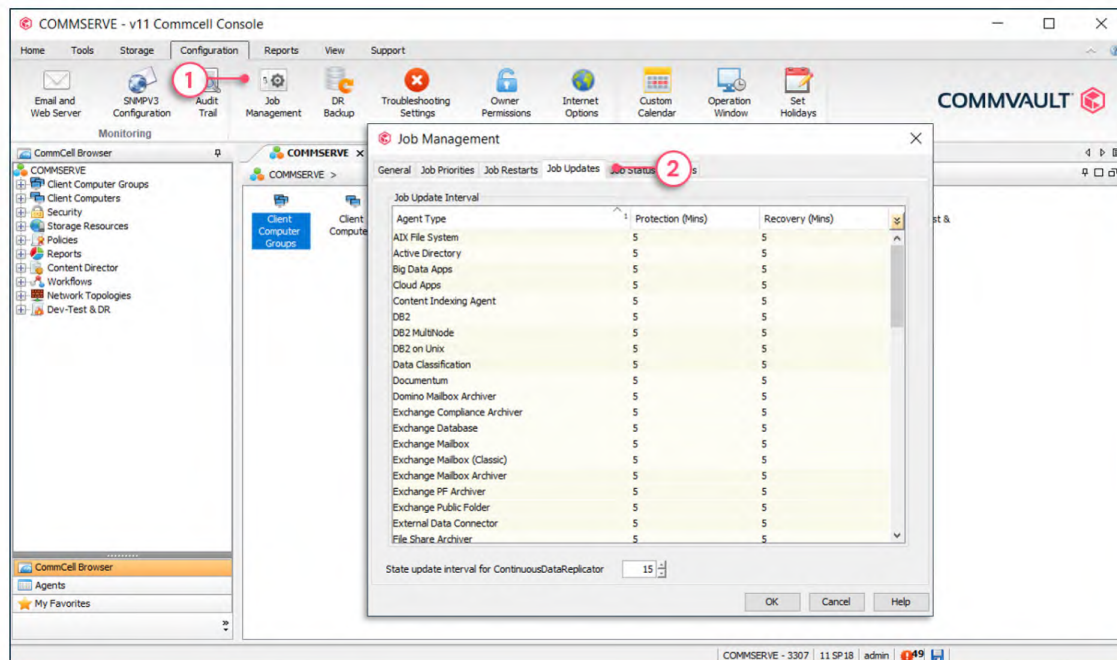
Set Job Controller Retention Time



By default, a job remains in the Job Controller for five minutes after it has completed successfully, failed, or was killed by the administrator.

1. **Tools** tab > **User Preferences**
2. Define how long you want to retain jobs

Modify Job Update Intervals

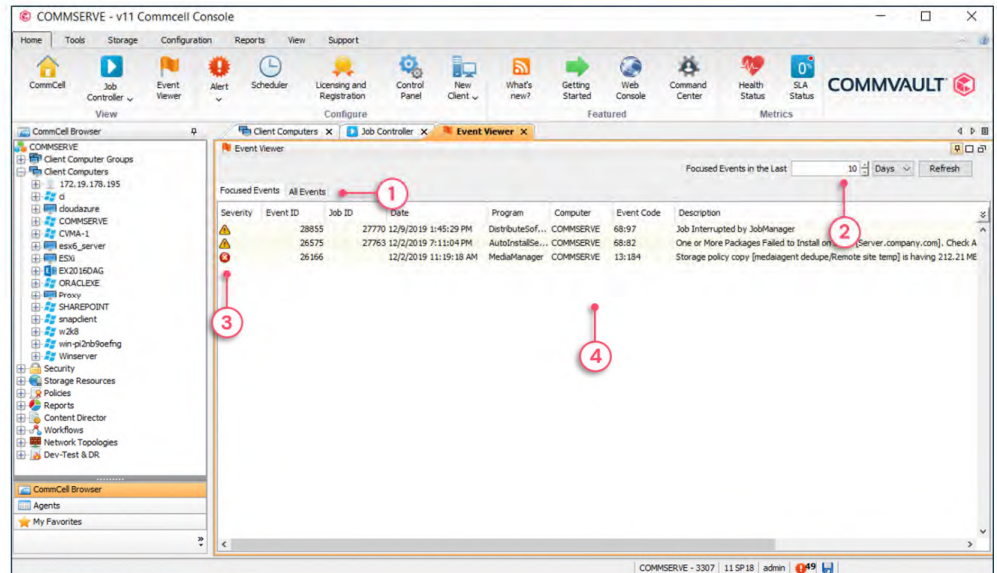


To reduce communication traffic during normal operations, the job status in the Job Controller updates every five minutes. Unless required for a specific job type, it is NOT recommended to decrease these values as it could have a negative impact on backup or restore performance. However, in certain situations where job status must be closely monitored, the update intervals can be modified.

1. *Configuration* tab > **Job Management**
2. Select the *Job Updates* tab to modify protection and recovery operations for different agents

Customize the Event Viewer

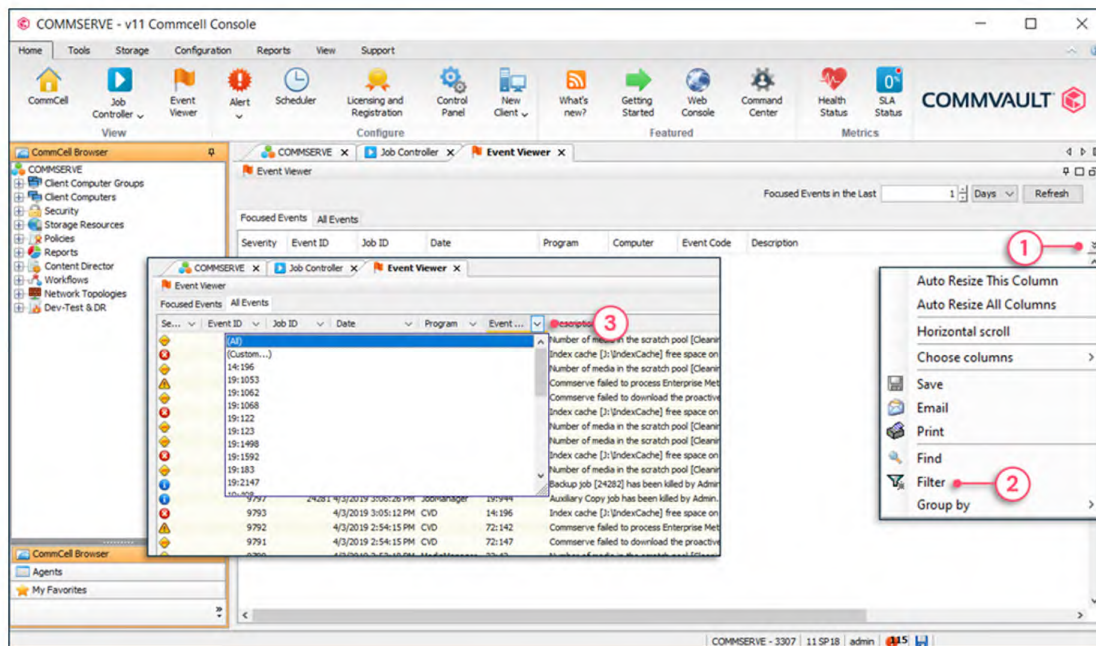
- Centrally displays all CommCell® events
- Focused vs. All Events
- Defaults to 200 events, but can be increased up to 1000
- Filters
- Search events
- Events retention



The Event Viewer window displays events reported based on conditions within the CommCell environment. By default, the event viewer displays the most recent 200 events. This number can be increased up to 1,000. The event log maintains up to 10,000 events or 7 days of events.

- Click to see all events or focused events
- Set a time range for the Focused Events tab
- Job severity icons allows to quickly gauge events
- Event details are displayed

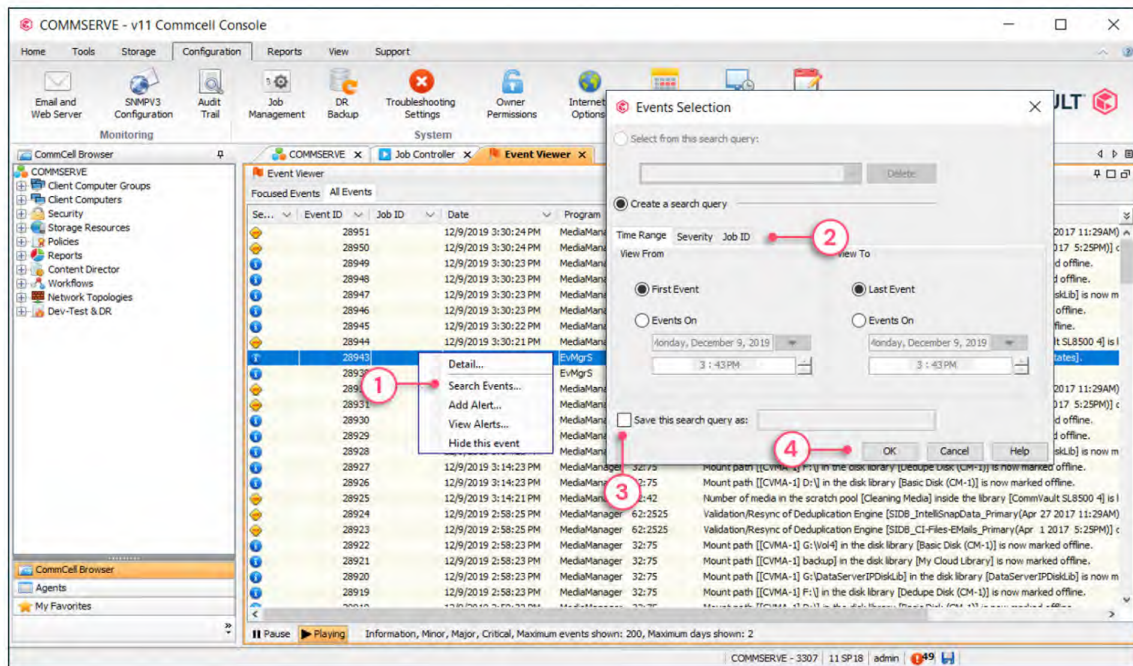
Filter the Event Viewer



The event viewer can be filtered based on the available fields to quickly locate specific events.

1. Click the **double down arrow** to expand Event Viewer menu
2. Select **Filter** to enable Event Viewer filtering
3. Filter fields using the drop-down arrow

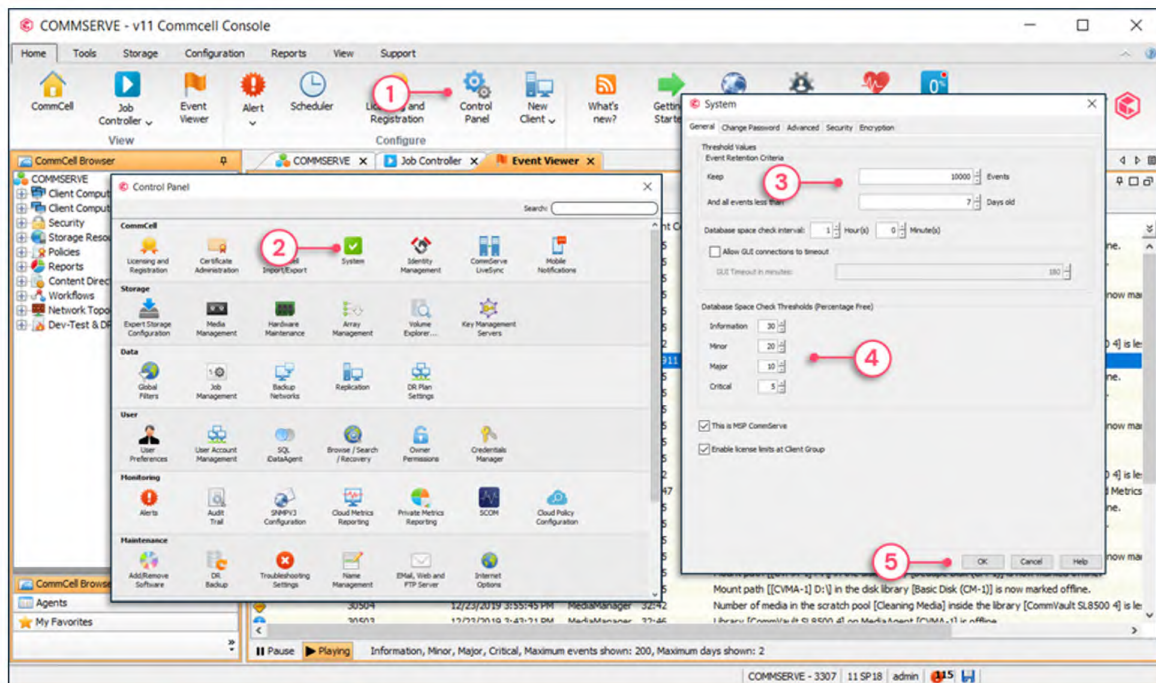
Search Event Log



The event viewer displays up to 1,000 events, but the entire event log can be searched from the event viewer. The default total number of events retained is 10,000. Search events by time range, severity, and job ID. Common searches can be saved as a query and run at any time.

1. Right-click in the *Events* window > **Search Events**
2. Define search criteria to display all relevant events in the event log
3. Check to save the query to run it as needed
4. Click **OK**

Set Event Log Retention



By default, the event log retains 10,000 events or 7 days of events. When the event logs reach their upper limit, the oldest events are pruned from the event logs. These options can be customized.

1. Home tab > **Control Panel**
2. In the *CommCell* category > **System**
3. Set the event retention criteria
4. Modify the Database space check interval
5. Click **OK**

Planning

Phases of a Deployment

- Planning
- Download software
- Install CommServe® Server
- Post Installation Tasks
- CommServe® Server Database Management
- Install MediaAgents



Installing Commvault software is a straightforward process. However, it is important to note that a full CommCell deployment must be well-planned and executed. This section presents the high-level steps and best practices for deploying a CommCell environment.

Planning a Deployment

- Assess the current environment
- Assess the data protection requirements
- Design the CommCell® environment
- Procure and install hardware
- Change management



Assess the Current Environment

If there is a backup environment already in place, it is important to collect as much information as possible. This helps to determine what is currently in place and identify gaps that the new solution must address.

It is also important to decide what will become of the old backup environment. Is there a cutoff date where once the new system is in place and the old system is decommissioned? Does the old system need to be phased out by keeping the old hardware for a period of time? Or will the tape library be partitioned between the old and the new systems to allow restores of old data? Considering all questions ensures a seamless transition of your current backup environment.

Assess Data Protection Requirements

The protection requirements must be properly established for all data types during the planning phase. This includes the following factors:

- Recovery Time Objective (RTO)
- Recovery Point Objectives (RPO) for systems
- Required retention and copies of the different datasets

To help design the new environment, metrics on the size of datasets requiring protection must also be collected. For instance, 50TB of virtual machines, 20TB of physical machines, 15TB of SQL databases and so on. This information is used in the design phase to accurately size CommCell resources. A common mistake is referencing sizing metrics from the old backup system. Quite often the old backup system, being limited in resources, is not protecting all systems which require protection.

Design the CommCell Environment

This phase determines what options, features, and protection methods will be used to design and document the new CommCell environment. Designing a CommCell environment that fits a company's needs is not an exact science. It requires experience and knowledge of the software and available options.

Options to address in the design:

- CommServe server, hardware requirements, and high availability
- Number and size of MediaAgents and related hardware requirements
- Libraries that will be used (disk, tape, cloud), and related hardware requirements
- Protection agents and methods that will be used (e.g., Virtual Server Agent, SQL agent)
- Storage policies and schedule policies

Procure and Install Hardware

Once the CommCell environment is designed, the next step is to establish hardware requirements of every component and to procure and install the required hardware. If hardware is missing, determine if the deployment phase should be postponed. For instance, if a MediaAgent is missing the SSD drives for the Deduplication Database (DDB), it might be preferable to wait instead of placing the DDB on the C:\ drive. Even if the database can be moved later, it could lead to a drive that fills up before the SSD drives are received. It can also significantly underperform, therefore impacting backup performances.

Coordinate with the Change Management Team

Completing any necessary paperwork and following procedures avoids major impacts, such as avoiding a reboot or inhibiting network usage during peak hours.

Download Commvault Software

Installing Commvault software is a straight-forward process. However, it is important to note that a full CommCell deployment must be well-planned and executed. Components such as the network, hardware, operating systems, and applications all play a significant role in creating a sound data protection strategy. This section presents the high-level steps and best practices for deploying a CommCell environment.

Deployment Sequence

- Planning
- Downloading the Commvault software
- Installing the CommServe server and executing post-install tasks
- Installing the MediaAgents and executing post-install tasks
- Configuring storage and deduplication
- Configure solutions
- Apply security

Commvault offers several methods for obtaining the software installation media required for installing a CommCell environment. The Commvault software installation media is available for download from the Maintenance Advantage website or from the Commvault Software Cloud Services download center.

The Commvault Maintenance Advantage website or Commvault Cloud website provides the latest version of the Commvault installation media. Select the **Downloads & Packages** tile to access the most current software installations and feature releases or select a previous version.

Download Options

- **Resumable Download Manager** – Based on the bootstrapper download manager; select the required files from the list and then click **Launch Download Manager** at the bottom of the screen.
- **Bootstrapper Direct Download** – This option reduces deployment time by selecting only the required Commvault software components and download of installation media.

Download from Maintenance Advantage

The screenshot shows the Commvault Maintenance Advantage portal. At the top, there is a navigation bar with the Commvault logo, a search bar, and user information (Patrick Shuff). Below the navigation bar is a row of six tiles: 'Create a Support Incident', 'Knowledge Search', 'Downloads' (highlighted with a red circle and the number 1), 'Log Upload Manager', 'Customer Support Forums', and 'Appliance Resources'. Below the tiles is the 'Incident Management' section, which includes tabs for 'Active Incidents' and 'Resolved Incidents'. Below the tabs is a table with columns: Incident, Brief Description, CommCell, Owner, Severity, Status, and Created. The table is empty, showing 'No data available in table'. Below the table is a 'Customer Support Forums' section, which includes tabs for 'Most Recent Posts', 'My Colleagues Recent Posts', and 'My Recent Posts'. Below the tabs is a table with columns: Topic, Thread, and Date. The table contains two rows of data: 'Oracle Compression' with thread 'Backup and Recovery' dated '30-Nov-2019', and 'Properly deleting retired Virtual Machines from Commvault' with thread 'Backup and Recovery' dated '29-Nov-2019'. On the right side of the portal, there are several sections: 'Welcome Back, Patrick.', 'Maintenance Contract' (stating 'Your maintenance contract expires on 01-Jan-2023'), 'Contact Support' (with a phone icon), and 'Your opinion matters!' (with a speech bubble icon). At the bottom right, there is a 'Chat Now' button.

1. Click the **Downloads** tile

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[Product Support](#)
[Downloads & Packages](#)
[Resources](#)

v11 Software, Feature Release and Maintenance Release Downloads

Please Note: "Feature Releases" were formerly called "Service Packs." "Maintenance Releases" were formerly called "Hotfix Packs."

Feature Release / Service Pack	Status	Release Date	End of Life Date	Software Distribution Kit	Maintenance Release/ Hotfix Pack	Recommendation	New Hotfix Support
v11.24 (Long-Term Support)	Technology Preview	15-Jun-2021	15-Jun-2024			v11.24 is best aligned for POCs and new customer installations. Maintenance Releases will be issued for this release. v11.24 is a designated Long-Term Support (LTS) release. Monthly maintenance releases will be issued for year one and year two, and then as needed. LTS will expire for this feature release after three years.	
v11.23	General Availability	15-Mar-2021	15-Mar-2022			v11.23 includes the most recent features and is the recommended target for any new upgrades. Maintenance Releases will be issued for this release.	
v11.22	General Availability	15-Dec-2020	15-Mar-2022			v11.22 is a general availability release. Upgrading is not required for normal operations, but you can upgrade to take advantage of new features. Maintenance Releases will be issued for this release.	
v11.21	General Availability	15-Sep-2020	15-Dec-2021			v11.21 is a general availability release. Upgrading is not required for normal operations, but you can upgrade to take advantage of new features. Maintenance Releases will be issued for this release.	
v11.20 (Long-Term Support)	General Availability	15-Jun-2020	15-Jun-2025			v11.20 is a general availability release. Upgrading is not required for normal operations, but you can upgrade to take advantage of new features. Maintenance Releases will be issued for this release.	

- Click the Software Distribution Kit *download* icon under to download the media kits including the most recent feature release
- Click the Maintenance Release/Hotfix Pack *download* icon to download the hot fixes that were launched after the latest feature release

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Search

11.0

Product Support

Downloads & Packages

Resources

Home

/ v11 Software Release Lifecycle

/ v11.24 Electronic Software Distribution (June 15, 2021)

v11.24 Electronic Software Distribution (June 15, 2021)

Please Note: This media kit includes v11 Feature Release 24 (11.24) Next Generation Platform software. Customers deploying their Commvault enterprise from this media kit do not need to perform a separate Feature Release 24 install. Customers who have implemented their v11 enterprise using a previously issued media kit can use the bootstrap installer in this kit to install Feature Release 24 in their enterprise.

For documentation on the installation of your Next Generation Platform ILO software, please visit the [Next Generation Platform ILO Books Online](#).

Using Bootstrapper Installers: The system time of the computer must be synchronized with the real time of your time zone. If the system time is ahead of or behind the real time, the installation of components using the Bootstrapper Download Manager will fail. The Bootstrapper Download Manager does not support downloads of all software and/or agents. Please refer to the ISOs listed below for access to the installation packages for this software and/or agents.

For additional detail regarding the management of package creation and distribution for the Bootstrapper Download Manager, please refer to [Next Generation Platform ILO Books Online](#).

If you are looking for details regarding an ESD kit that you have previously downloaded, please visit the [ESD Kit Information Archive](#) for information on each of the files in our previously issued ESD kits.

Download Packages

Maintenance Releases

4

Download

v11 Download Manager with Feature Release 24

MD5 Sum	82807f64fcdc5139cedf30d387956661
Download File Size	22.66MB

All Windows, Unix, Linux and Macintosh components, which includes the following:

- CommServe® (Windows x64 only)
- Microsoft SQL 2016 software (with Service Pack) for the CommServe® Database Engine
- Commvault® I-Touch Server (Windows)
- Media Agent (Windows, AIX, HP-UX, Linux and S)
- CommCell® Console (Windows, Linux and Maci

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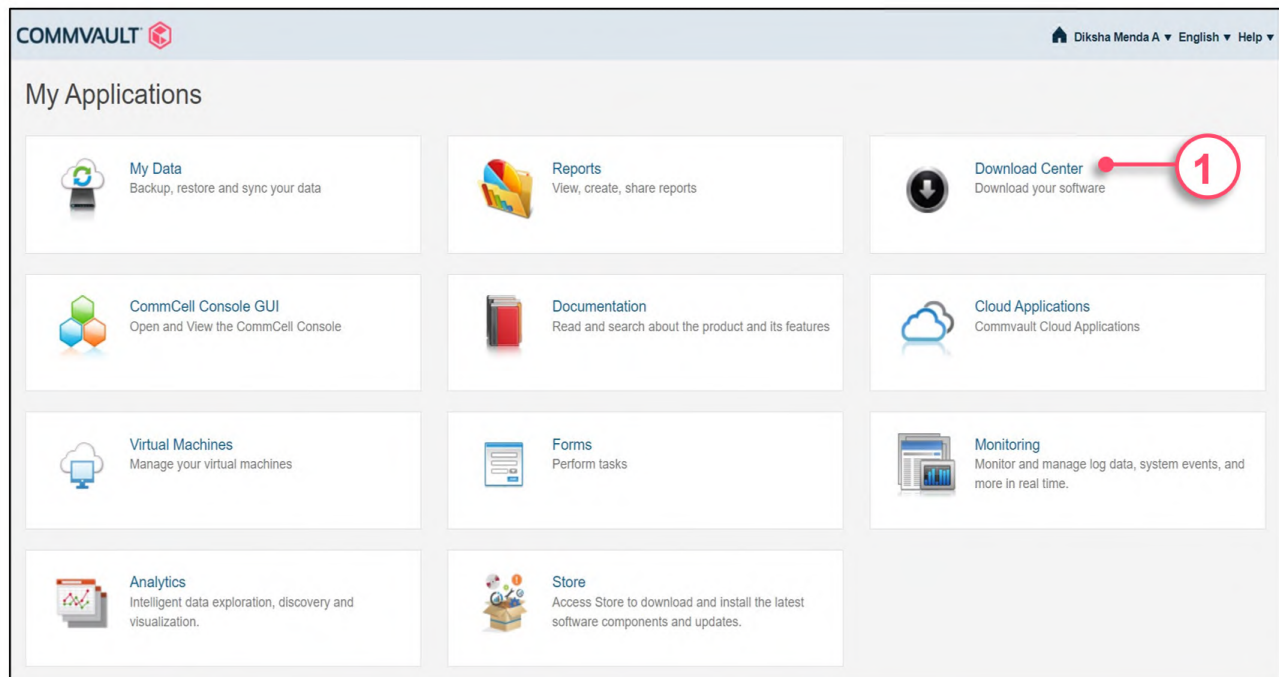
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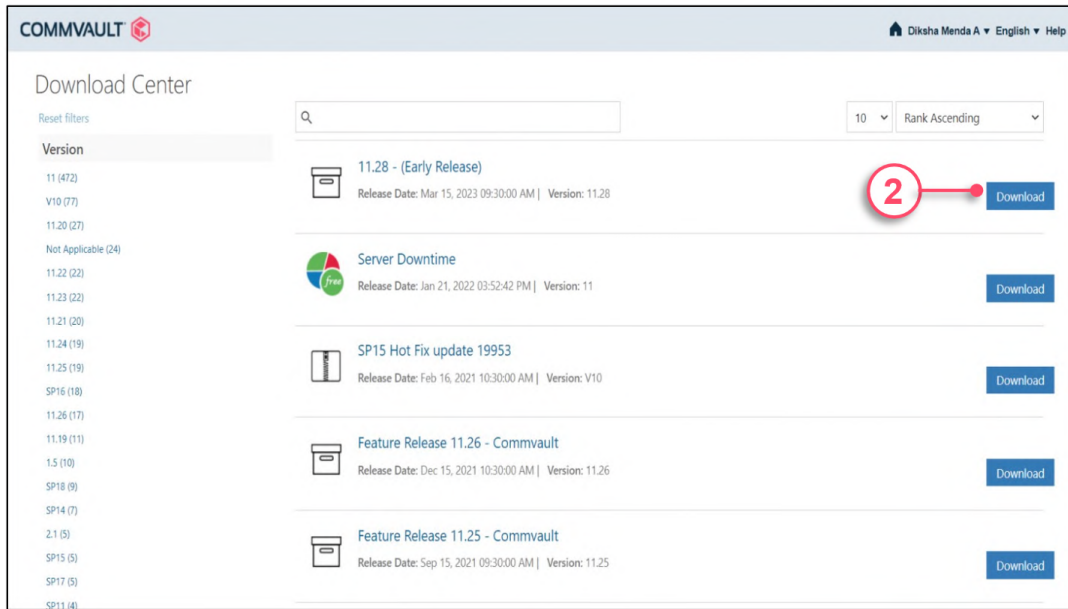
4. Click to launch the downloader

Page 37 of 106

Module 1 - Deployment

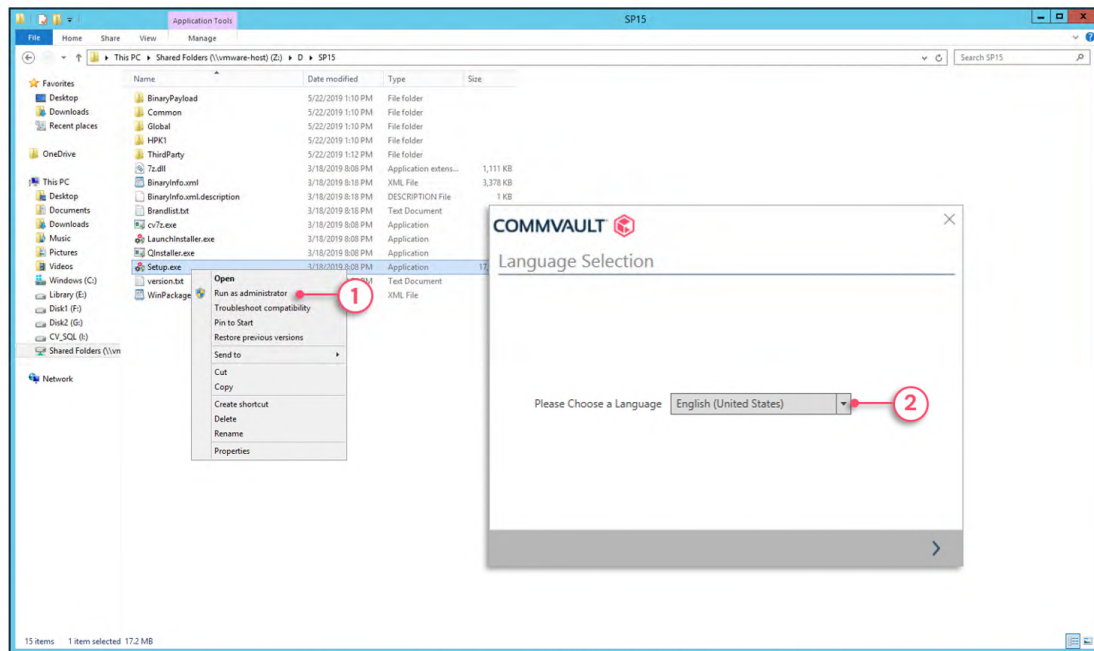


1. Click the **Download Center** tile

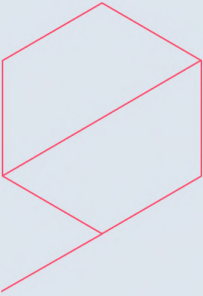


2. Click **Download** for the desired version

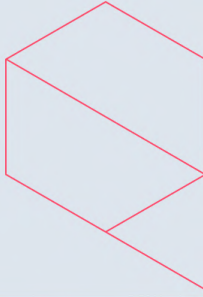
Run the Setup.exe Command



1. Once the file is downloaded, extract the contents into a directory and run the Setup.exe as administrator
2. Follow the prompts to install the CommCell, MediaAgent, and optional agents

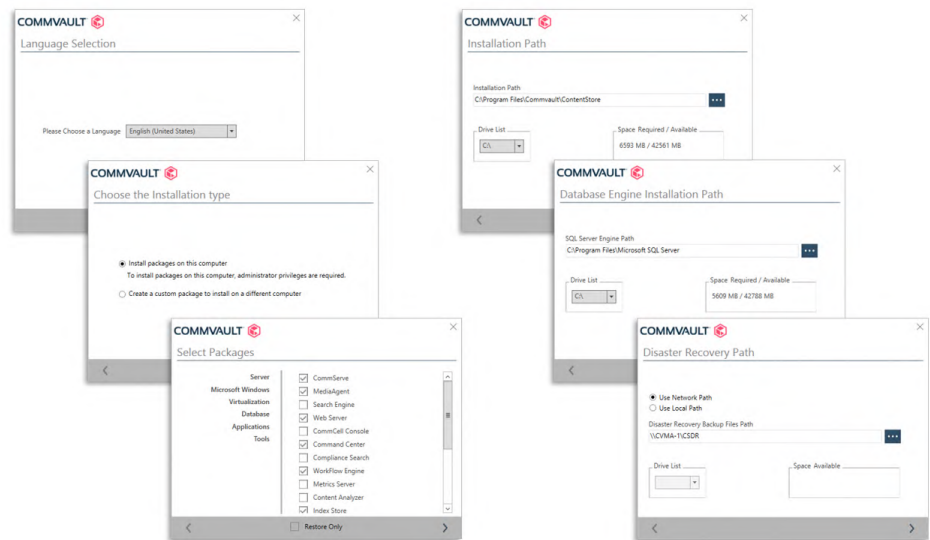


CommServe® Server Installation



CommServe® Server Installation Overview

- Installation information
 - Additional packages
 - SQL install directory
 - CommServe® database directory
 - Commvault® software install directory
 - DR share



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When deploying Commvault software, it is important to note that every environment is different relative to the available infrastructure, technology, budget, culture, and requirements of the organization. Whether performing a new installation, an upgrade, or expanding an existing environment, a good amount of planning should take place prior to installing Commvault software. The more emphasis put into planning, the more likely the deployment will go smoothly.

- Commvault software must be downloaded prior to installation. Plan for routinely downloading updates and for uploading log files from the CommServe host. If the CommServe will not have internet access, alternate methods should be discussed and documented.
- Determine the location for a local and remote software cache. The software cache is a directory where Commvault updates and software packages are stored. These position the software to be routinely accessible throughout the organization and are essential for disaster recovery.
- Verify the hardware and system requirements.
- Assess the size of the environment to allow for adequate resource availability for the CommServe server.
 - o Based on the sizing assessment, determine if the CommServe server will be physical or virtual.
 - o Determine if the CommServe server needs to be deployed in a clustered configuration for high availability.
- Ensure the operating system meets the Commvault specifications and patched with updates prior to the installation.
- Determine if the method of deployment requires additional considerations for Disaster Recovery. For example, configuring a 'Floating Host Name' for the CommServe server.
- Evaluate additional components such as metrics reporting or if the workflow engine will be installed on the CommServe server.
- Determine the methods for accessing the CommCell console and/or Commvault Command Center. These are installed by default along with the CommServe components.

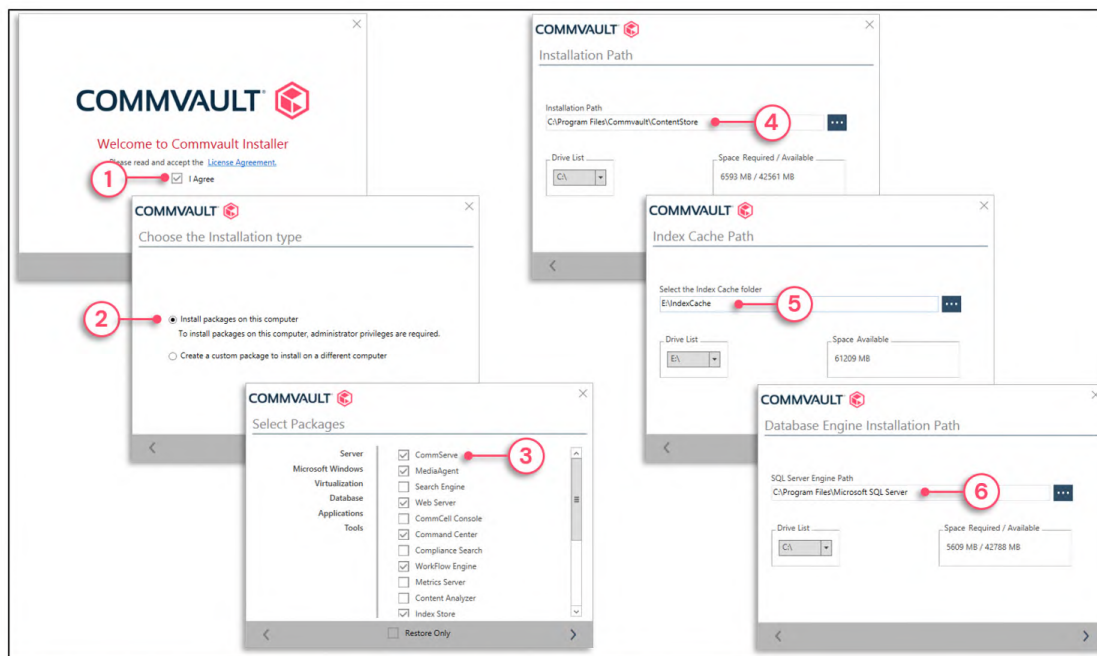
- IIS is required for the Web Server and Web Console and are automatically installed when IIS is enabled on the CommServe server.
- Although not always required, reboots may be required to complete an installation or update. The organization may require the changes be implemented after hours.
- Document the firewall and network configurations. Unless performing a decoupled install, all software components must communicate with the CommServe server during installation.
- Identify any monitoring, security, or anti-virus software that will be installed on the same systems as Commvault software. Installation and Operations may be blocked, or performance severely degraded by such software. This can be avoided by applying the appropriate exceptions or filters for the Commvault software in advance.
- Configure service and administrative accounts before the installation. The account type and permissions required are determined by the components being deployed. A thorough review of the deployment will help determine the needs.
- For the CommServe server, an account with local Administrator privileges is required for the software installation.
- A password for the CommCell 'admin' account is configured during the installation. This password should be a complex password and the primary administrator should always use this account when managing the environment.
- A permanent license file must be applied after the CommServe software is installed. Ensure that any pending purchase agreements are completed prior to the deployment of the Commvault software.

Prepare for Installation

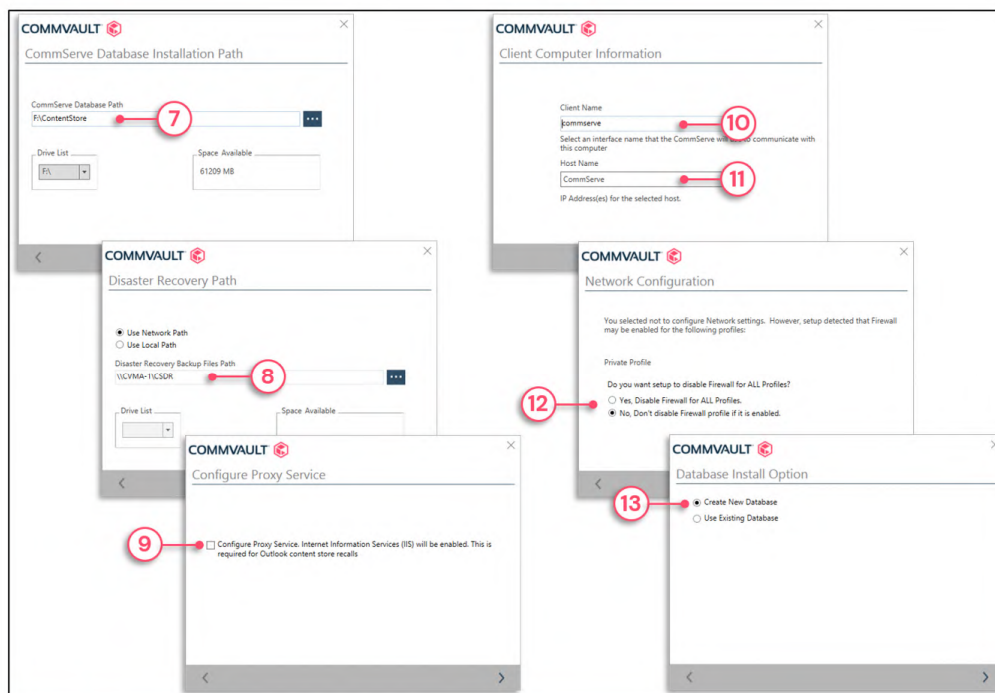
Attribute	Value (Examples)
Additional Packages	File system agent, MediaAgent, Workflow engine, Web Server
SQL Install Directory	Drive:\Program Files\Microsoft SQL Server
SQL Database Directory	Drive:\Commvault\Database
Commvault Software Directory	Drive:\Program Files\Commvault\ContentStore
DR Share	\\DRCommserve\CSDR
Software Cache	Drive:\Program Files\Commvault\ContentStore\Software

Proper documentation of the CommCell components being installed is essential for a smooth deployment. The following chart provides a sample of the information that must be obtained for the CommServe deployment. Having this information in advance will help the deployment go quicker, bring any shortcomings to the surface, aid in verifying site readiness, and serve as a template for post deployment documentation.

Install the CommServe® Server



1. Accept the license agreement
2. Select to install packages on this computer
3. Select the CommServe package as well as any other required components
4. Define the installation directory for Commvault software
5. If the MediaAgent package was also selected, enter the location for the Index Directory
6. Enter the location where the Microsoft SQL software will be installed



7. Select the folder in which the CommServe Server database will be created. It is recommended to use a dedicated volume
8. Define the location to store the CommServeDR backup exports. It is recommended to use a network location, preferably offsite
9. If you plan on using email archiving, check to configure IIS for recalls
10. Define the CommServe server name as displayed in the GUI
11. Provide the host name to use to reach the CommServe server
12. If Windows Firewall is enabled, define if it must be disabled or not
13. During a new install, select **Create New Database**, or select **Use Existing Database** during a disaster recovery or staging scenario



14. Enter an email address and a password for the built-in admin account

15. Click **Finish**

Software Cache Options

During the installation, the software packages and updates are copied to local disk. This software cache can be leveraged to push Client Agent software and updates to other servers in the environment. These settings can be changed at any time.

Cluster Setup Options

If the CommServe server is being installed in a cluster, log into the active node with an administrative account and then run Setup.exe from the installation media. The Cluster Setup Install Option page is displayed during the installation. After completing the selections on the active node, it may be necessary to log into the remaining cluster nodes and repeat the installation process. The installation will apply the missing components to the cluster node.

To assist in troubleshooting installation errors, check the following log:
%allusersprofile%\Commvault Systems\Galaxy\LogFiles\Install.log.

If the error occurs after the summary page, check the installation logs in the Software_Installation_Directory\Log Files directory.

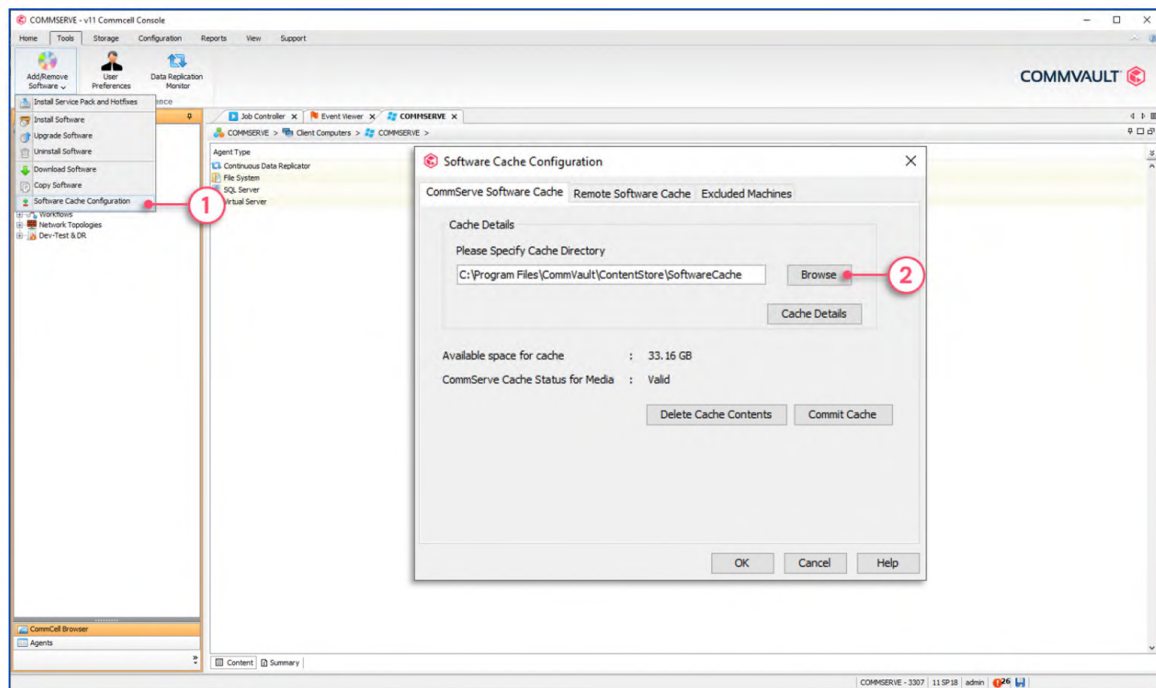
CommServe® Server Post Installation Tasks

Platform releases and maintenance updates are available for download from the Maintenance Advantage website. They are also included as automatic updates via the Commvault software update cache process approximately two weeks after initial availability. Platform releases should be deployed when available.

Additional hotfixes may follow the release of a feature release to address critical issues. The hotfixes, included in maintenance releases, should also be applied as soon as they are available. Subsequently, if a new platform release or maintenance release hotfix conflicts with an older installed hotfix, the installation process removes them automatically.

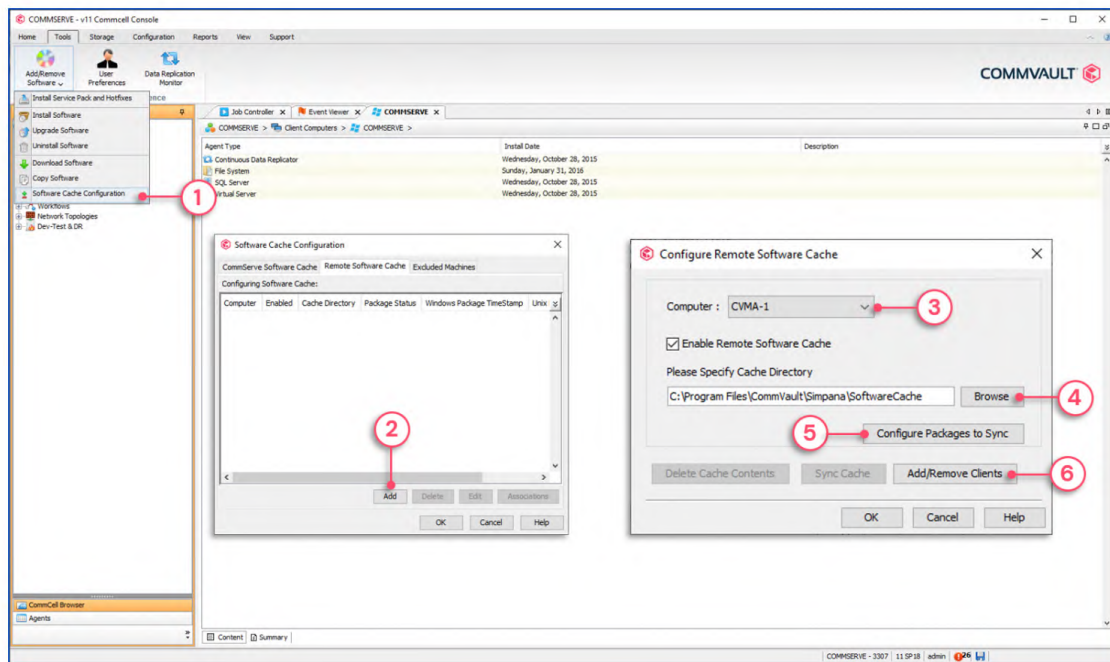
By default, the system creates automatic schedules that download and install updates on Commvault servers, as well as on clients. These schedules can be modified as needed.

Configure the Main Software Cache



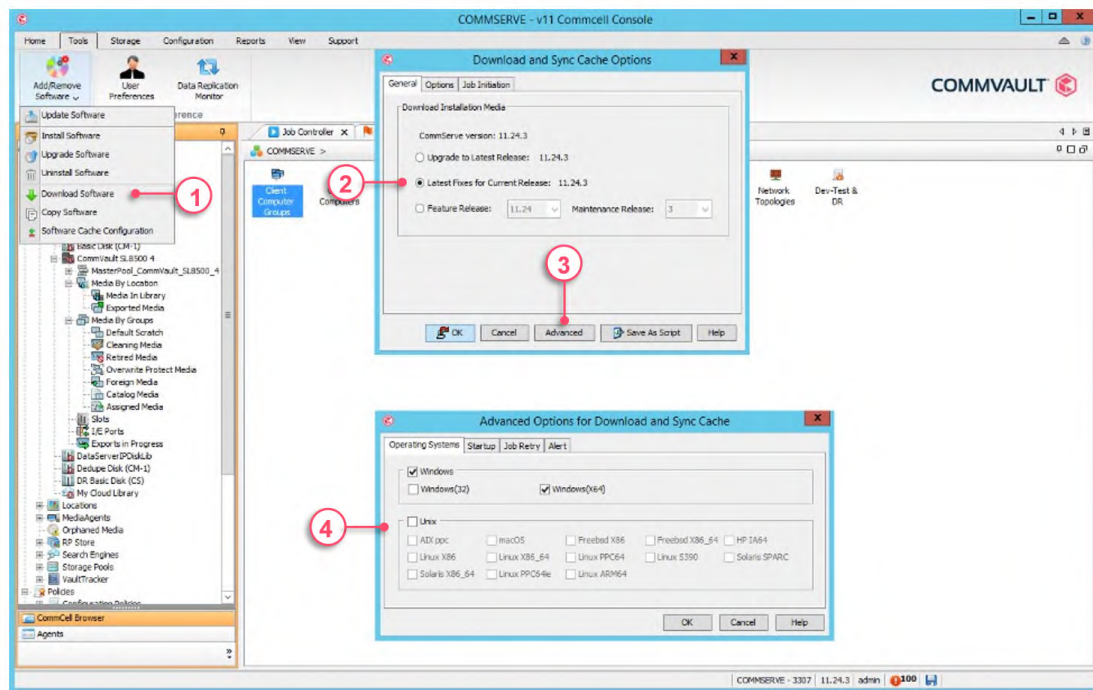
1. *Tools* tab > **Add/Remove Software** > **Software Cache Configuration**
2. *CommServe Software Cache* tab > **Browse** to specify the local path to store software and updates

Configure a Remote Software Cache



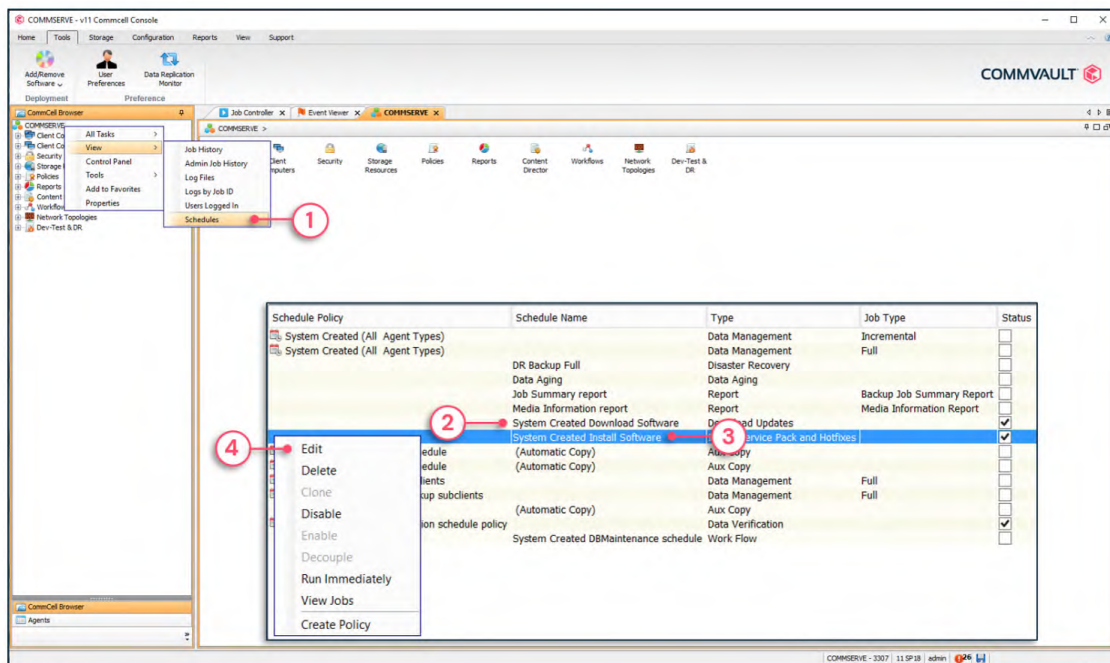
1. *Tools* tab > **Add/Remove Software** > **Software Cache Configuration**
2. *Remote Software Cache* tab > **Add** to add the machine used as a remote cache
3. Select the client that will host the remote software cache
4. Click **Browse** to define the folder in which to store the packages
5. Configure the packages to sync
6. Define clients that will get updated from this cache

Populate the Software Cache



1. **Tools** tab > **Add/Remove Software** > **Download Software**
2. Choose the platform release level
3. Click **Advanced** to open the advanced download and sync cache options
4. Choose agent types to download

Configure Automatic Updates



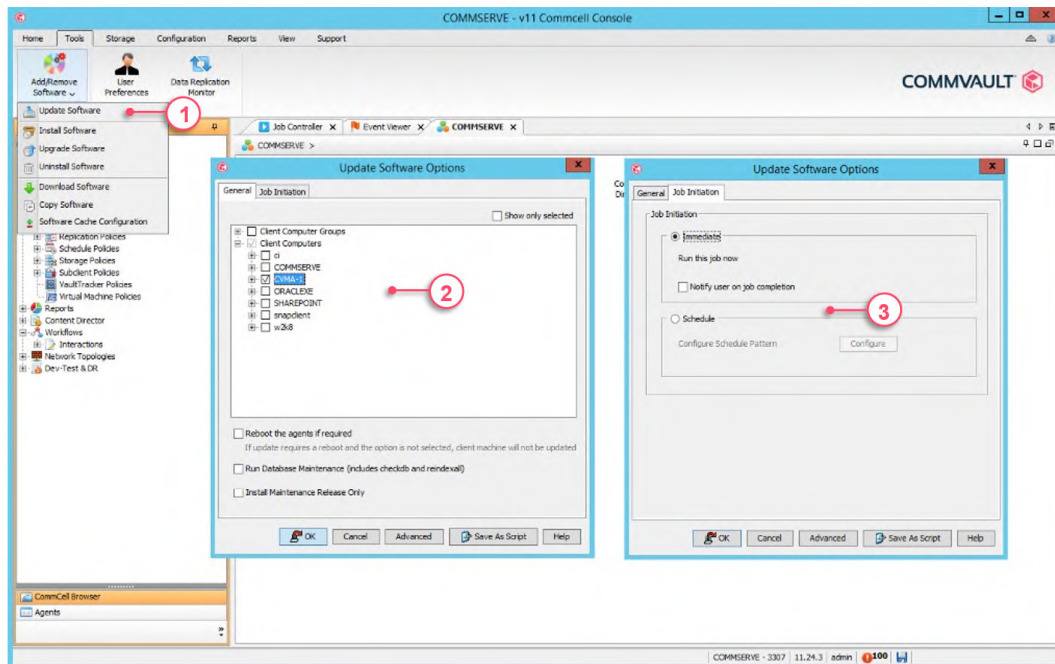
Automatic update schedules are as follows:

- **System Created Download Software** – Download the updates automatically in the software cache once a week if new updates are available.
- **System Created Install Software** – Automatically install updates on Commvault servers and clients once a week if required. For instance, many companies have change control procedures in place. Installing updates automatically on servers might go against these procedures. In this case, the System Created Install Software schedule can be modified or simply disabled.

Steps for Configuration

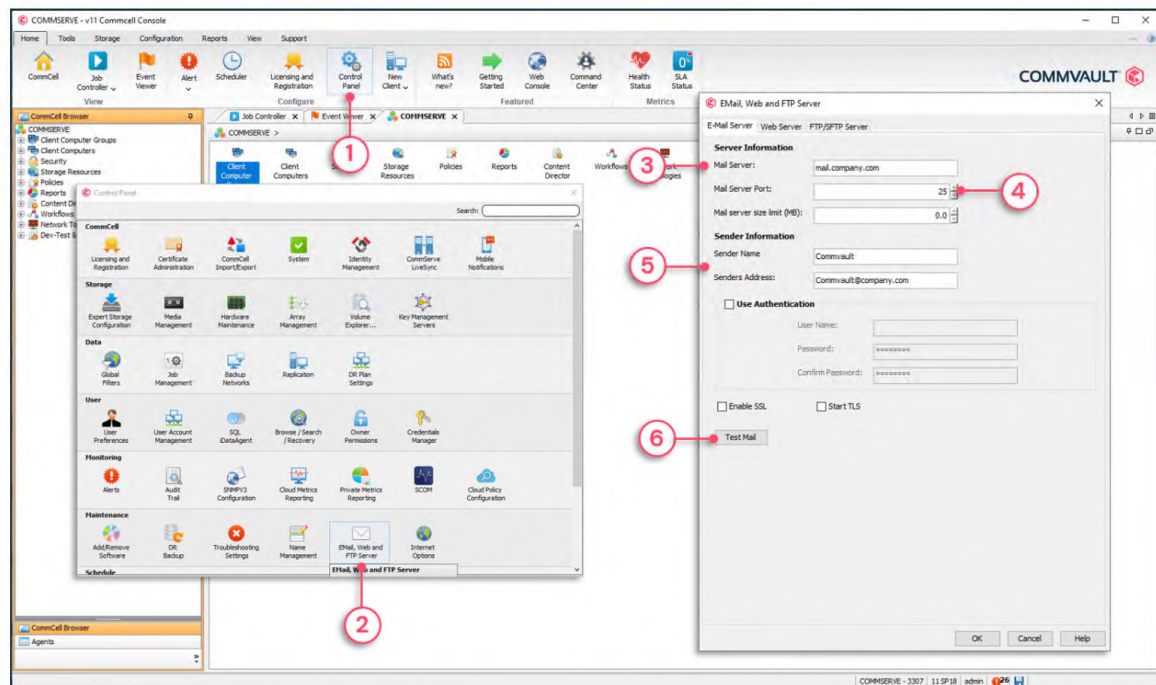
1. Right-click the **CommServe Server > View > Schedules**
2. Indication of schedule that automatically downloads updates in the cache
3. Indication of schedule that automatically installs updates on servers
4. Right-click to edit, disable, or execute immediately

Install Updates to the CommServe® Server



1. **Tools** tab > **Add/Remove Software** > **Update Software**
2. Choose the clients or client group
3. Choose either to install immediately or schedule for later

Email Settings



Configure the email server to allow Commvault software to send alert notification or reports by email. If your corporate mail server is secured, it is important to understand the level of security. Commvault software uses SMTP which must be allowed on the mail server for the CommServe server IP address. Refer to your software vendor documentation for more information about SMTP relay and the mail server.

1. Home tab > **Control Panel**
2. Maintenance section > **Email, Web, and FTP Server**
3. Enter the email server name
4. Define the SMTP port to use
5. Enter the sender's name and email address displayed to recipients
6. Click **Test Mail** to send a test email to the user currently logged in

User Accounts and Security

Commvault software uses two types of accounts to login to the CommCell console: Commvault user account or Active Directory integrated account. When the system is installed, a Commvault administrative account, named 'Admin' is created. This account has all privileges and permissions in the Commvault software. It cannot be disabled, and its permissions cannot be modified. This account is used to login to the CommCell environment in case of a disaster of the domain controller that prevents the use of Active Directory integrated accounts.

Additional accounts and roles can be created to allow users to perform specific tasks in the system. For instance, a role allowing only backup and restore operations can be created and assigned to the SQL servers and DBA Active Directory account. The result is that a DBA can now connect to the console and only see SQL servers, on which he or she could only run backups and restores.

Security Best Practices

- In addition to the Admin account, create at least one CommCell account having all privileges.
- Ensure primary administrators always use local accounts when managing Commvault software.
- Document the passwords of these two accounts with the DR plan.
- Create an Active Directory connection to allow the use of Active Directory integrated accounts.
- When creating a role and security rule, use a test user first to validate if the rule is properly created and gives the expected result.

Each administrator should have their own account. This helps keep track of changes made by each user.

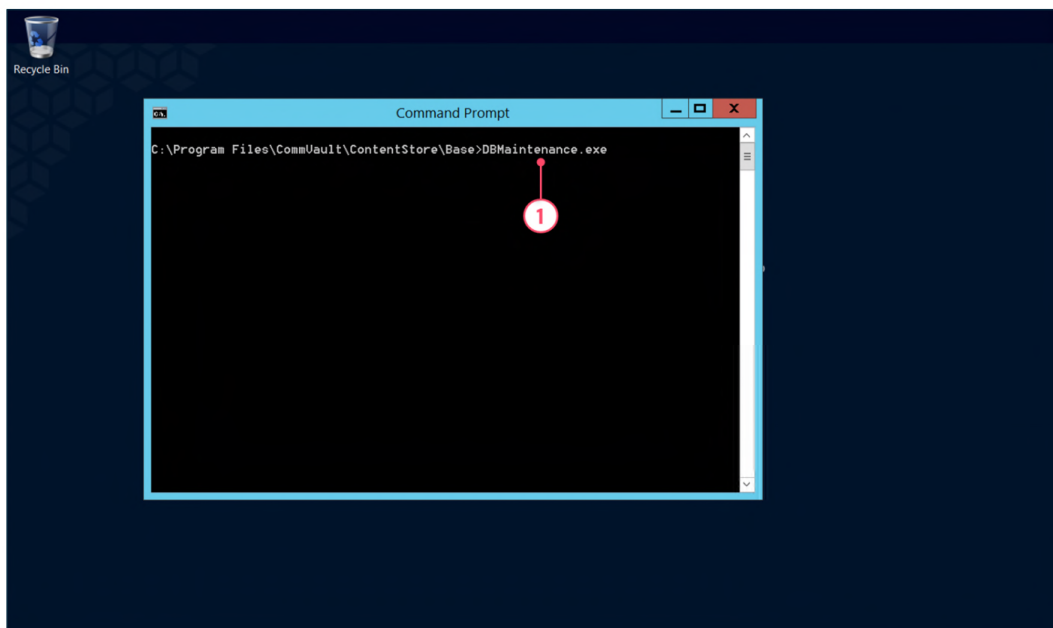
CommServe® Server Database Maintenance

It is crucial to ensure that the CommServe server is maintained for maximum productivity. Access DBMaintenance.exe to optimize the CommServe server database.

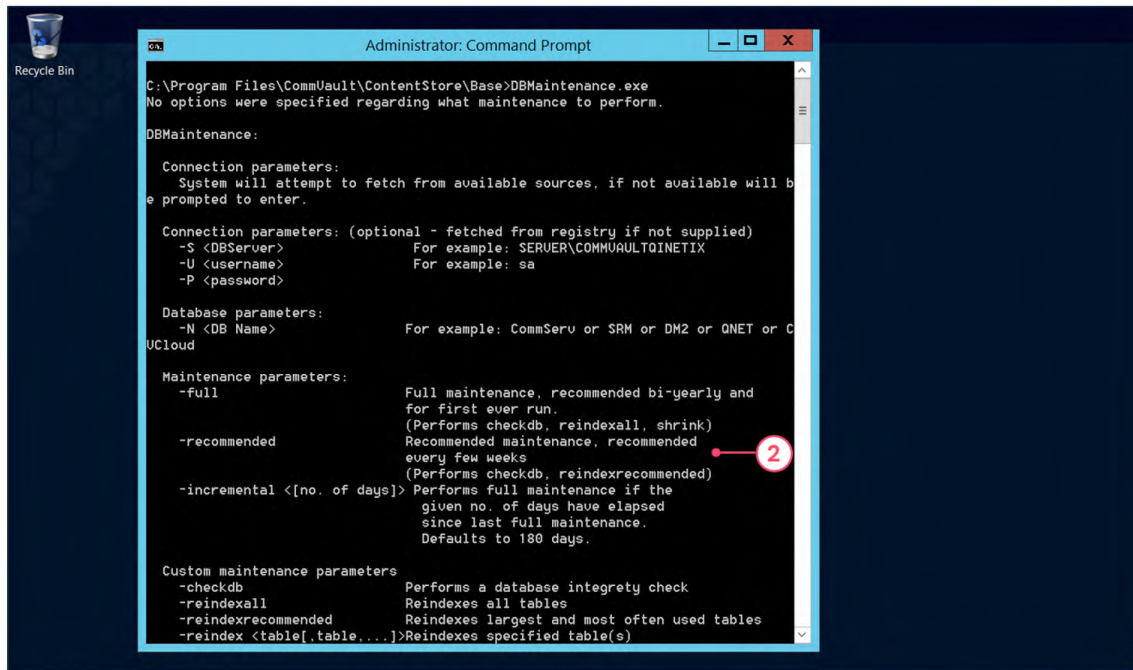
Maintenance Modes

- **Full** – Performs a full maintenance on the database. It includes CheckDB, ReindexAll and ShrinkDB commands. It is recommended to run this twice each year.
- **Recommended** – Performs a recommended maintenance including ShrinkDB and ReindexRecommended commands. It is recommended to run this maintenance mode every two weeks. By default, a system created schedule will execute it on every other Sunday.
- **CheckDB** – Validates the consistency of the CommServe database by running an integrity check.
- **ReindexRecommended** – Re-indexes the largest and most frequently used tables of the database.
- **ReindexAll** – Re-indexes all tables of the database.
- **ShrinkDB** – If table re-indexing creates a significant amount of fragmentation, the ShrinkDB command will reclaim that space by shrinking the database.

Run the DBMaintenance Tool



1. From the base directory, execute DBMaintenance.exe



```
C:\Program Files\CommVault\ContentStore\Base>DBMaintenance.exe
No options were specified regarding what maintenance to perform.

DBMaintenance:

Connection parameters:
  System will attempt to fetch from available sources, if not available will be
  prompted to enter.

Connection parameters: (optional - fetched from registry if not supplied)
  -S <DBServer>           For example: SERVER\COMMVAULTQINETIX
  -U <username>           For example: sa
  -P <password>

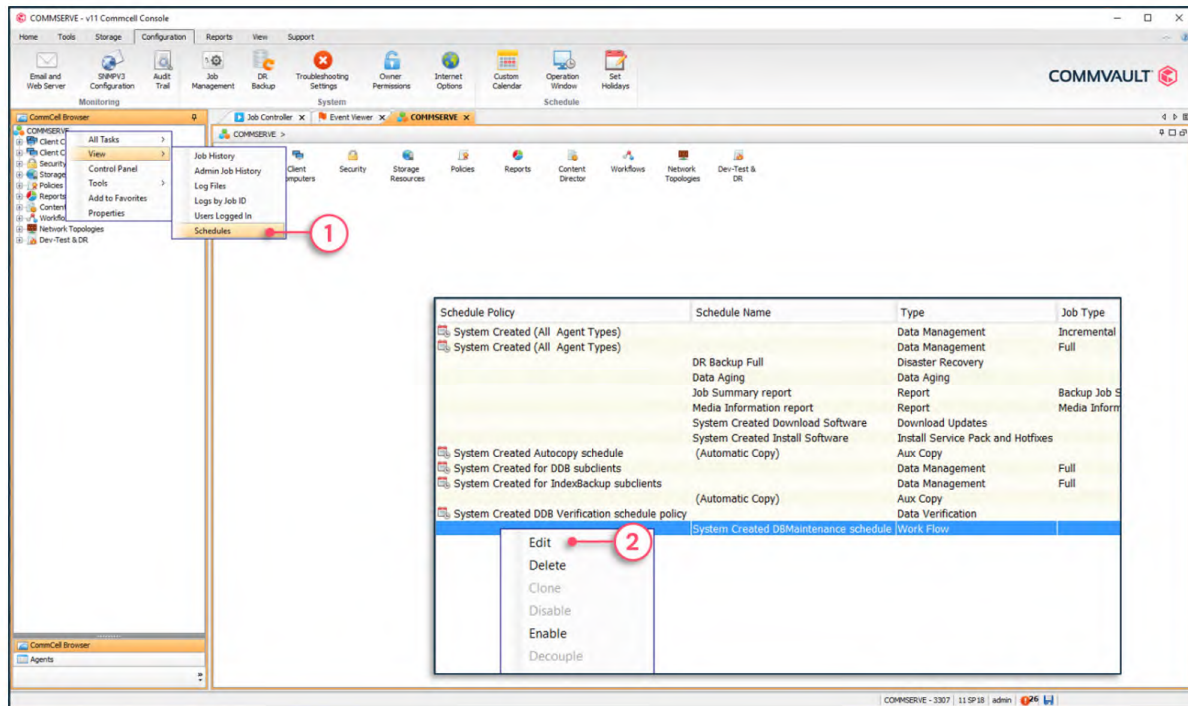
Database parameters:
  -N <DB Name>           For example: CommServ or SRM or DM2 or QNET or C
  UCloud

Maintenance parameters:
  -full                   Full maintenance, recommended bi-yearly and
                        for first ever run.
                        (Performs checkdb, reindexall, shrink)
  -recommended           Recommended maintenance, recommended
                        every few weeks
                        (Performs checkdb, reindexrecommended)
  -incremental <[no. of days]> Performs full maintenance if the
                        given no. of days have elapsed
                        since last full maintenance.
                        Defaults to 180 days.

Custom maintenance parameters
  -checkdb               Performs a database integrity check
  -reindexall            Reindexes all tables
  -reindexrecommended    Reindexes largest and most often used tables
  -reindex <table[,table,...]> Reindexes specified table(s)
```

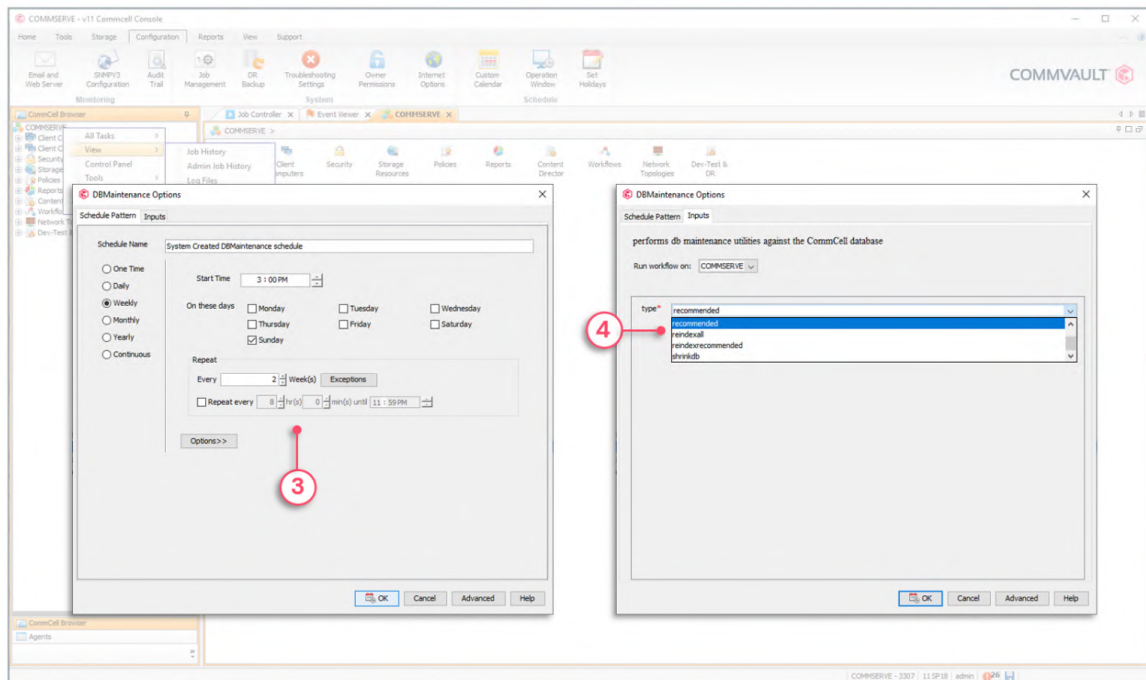
2. It provides the correct syntax and options for the tool, such as the maintenance mode

DB Maintenance Workflow and DB Maintenance Schedule



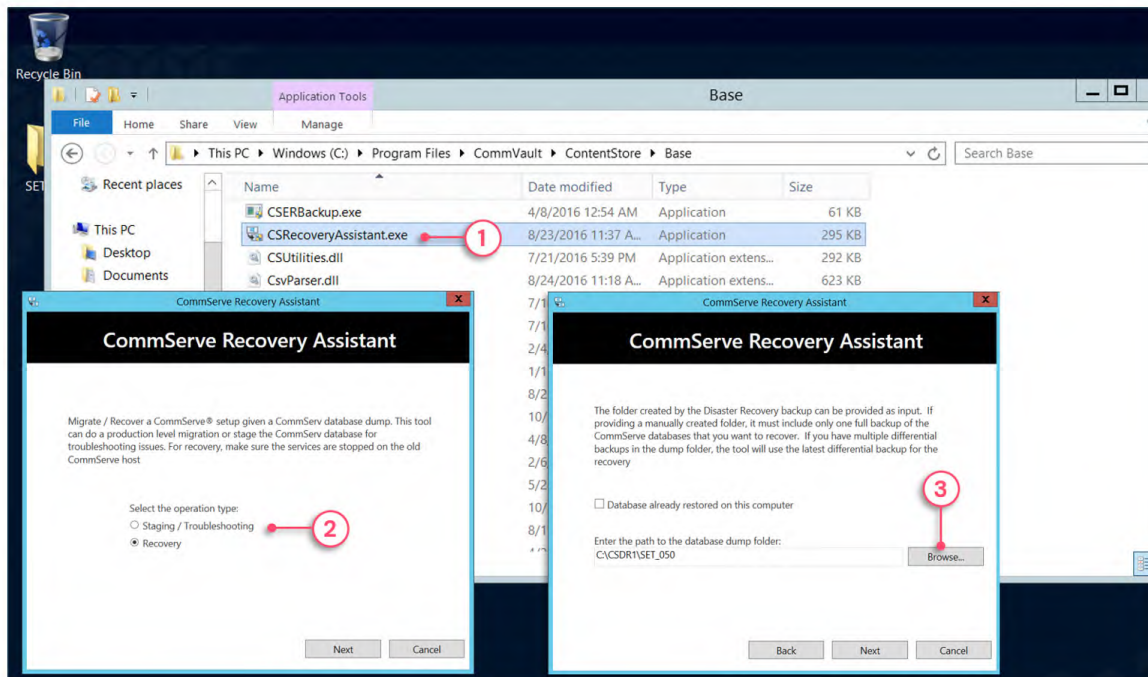
The System Created DB Maintenance schedule runs every other Sunday at 3:00 PM and executes a recommended maintenance. The Full maintenance is not automatically scheduled. It is therefore recommended to either run it manually or schedule it twice a year. This schedule executes a workflow called 'DBMaintenance,' which executes the maintenance based on the mode that is selected in the schedule. The workflow also contains email components that can be modified to send a result notification on failure or success.

1. Right-click the **CommServe server > View > Schedules**
2. Right-click the **System Created DBMaintenance schedule > Edit**



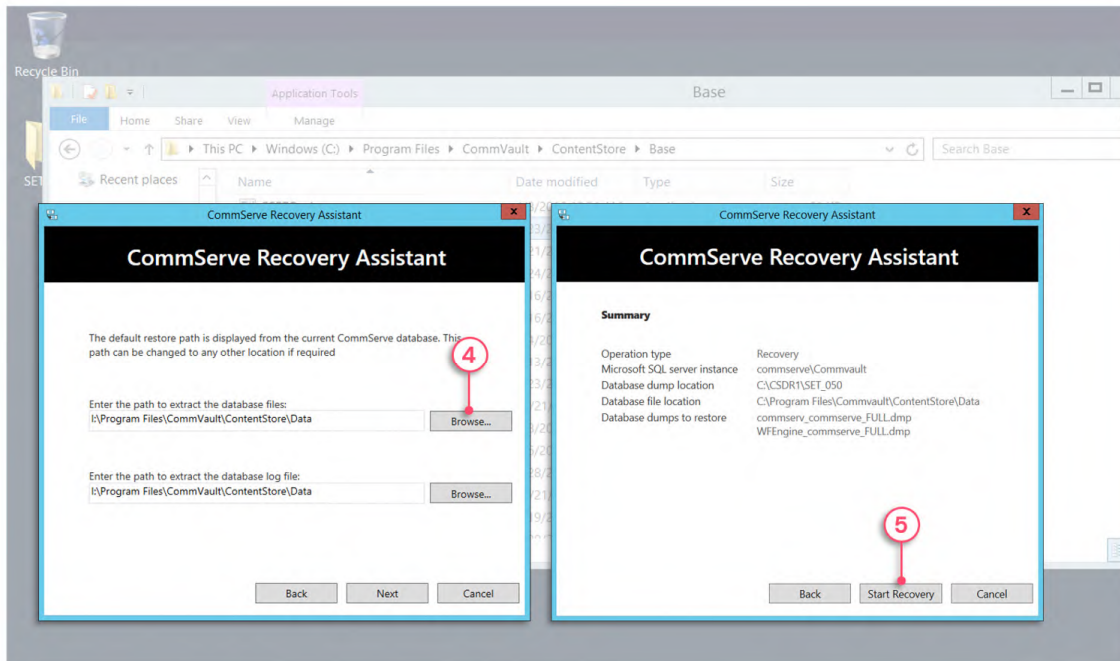
3. The DBMaintenance runs every two weeks on Sunday afternoon by default
4. By default, it runs in recommended mode, but a different mode can be manually selected

Recovery Assistant

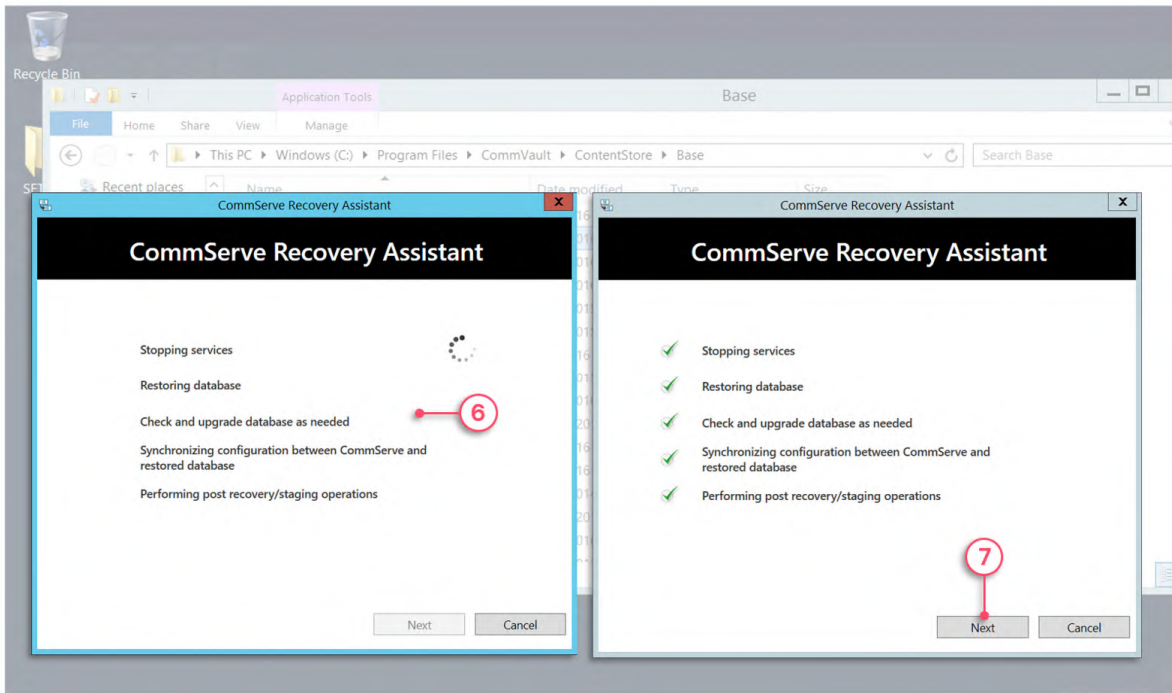


The CommServe Recovery Assistant tool is used to restore the CommServe database from the DR backup. The tool is used to rebuild the CommServe server on the same or different computer, change the name of the CommServe host, and update the CommCell license.

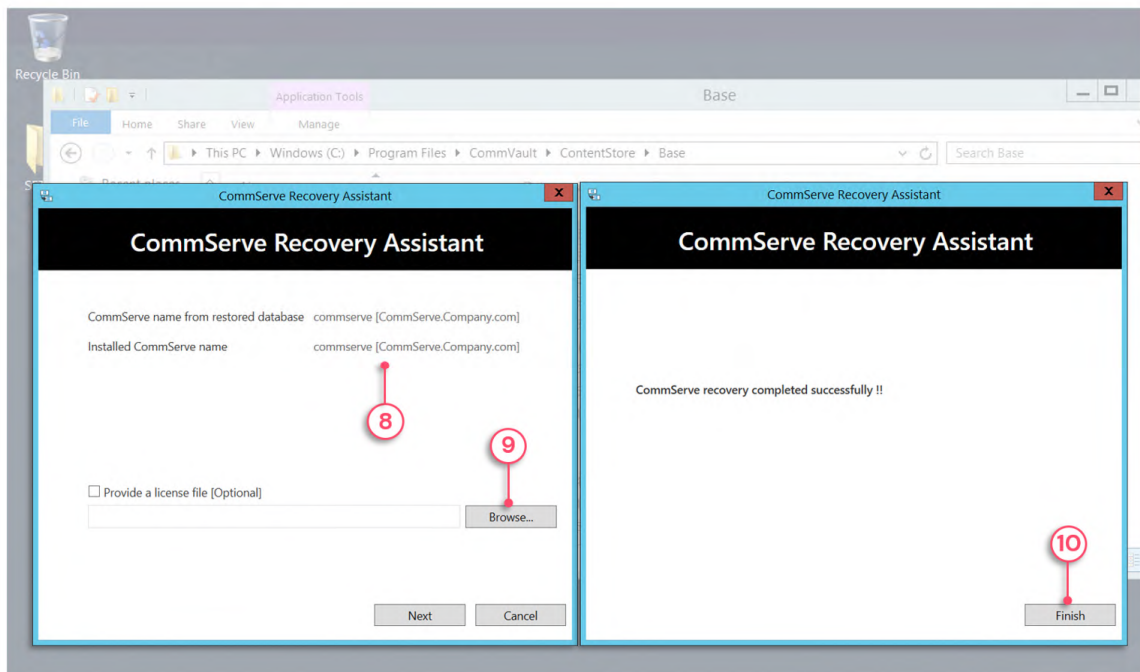
1. From the installation directory, execute CSRecoveryAssistant.exe
2. Choose the operation type
3. Click **Browse** to provide the location of the DR Backup



4. Click **Browse** to provide the location where to restore the database and logs
5. Click **Start Recovery** to start the recovery



6. Status of recovery steps are displayed
7. Once recovery is completed, click **Next**



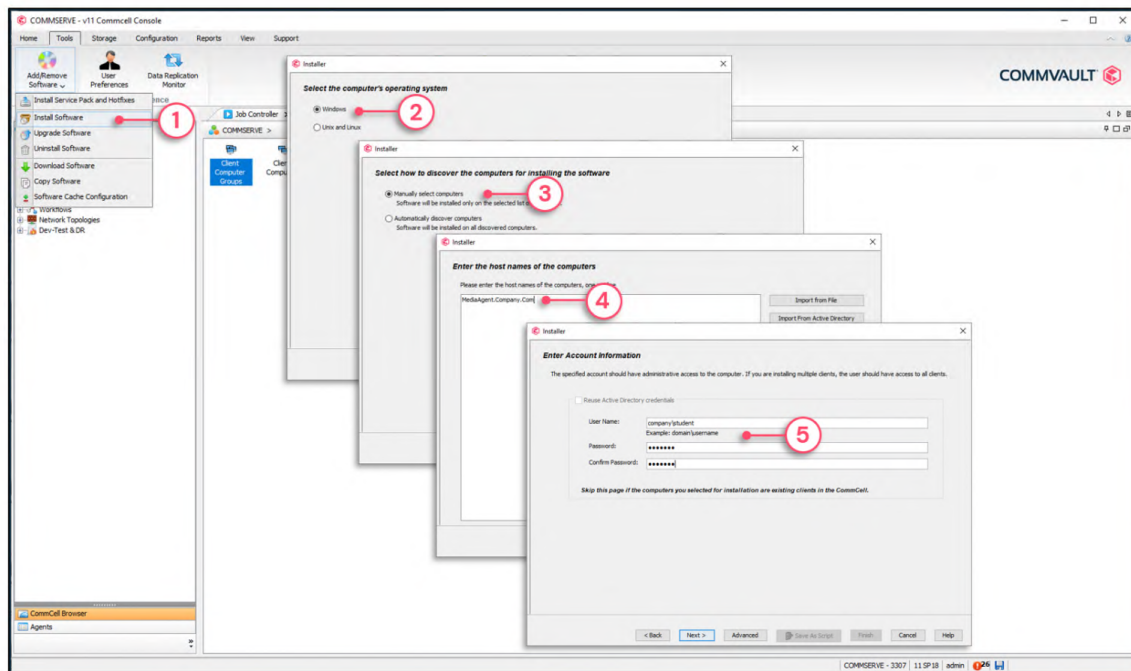
8. If restoring to an alternate server, the CommServe server name is changed
9. A license file can be provided
10. Once completed, click **Finish**

MediaAgent Installation

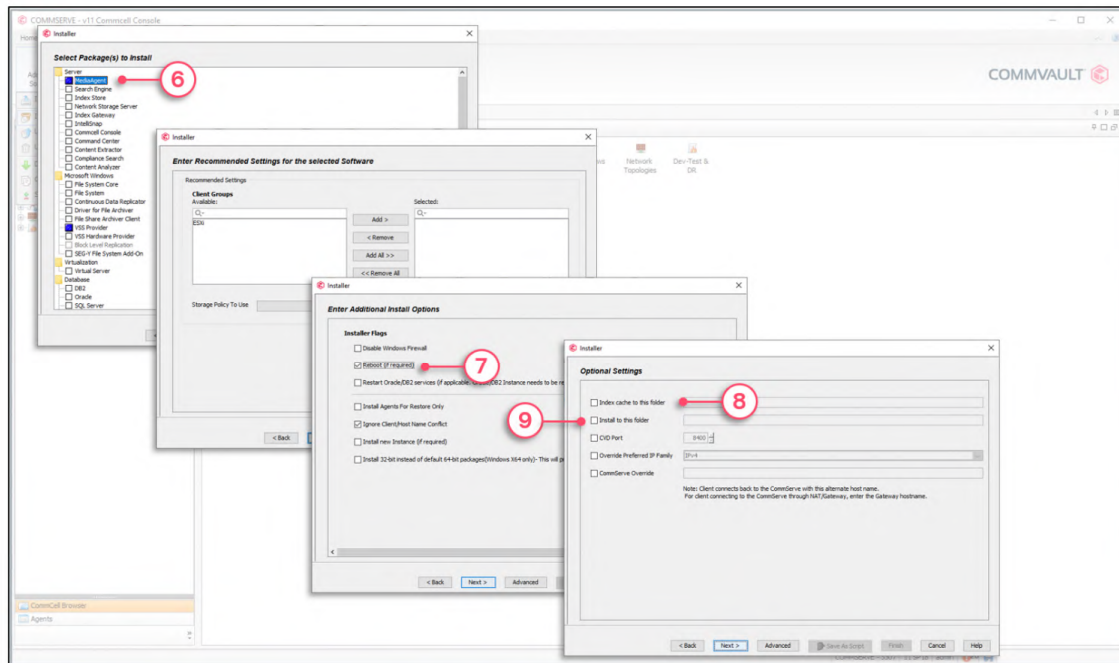
MediaAgent Installation Overview

The next component to install is the MediaAgent. MediaAgents move and retrieve protected data from disk, tape, and cloud storage. The MediaAgent software is pushed to a server directly from the software cache using the CommCell Console.

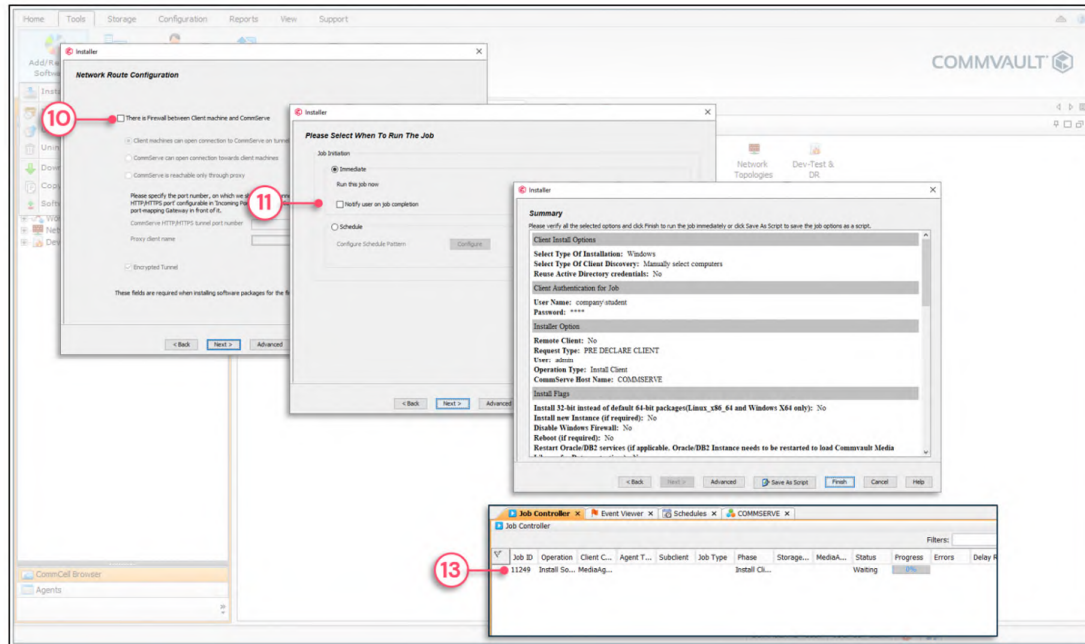
Install the MediaAgent Package



1. *Tools* tab > **Add/Remove Software** > **Install Software**
2. Choose the platform to use
3. Choose **Manually select computers**
4. Enter the MediaAgent server names, one per line
5. Enter credentials that have administrative rights on the systems

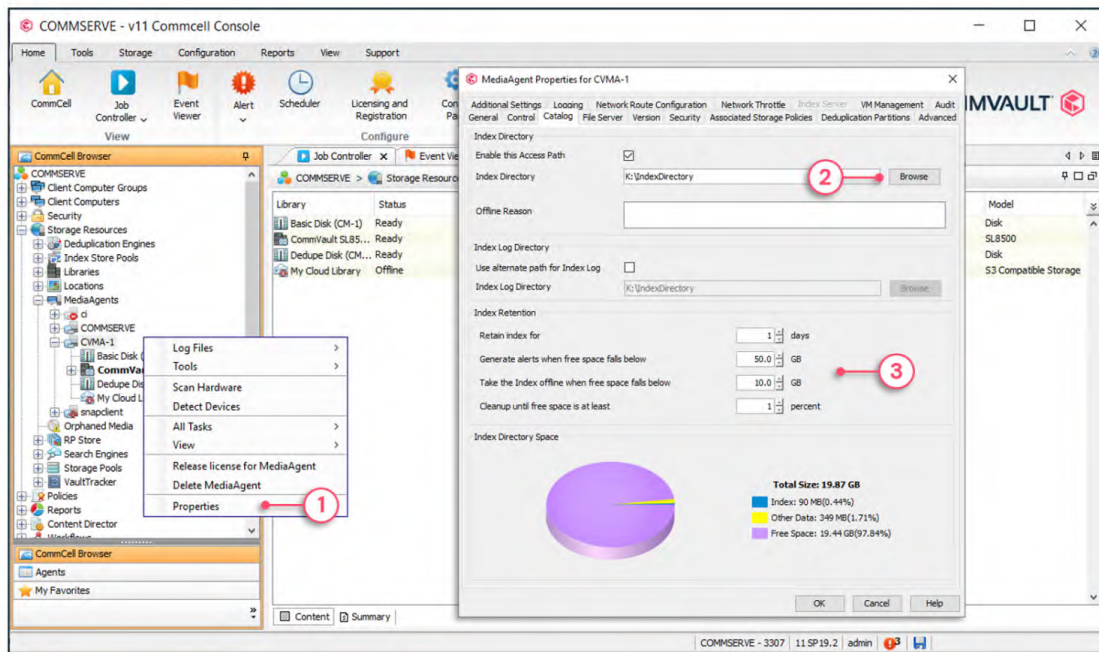


6. Check the box for **MediaAgent**
7. Choose **Reboot** if required
8. Provide the location for the Index directory
9. Change the installation directory



10. Check this box if there is a firewall between the MediaAgent and the CommServe server, and network routes are configured
11. Installation can be executed immediately or scheduled
12. Click **Finish**
13. Installation job appears in the Job Controller window

Access the Index Directory

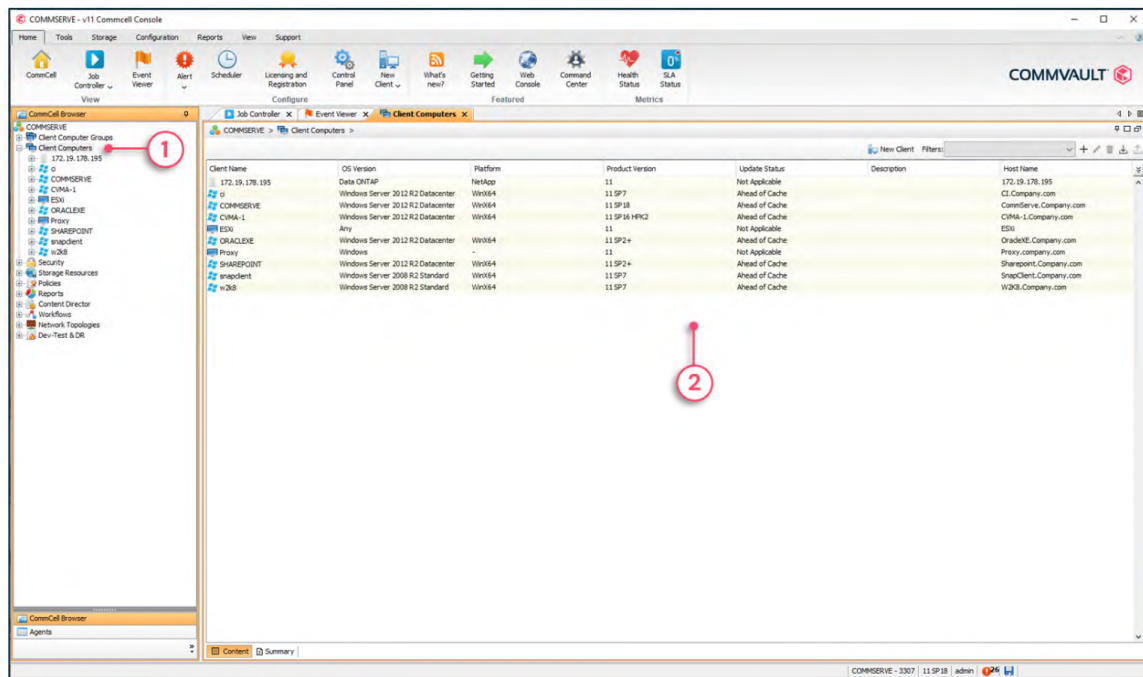


The index directory must be located on a dedicated high-speed disk, preferably solid-state drives. Index directory performance is critical when streaming a high number of jobs to a MediaAgent and when conducting DASH full operations. When MediaAgent software is first deployed to a server, the location of the index directory will be on the system drive. It is recommended to change the location to a dedicated drive prior to any jobs running.

1. Right-click the **MediaAgent** hosting the Index Directory > **Properties**. From the *Properties* window select the *Catalog* tab
2. Click **Browse** to change the location of the Index Directory
3. Set the index retention thresholds

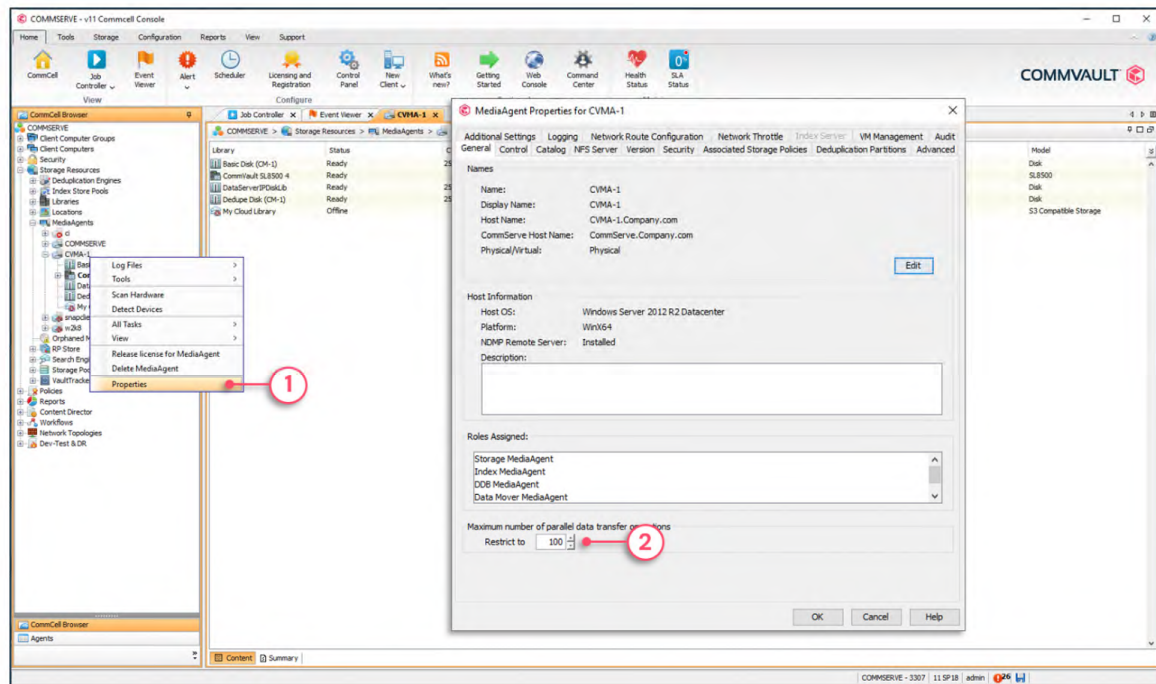
Note: Days and percentage apply to V1 indexing clients only.

Apply Updates to the MediaAgents



1. Check the status of updates by selecting the client computers in the CommCell Browser
2. Product version, Service Pack or Platform Release level, and update status will be displayed in the Contents / Summary window

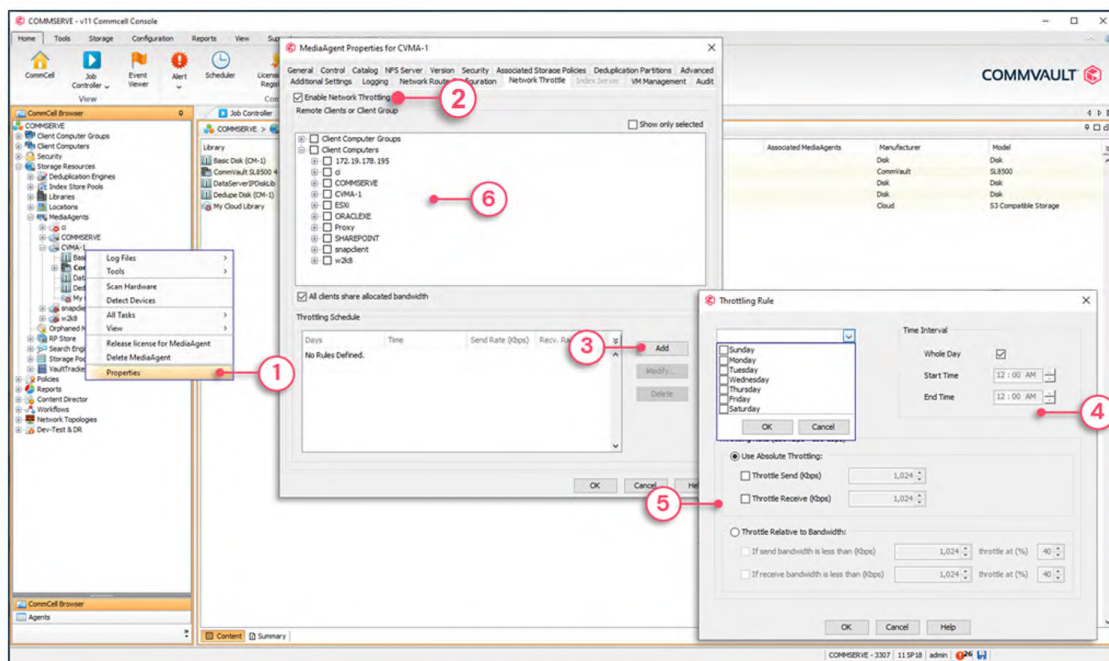
Configure Parallel Data Transfer Operations



A MediaAgent can receive and handle a certain amount of parallel data transfer operations, or streams. This is based on hardware resources of the MediaAgent, as well as the Commvault Online Documentation. By default, a MediaAgent has 100 set as a maximum of streams it can receive. After proper evaluation, this number can be increased or decreased.

1. Right-click the **MediaAgent > Properties**
2. Defines the maximum number of streams received by the MediaAgent

Network Throttling Rules



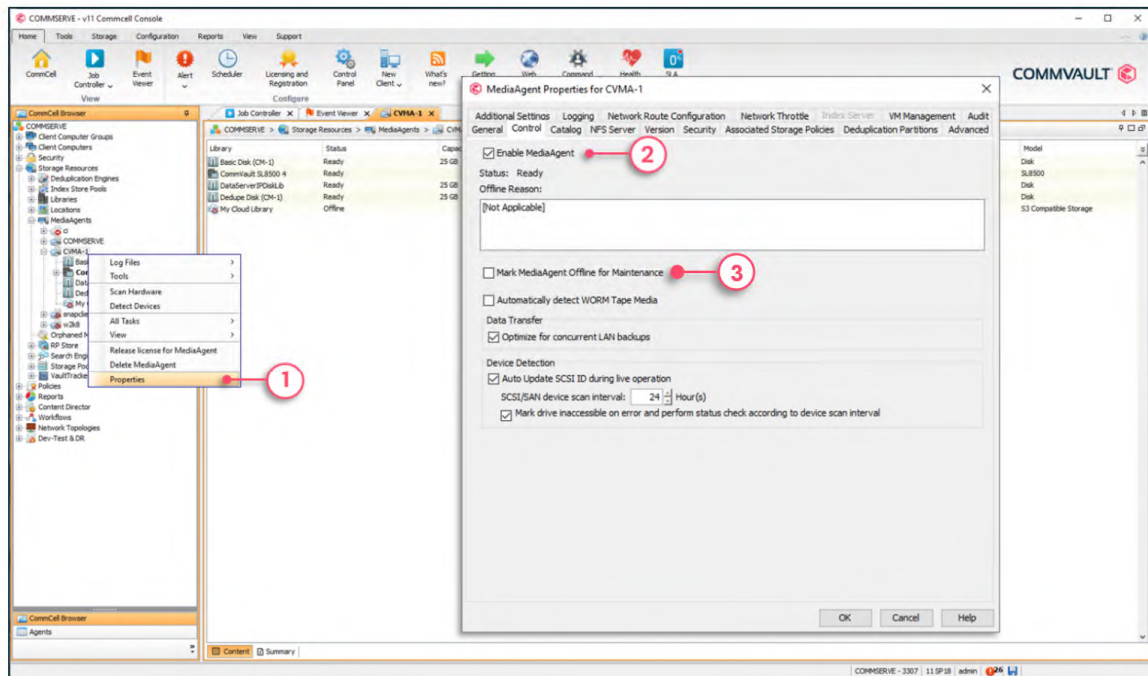
In some cases, it can be useful to limit the bandwidth used by a MediaAgent when it communicates with clients during backups and/or another MediaAgent to replicate data, especially when the network is shared with other systems and users. A network throttling rule determines how the bandwidth is controlled and the time range of the day or the days of the week for which the rule is enforced.

1. Right-click the **MediaAgent > Properties**
2. Check the box to enable the use of network throttling
3. Click **Add** to create a rule to limit bandwidth usage
4. Define when throttling will be enforced
5. Set an absolute or a relative throttling rule
6. Choose computers or computer group to which the rule will be applied

Disable a MediaAgent vs. Mark MediaAgent Offline

Option	Result
Disable a MediaAgent	No client nor administrative tasks can be executed. The MediaAgent is unavailable and unusable by the CommCell® console.
Mark a MediaAgent offline for maintenance	Temporarily prohibits data protection, data recovery, and auxiliary copy jobs using the MediaAgent. However, administrative tasks such as 'Clean Drives,' 'full scan,' or 'disk library pruning' may still be conducted.

If maintenance is required on the MediaAgent or hardware, it is possible to disable or mark a MediaAgent offline for maintenance. It is important to understand the differences between disabling a MediaAgent and marking it offline for maintenance.



1. Right-click the **MediaAgent > Properties**
2. Choose to enable or disable the MediaAgent
3. Choose to mark the MediaAgent as offline for maintenance

Agent Installation

Agent Installation Overview

Deploying agents can be achieved with a push install, interactive install, or custom package.

Agent Deployment Best Practices

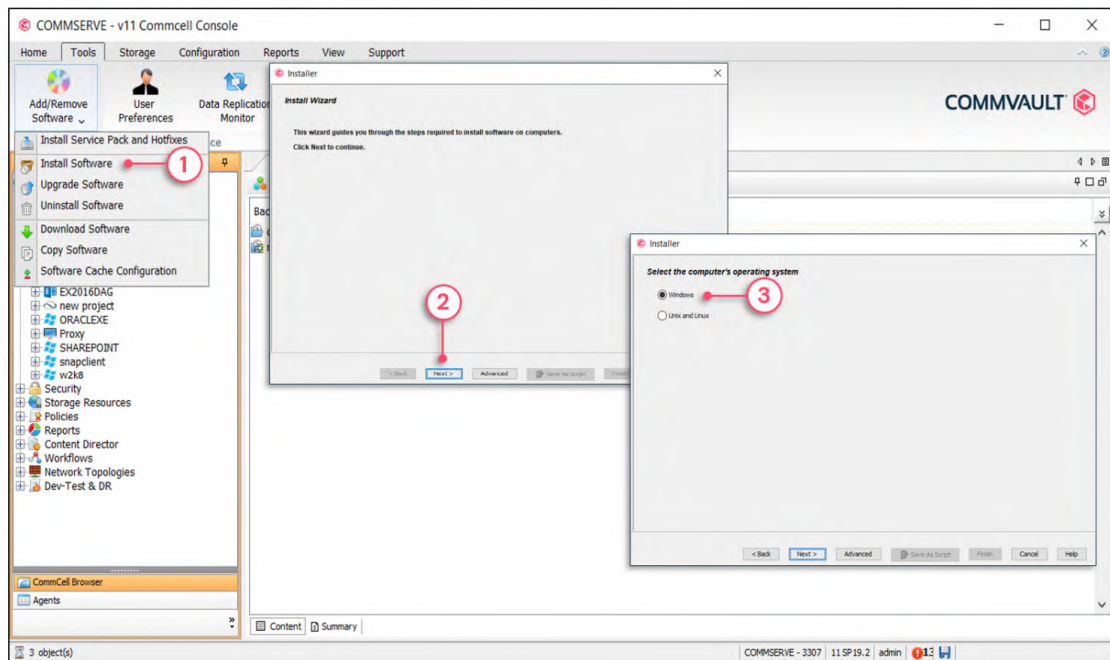
- If DNS names are used, ensure the DNS is properly resolving the name forward and reverse.
- If deploying an agent on a remote site, consider using a remote software cache or transfer a custom package.
- If the client is behind a firewall blocking ports, set network configurations to tunnel communication in a port.

Agent Requirements

Prerequisites will vary. Always confirm as it may change when a new service pack is released, even for components that you frequently deploy. Pay particular attention to:

- Operating system version
- Application version (for application agents)
- Service account with specific privileges

Push Installation

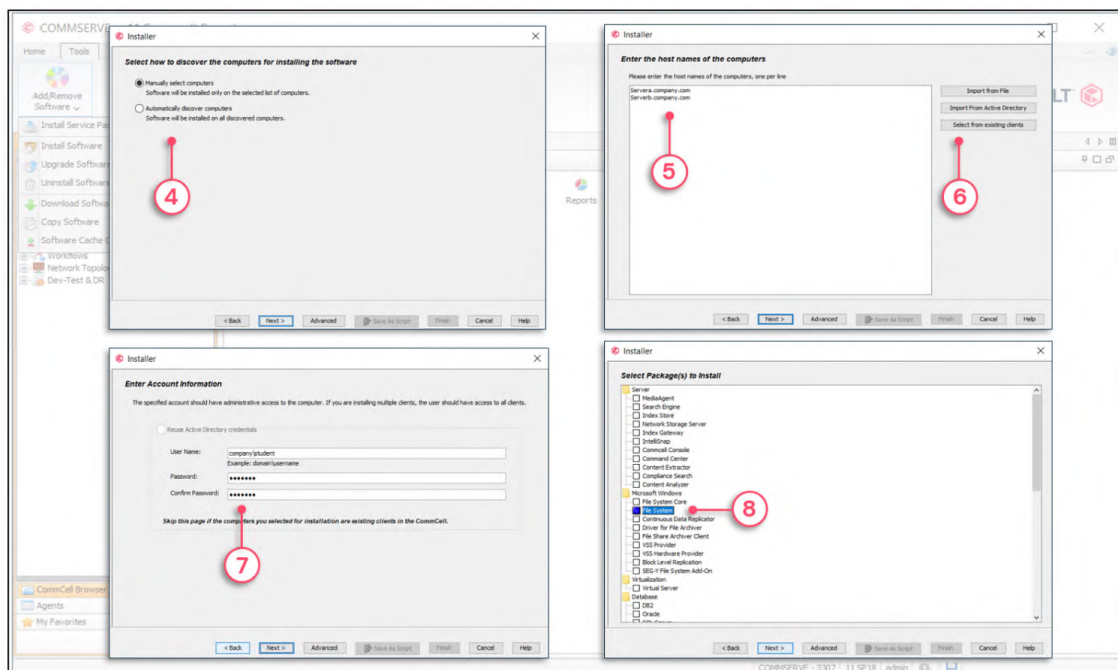


The CommCell console is used to push the Commvault software to clients. The following ports are used to achieve the install:

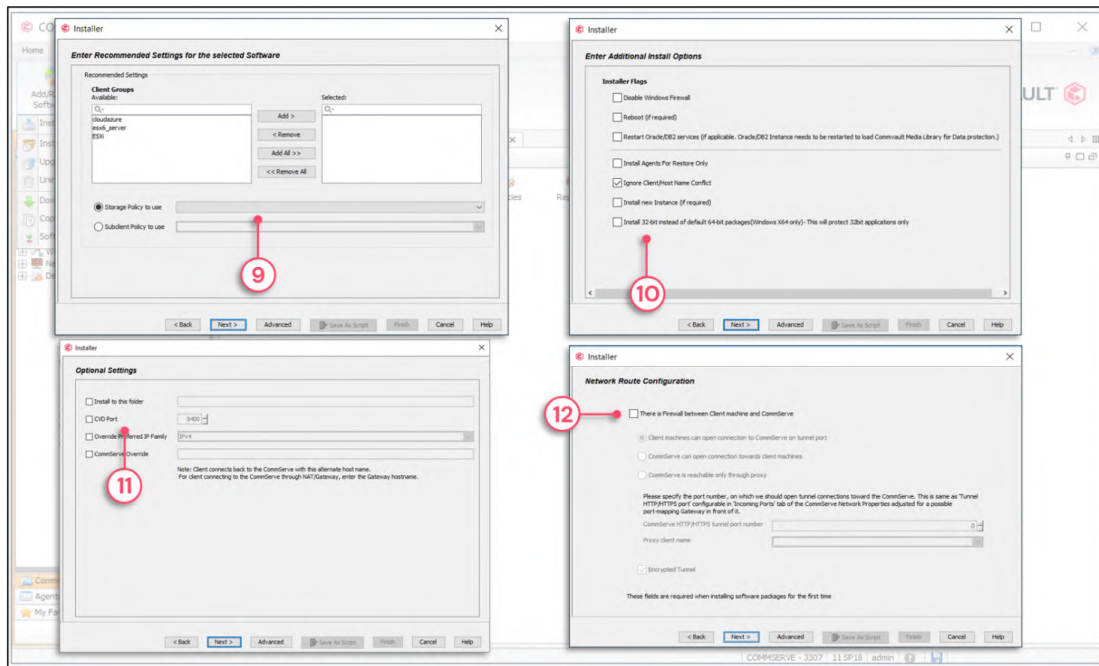
- **Commvault base services** – Ports 8400 and 8401
- **Linux** – port 22
- **Windows** — ports 135, 139, and 445

Execute a Push Install

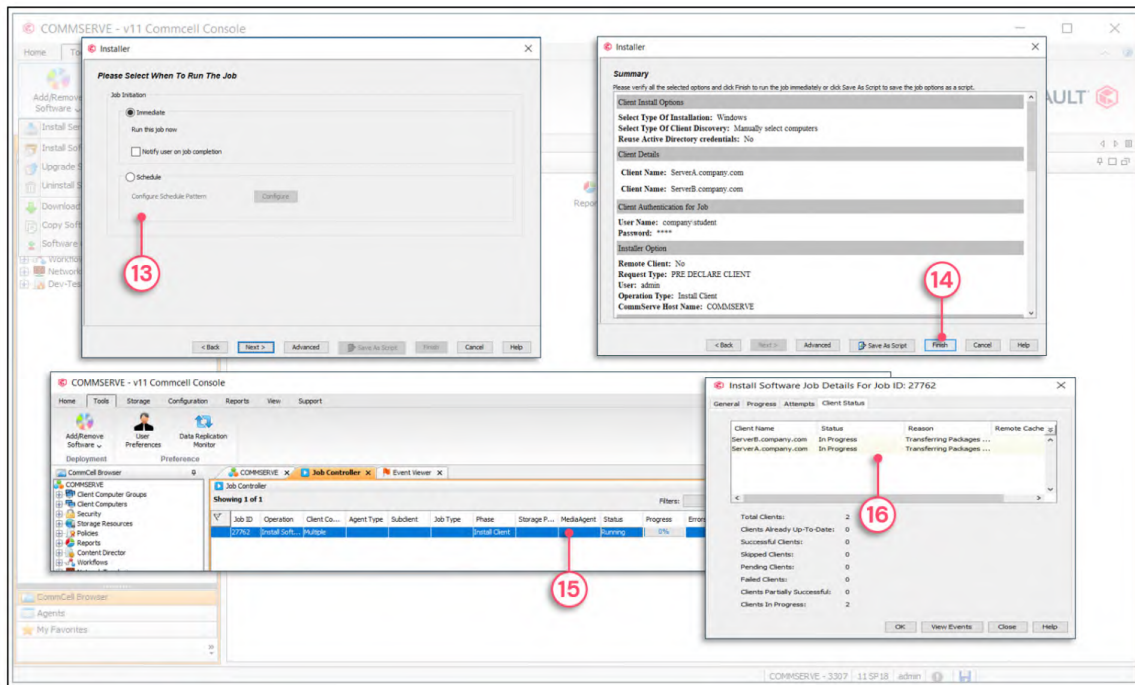
1. **Tools** tab > **Add/Remove Software** > **Install Software**
2. Click **Next** to continue
3. Choose the operating system



4. Select option to manually specify computers or to automatically discover servers
5. Enter the computer names or IP addresses, one per line
6. Use the tools to select computers in bulk
7. Enter a domain account that has administrative privileges on the target systems to install software
8. Select the agent(s) and component(s) to install

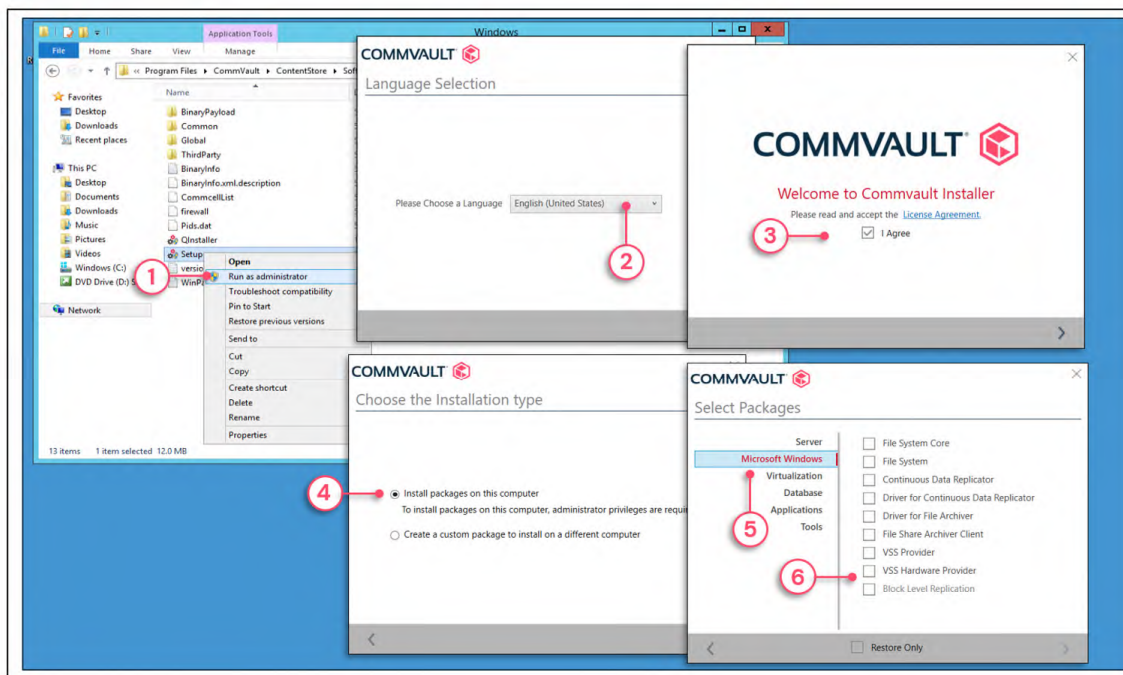


9. If required, choose client group and storage policy to use for default subclients
10. Review and select any additional install options
11. If required, change the default install path or default port
12. If there is a firewall, set network route to communicate with the CommServe server



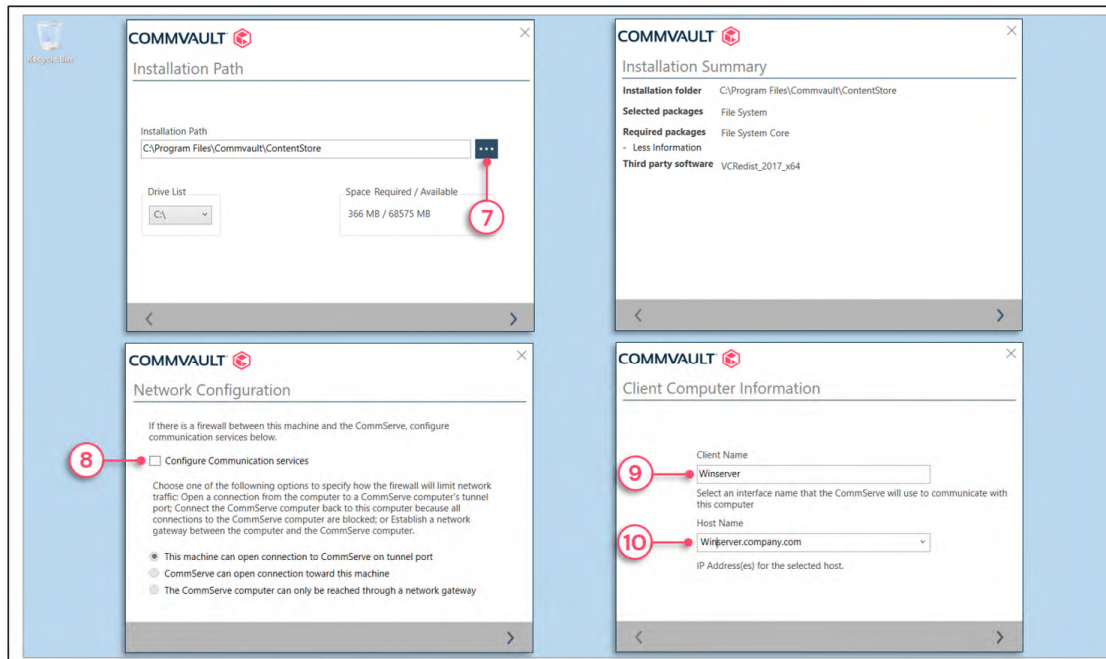
13. The installation task can be executed immediately or scheduled
14. Review the summary and Click **Finish** to launch the job
15. From the Job Controller window, double-click the job
16. Choose the client status tab to see every client's status

Interactive Installation

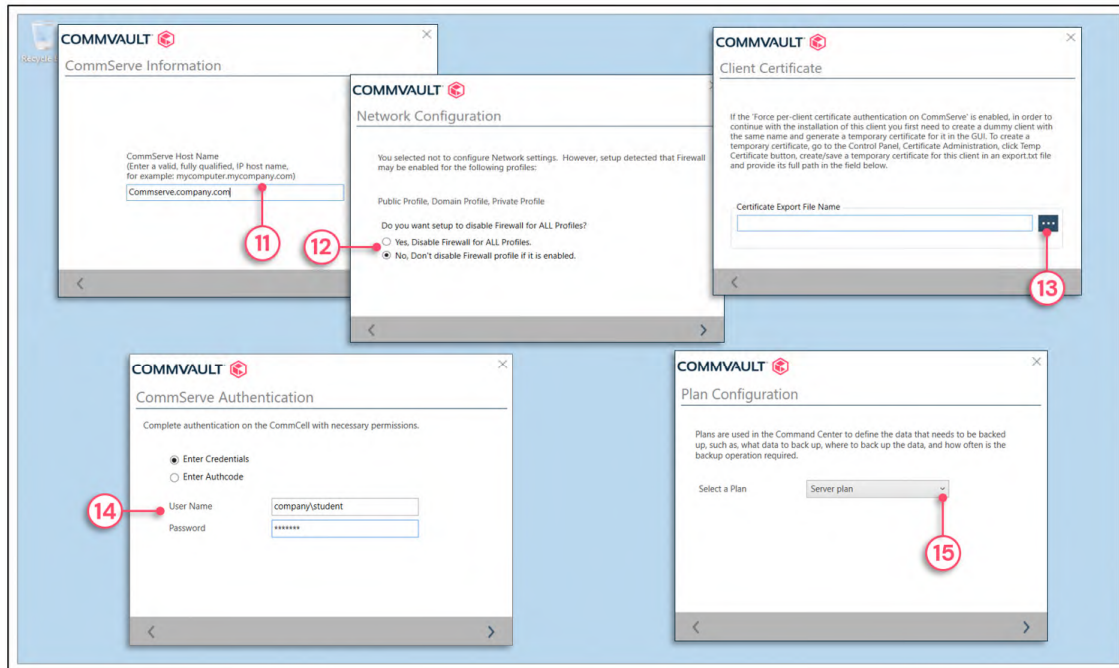


It is possible to download the packages on a client machine by using the download bootstrapper. Once downloaded, start the installation by executing Setup.exe.

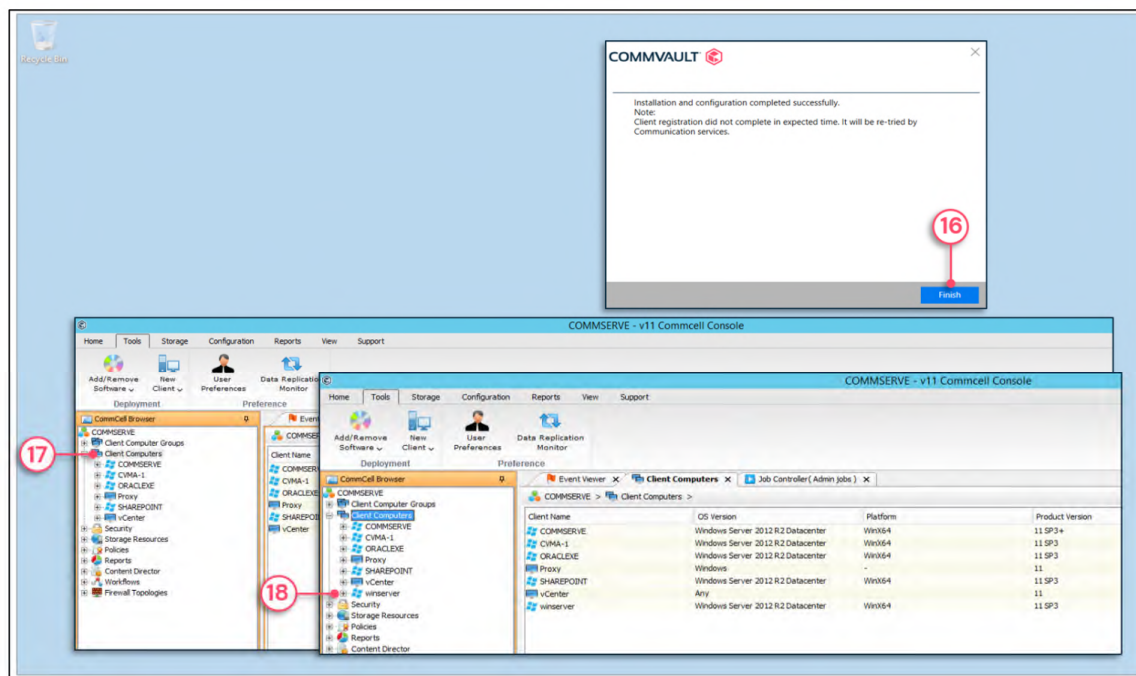
1. Download the packages or remotely reach the Software Cache > Execute Setup as an administrator
2. Select the Language
3. Check the box to agree with the License Agreement
4. Choose to install packages on this computer
5. Browse through categories to display the different agents
6. Select agent(s) and component(s) to install



7. Use the default path or enter a different drive and installation path
8. If there is a firewall, set rules to communicate with the CommServe
9. Enter the client display name
10. If there are multiple network interfaces, select the DNS name or type the IP address used to communicate from this server to the CommServe

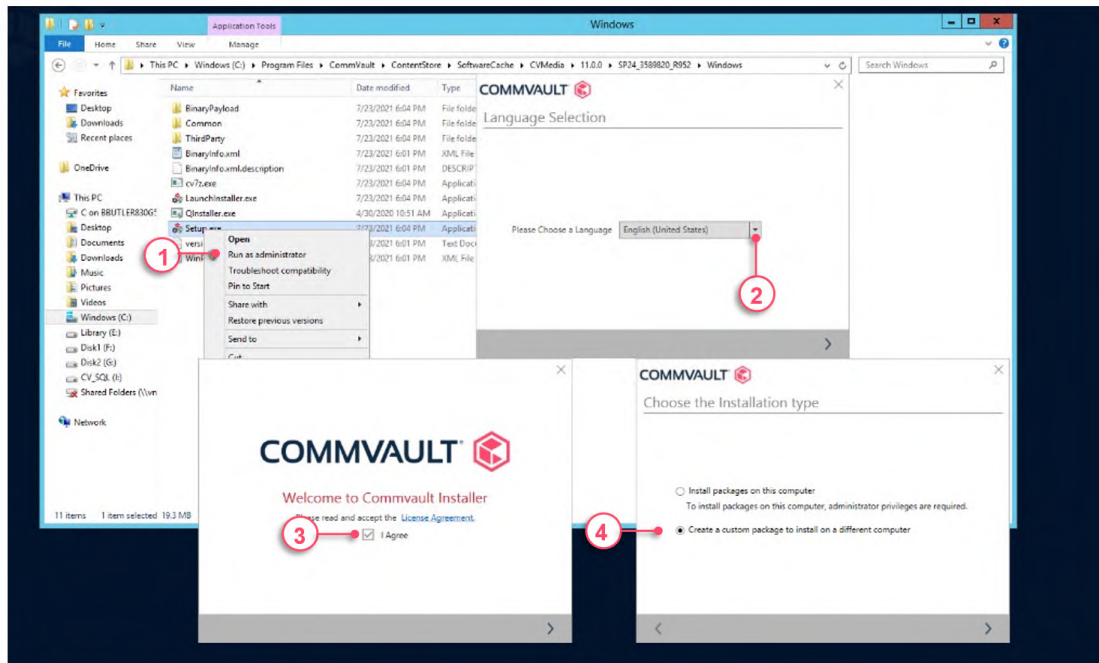


11. Enter the CommServe DNS name or IP address, and leave empty for a Decoupled Install
12. Select to disable local server firewall if required
13. If it is a locked down CommCell, provide the certificate file
14. Provide user account with CommCell Security Role having installation permissions
15. Select the appropriate data management plan



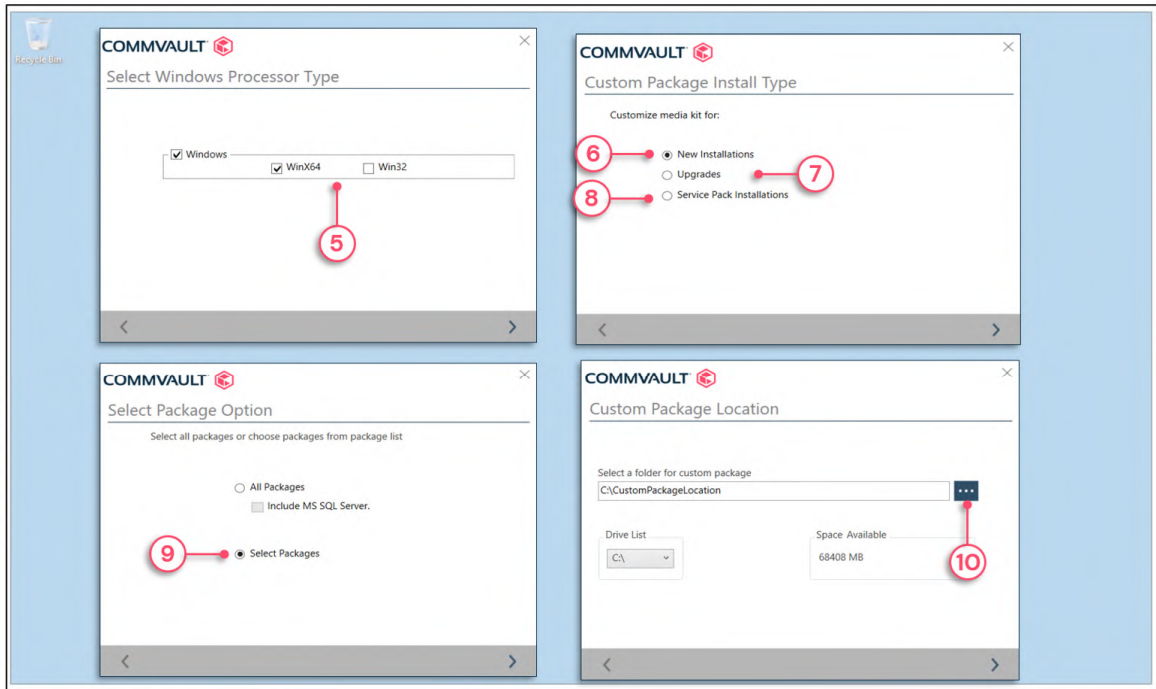
16. Click **Finish** to complete the installation
17. Click **Client Computers** and press [F5 key] to refresh the view
18. View the newly installed client in the CommCell Browser window

Create a Custom Package

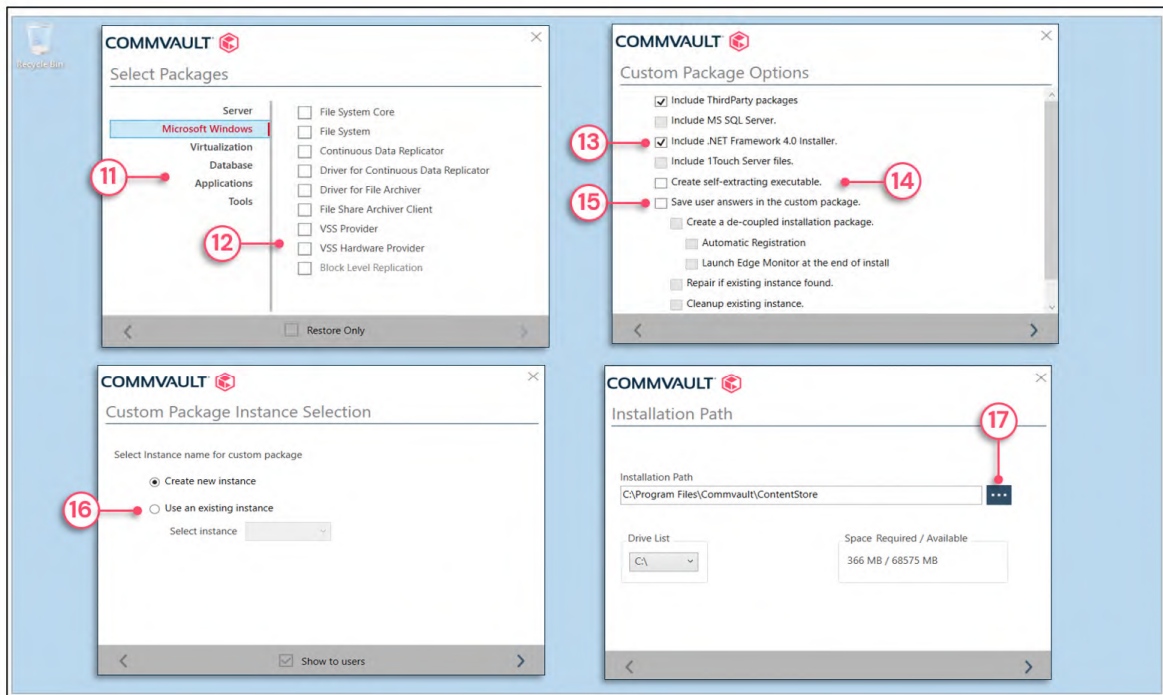


A custom package is a lightweight agent package created by the Commvault administrator. A typical agent installation requires many questions to be answered. The custom package has all these questions pre-answered. It also is useful to run silent installations. If an enterprise level deployment solution is in place, such as Microsoft SCCM, it can be leveraged to silently push the package to multiple machines.

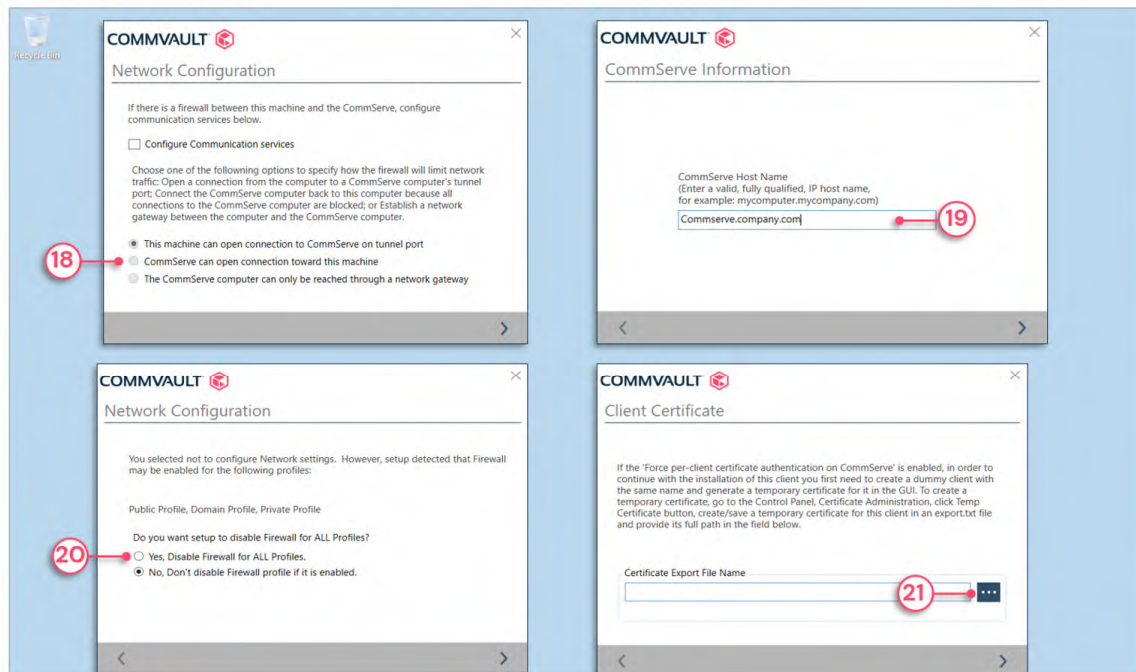
1. Download the packages or remotely reach the Software Cache > Execute Setup as an administrator
2. Select the Language
3. Check the box to accept the license agreement
4. Choose to create a package for installation on a different computer



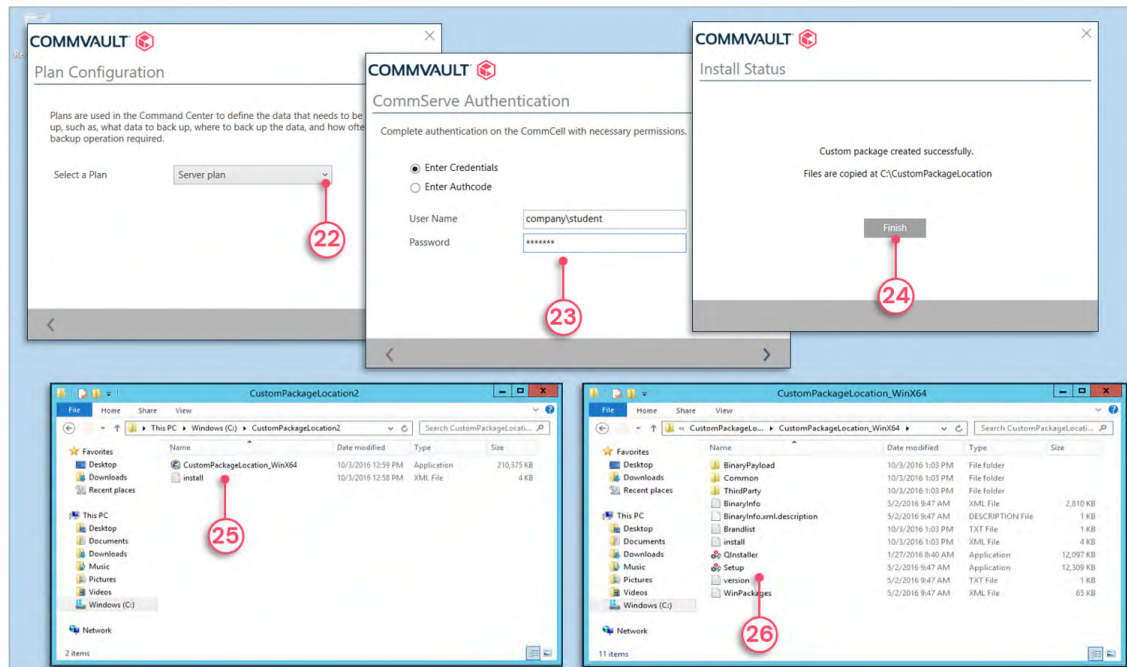
5. Choose the server architecture
6. Select **New Installations** to install new agent(s) on a server
7. Select **Upgrades** to upgrade installed agents to a new version
8. Select **Service Pack Installations** to update agent(s) to the latest service pack
9. Choose **Select Packages** to select the agents to include in the package
10. Choose the folder to store the created custom package



11. Browse through categories to display the different agents
12. Check each box to select agent(s) and component(s) to install
13. Select to include .NET Framework
14. Select to create a self-extracting custom package, or deselect to create a silent install package with response file
15. Select to include user answers in the package
16. Choose to create a new instance or use existing instance
17. Use the default installation folder or type or browse to a different installation directory



18. If there is a firewall in place, set rules to communicate with the CommServe to provide interaction for this section at time of install
19. Provide CommServe DNS name or IP address, or leave empty for a Decoupled Install to provide interaction for this section at time of install
20. Select to disable local server firewall if required or to provide interaction for this section at time of install
21. If it is a locked down CommCell, provide the certificate file to provide interaction for this section at time of install



- 22. Select a server plan to manage client jobs
- 23. Enter a user account with CommCell permissions to add client to CommCell
- 24. Click **Finish** to create the package
- 25. If the 'Create a self-extracting executable' option was selected, it will create a custom package
- 26. If the 'Create a self-extracting executable' option was not selected, it will create a silent install folder structure

Encryption

Encryption Overview

Type	Where Encryption is Performed	How it is Enabled / Disabled	Advantages	Disadvantages
Inline	Client or MediaAgent	Enabled on the client and configured on the subclient.	Allows encryption of data during primary movement and storage.	Can slow primary data protection job. Consumes CPU & memory of client or MediaAgent.
Offline	MediaAgent	Configured in the storage policy secondary copy.	Allows encryption of data during secondary movement and storage. Does not slow primary data protection job.	Consumes CPU & memory of MediaAgent.
Hardware	LTO4 and above drive with encryption support	Turned on/off at storage policy copy.	Enables faster encryption for tape media with minimal CPU or Memory consumption on client or MediaAgent.	Requires dedicated hardware for backups and restores. Decryption takes place on the drive potentially exposing data during transmission.

Encrypting data is an essential part of data protection, especially when data is entrusted to a third party for storage. Equal consideration must be paid to backup and archive data. If an unencrypted tape is stolen with sensitive information on it, there is no way to prevent someone from accessing that data. Simple security measures such as password protection may only delay access to the data. Preventing intrusion into your production environment requires a front-line defense that is costly and time consuming.

Commvault software offers three methods to encrypt data:

- **Inline encryption** – Secures data during the data protection job. Inline encryption is performed on the client or MediaAgent or is used for network transmission.
- **Offline encryption** – Secures data during auxiliary copy jobs. Offline encryption is performed on the source MediaAgent.
- **Hardware encryption** – Encrypts data when writing to tape drives that support hardware encryption. Hardware encryption is supported on LTO 4 and above tape drives which support and are licensed for encryption by the vendor.

Commvault software is certified for the US DoD/Canadian DND FIPS encryption accreditation for information security.

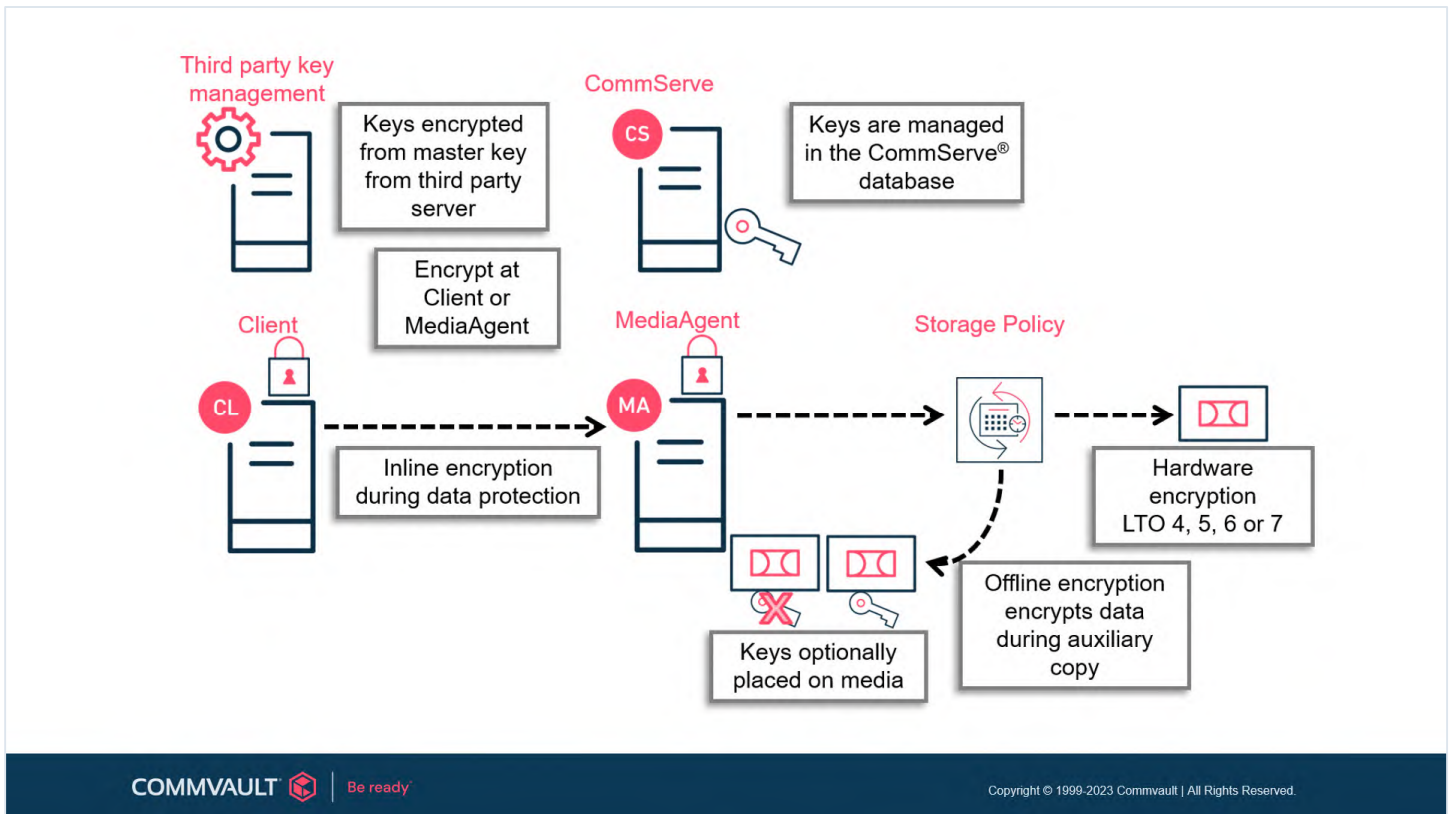
Encryption Key Generation and Management

Both software and hardware encryption keys use a proprietary algorithm and are maintained in the CommServe database. Encryption keys can optionally be written to storage media. In the event of the loss of the CommServe database, encrypted data may be recovered using tools provided by Commvault Support. Encryption keys are destroyed when the job is aged and deleted from the CommServe database. This provides complete end-to-end encryption key management.

Third Party Key Management

Commvault software also supports third party key management. Data is encrypted using Commvault encryption keys, which are stored in the CommServe database. These keys are encrypted using a third-party master key. The third-party system and the keys are required for any restore operations.

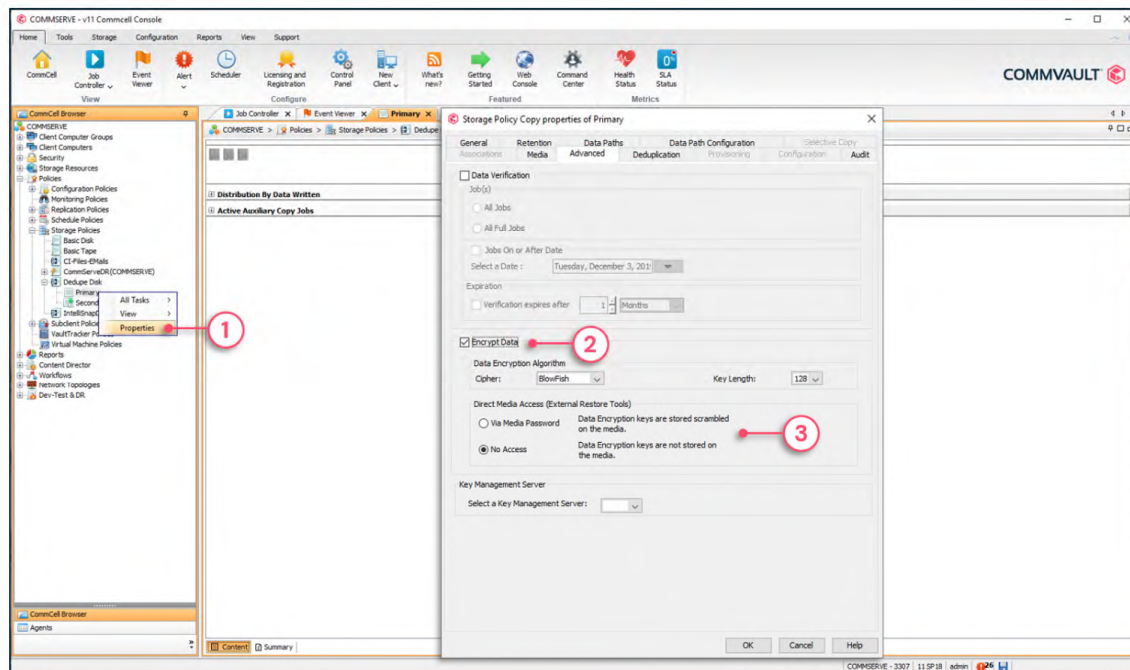
Software Encryption



Advantages

- Data can be encrypted on the client during initial data protection providing complete end-to-end security.
- Different encryption ciphers are used based on security requirements.
- In certain cases, software encryption can provide a performance benefit by distributing the load of data encryption to multiple systems as opposed to hardware encryption, where all data encryption is handled on the tape drive.
- Data can selectively be encrypted using inline encryption by configuring encryption settings at the subclient level. This can further improve performance by only encrypting data that requires encryption.
- Restore operations always decrypt data at the destination location.

Inline Encryption



Inline encryption is used to encrypt data during primary protection operations. The encryption can take place on the client or the MediaAgent. Encryption is enabled for Commvault software through the storage policy primary copy or at the client level. Encryption can further be configured at the subclient level. Subclient level encryption provides the flexibility of defining only that data which requires encryption. By default, when encryption is enabled on a client, encryption is enabled on all subclients.

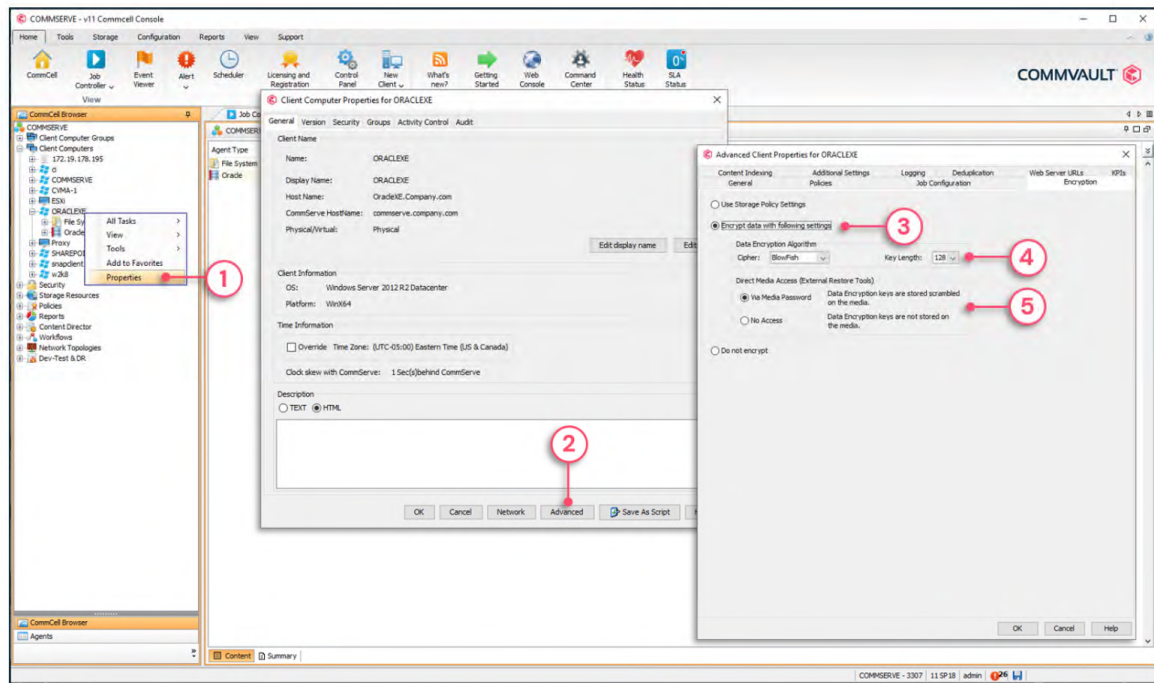
Inline Encryption Best Practices

- Only encrypt the data that has such a requirement.
- Isolate encrypted data in a different storage policy than unencrypted data.
- To achieve these goals, turn off encryption on the default subclient and create a dedicated subclient with the folders or files requiring encryption defined as content.
- Turn on encryption on that subclient only and associate it with the dedicated storage policy.
- A storage policy primary copy is used to enable encryption on all subclients associated with the storage policy. Ensure the client's encryption settings are configured to 'Use Storage Policy Settings'.

Configure Inline Encryption

1. Expand the storage policy > Right-click the **storage policy Primary copy > Properties**
2. On a primary copy, enable encryption
3. Set cipher, key length, and location for storing keys

Enable Encryption for a Client

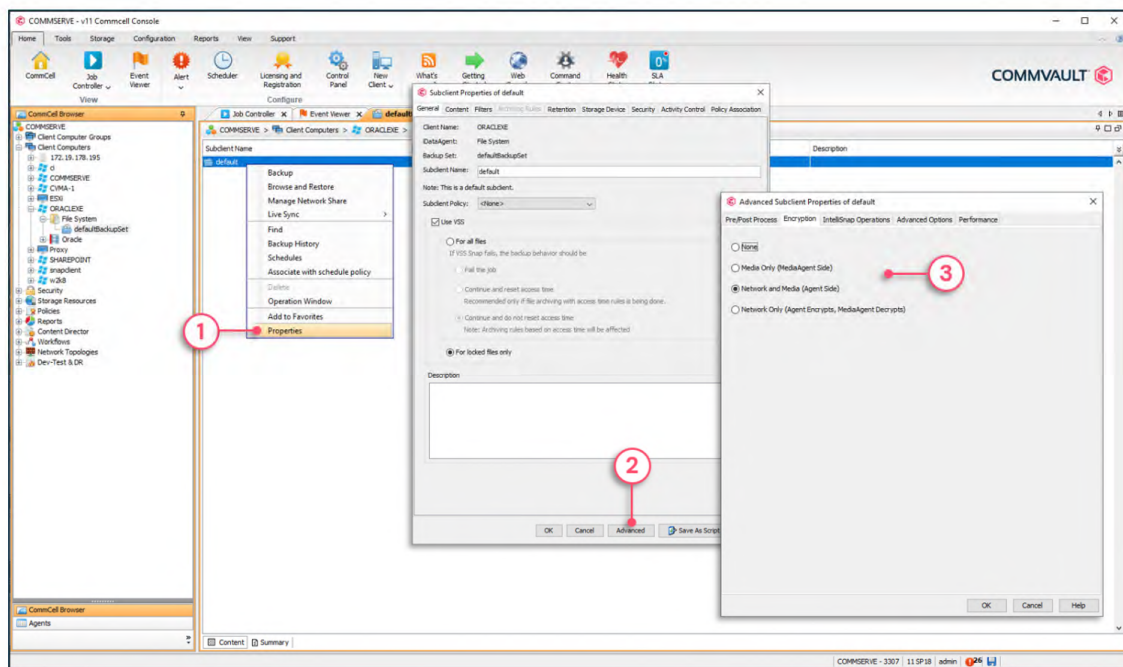


When encryption is enabled on a client, the cipher and key length must be set. The default cipher used is blowfish 128 bit. The 'Direct Media Access' setting determines whether encryption keys are stored on the media. The 'Via Media Password' option puts the keys on the media. The 'No Access' option only stores the keys in the CommServe database. If the keys are stored on the media, data can be recovered using Commvault software's 'catalog' feature, or in the case of Disaster Recovery data, the Media Explorer tool. Encryption keys are always stored in the CommServe database.

DR Data recovery using Media Explorer requires the user to provide the Media Password used when the data was written. The default Media Password is blank. If the media password is not known, contact Commvault Support to assist in recovering the password.

1. Expand Client Computers > Right-click the **client** > **Properties**
2. Click **Advanced** to configure client properties
3. Enable the use of encryption
4. Define the cipher and key length
5. Define if the encryption keys should be stored on the media

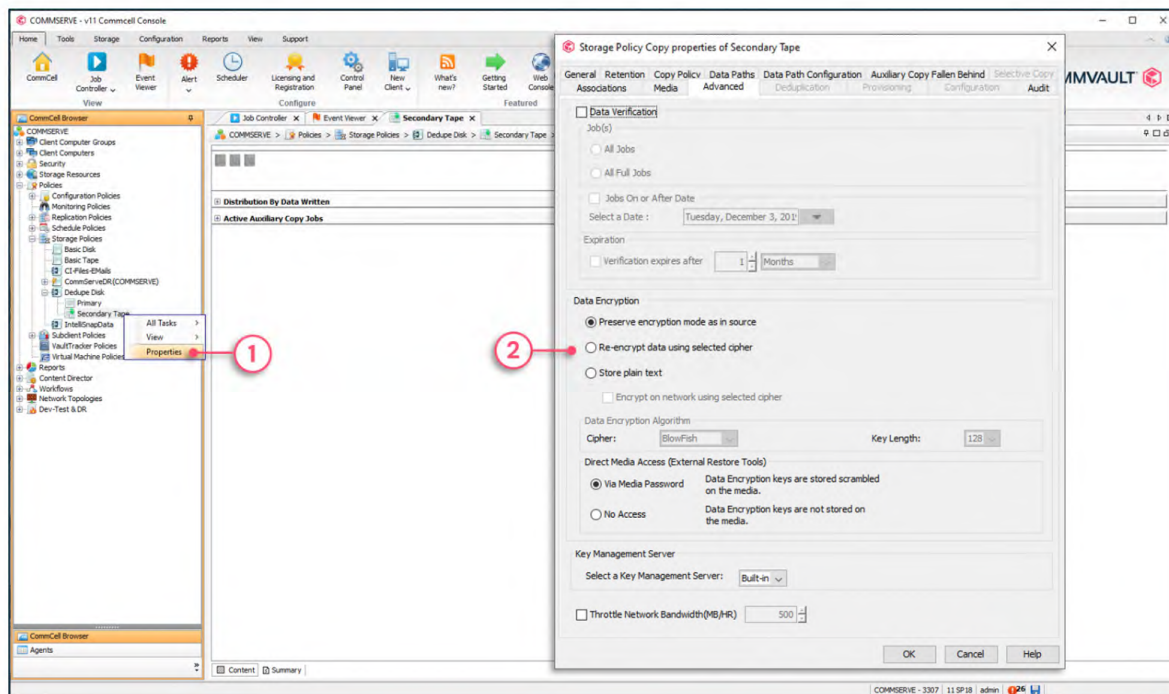
Subclient Encryption Settings



When encryption is enabled for a client, the default subclient encryption setting 'Client and MediaAgent' encrypts all data on the client and the data remains encrypted when written to storage.

1. Expand client > Right-click the **subclient** > **Properties**
2. Click **Advanced** to configure the subclient properties
3. Define where to apply encryption

Offline Encryption

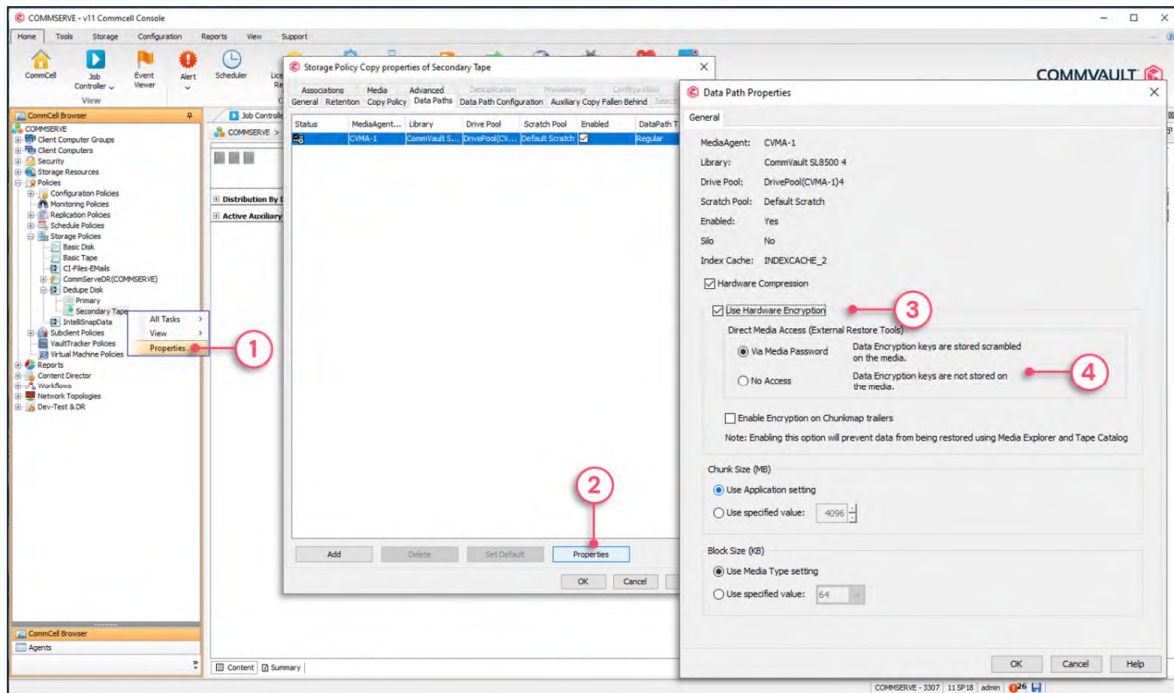


Offline encryption uses Commvault software encryption to secure data during auxiliary copy jobs.

In some cases, encrypted source data will be decrypted first then re-encrypted when storing deduplicated data or changing encryption ciphers. By default, encrypted data is preserved during an auxiliary copy operation.

1. Expand the storage policy > Right-click the **storage policy copy** > **Properties**
2. *Advanced* tab > **Re-encrypt Data** and configure the cipher and media access key

Hardware Encryption



Commvault software can enable or disable an encryption operation on the drive and manage encryption keys for tape drives that support hardware encryption. Keys are stored in the CommServe database.

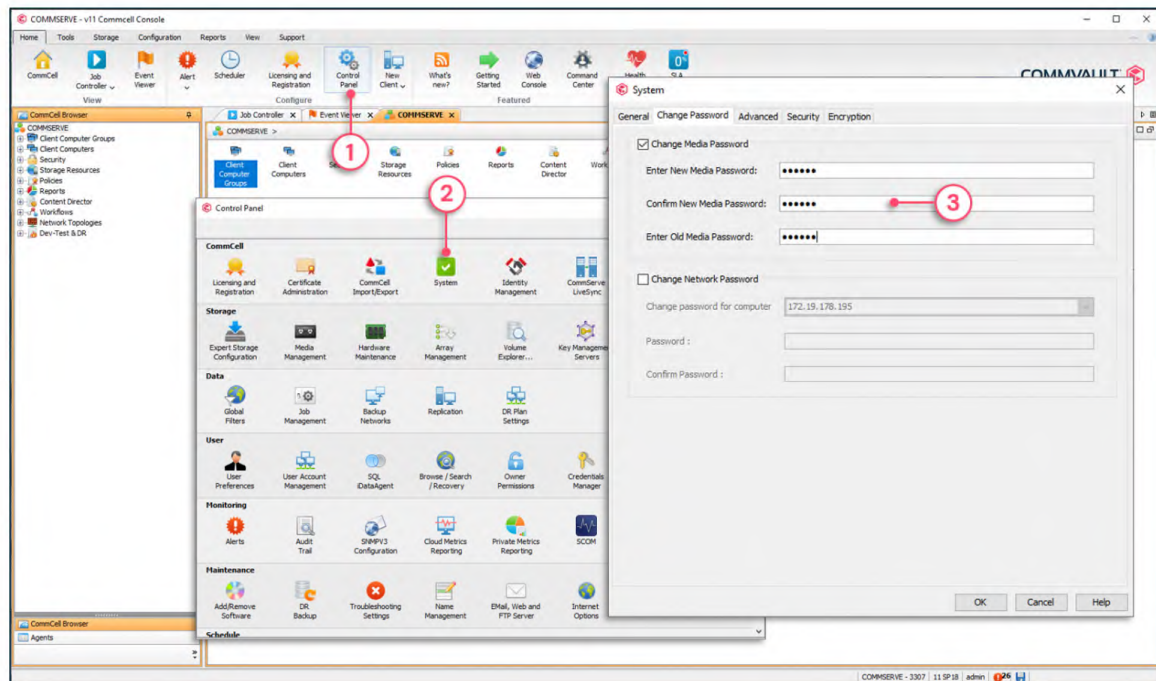
Commvault software writes data in chunks. Tape media uses 8GB chunks for indexed-based backups and 16GB chunks for database backups. When encryption is enabled for data protection jobs writing to tape media with 'hardware encryption' enabled, each chunk has a separate encryption key seeded by a random number generator and other factors. Generating keys at the chunk level provides an enhanced level of security and greatly reduces the potential of data compromise.

Configure Hardware Encryption

1. Right-click the **storage policy tape copy > Properties**
2. Select the path to the drives and click **Properties**
3. Enable the use of hardware encryption
4. Define if the encryption keys should be stored on the media

Encryption Considerations

Media Password

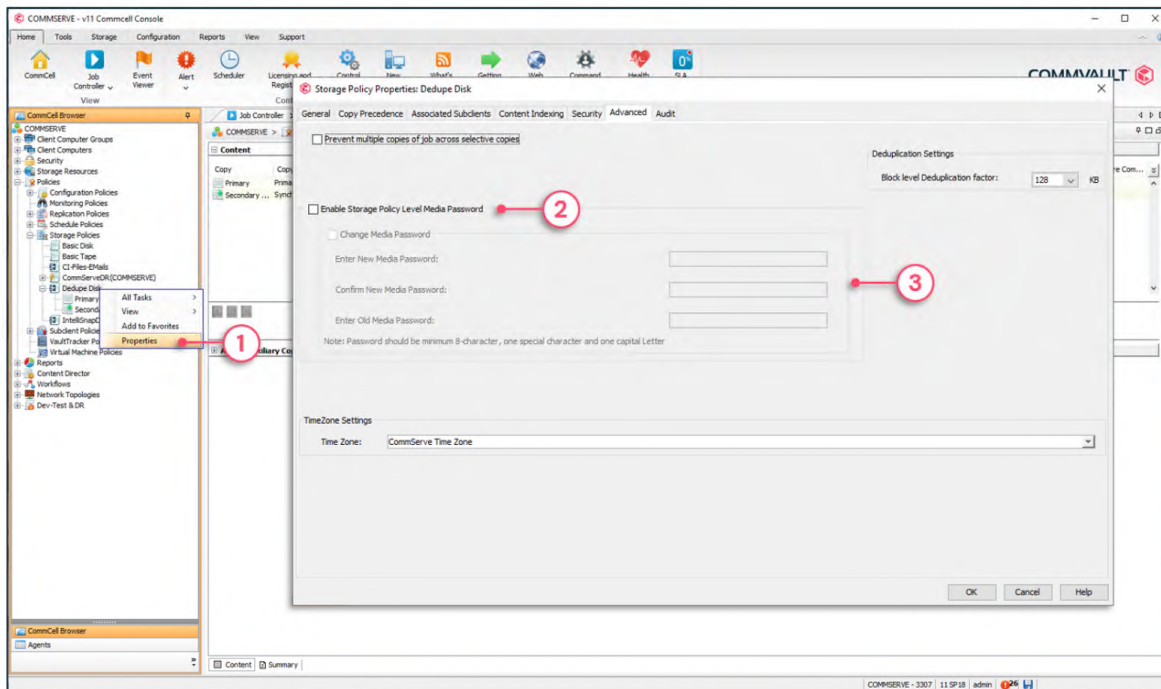


The Media Password is used when recovering data through Media Explorer or when the 'Catalog Media' option for tape is used. A media password is essential when using hardware encryption or Commvault software copy-based encryption (with the 'Direct Media Access' option set to 'Via Media Password').

By default, the password is set for the entire CommCell environment in the System applet in Control Panel. The default password is blank. Storage policy level media passwords can be set to override the CommCell password settings. For higher level of security or if a department requires specific passwords, use the 'Policy level' password setting which is configured in the Advanced tab of the storage policy properties.

1. Home tab > **Control Panel**
2. *CommCell* section > **System**
3. Enable and set Media Password

Enable Storage Policy Media Password



1. Expand Storage Policies > Right-click the **policy** > **Properties**
2. Enable the use of Media Password for the storage policy
3. Define the Media Password

Erase Data Feature

Commvault administrators can mark individual objects as unrecoverable within the CommCell environment. Administrators can browse or search for data through the CommCell console and mark the data as unrecoverable.

Erase Data is a CommCell-level license. Once it is applied you need to enable 'erase data' in the General tab of the storage policy. Once this license is applied, the 'Catalog' option cannot be used to recover backup data managed by the storage policy. This is because random binaries are written to the OML header in the media password location. The password cannot be used, and you always get a decryption error when using these tools.

If the erase data operation is something that would be of value to your organization, then it may be worth the risks previously described. If you are not sure, then it is recommended that you do not use it. If you have capacity-based licensing arrangements with Commvault software, check to see if this license is installed. If it is, you may want to disable 'erase data' in all storage policies within the CommCell environment.

How Erase Data Works

When the erase data license is implemented, it is only effective when writing jobs to new or recycled media. The license cannot be retroactively applied to jobs already in storage. If the license is removed, it is only effective when writing to new or recycled media. All jobs written to media for the storage policy when the license was being used cannot be recovered through Media Explorer or the 'Catalog' option.

It is technically not possible to erase specific data from within a job. Erase data works by logically marking the data unrecoverable. If a Browse and Restore or Find operation is conducted, the data does not appear. For this feature to be effective, any media managed by a storage policy with erase data enabled will not be able to be recovered through Media Explorer or the 'Catalog' option.

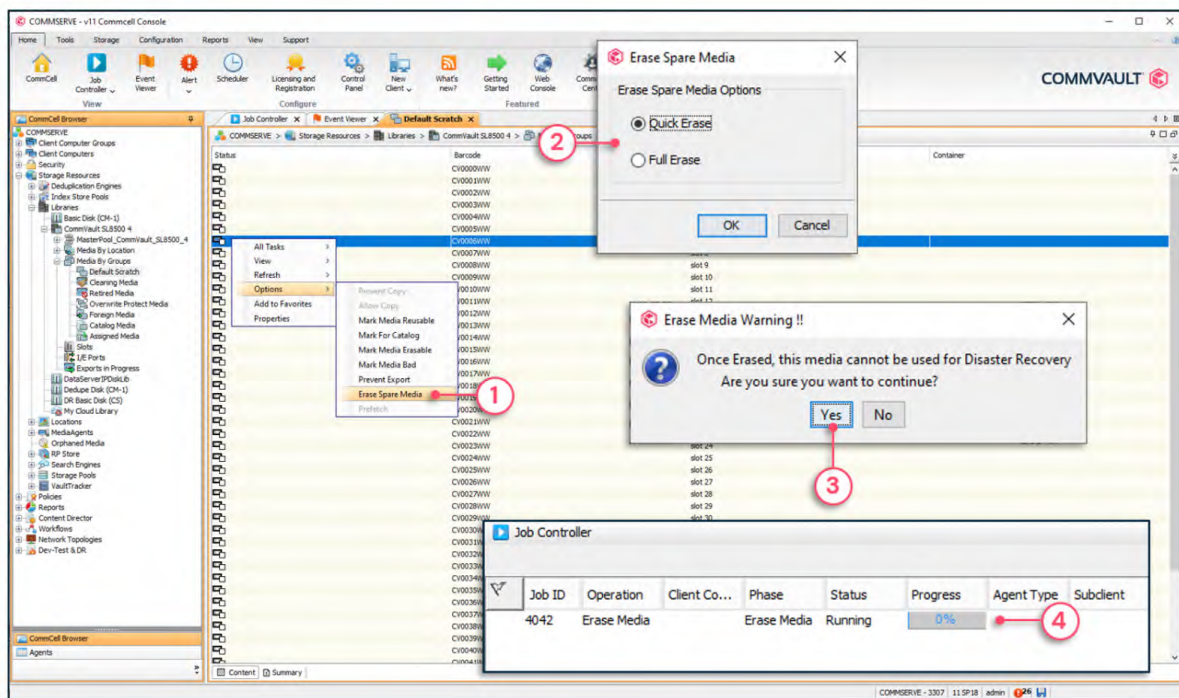
Erase Media

To ensure encryption keys are destroyed in the CommServe database when tapes are aged, the 'Erase Media' option is used. Erase Media is a physical operation that mounts the tape and overwrites the OML header. Once the header is overwritten, data cannot be recovered using any method Commvault software provides. This is considered a destructive operation so it cannot be performed on any tapes where jobs are actively being retained. The option to erase media is available in all logical media groups except the Assigned Media group.

There are two methods for erasing a tape:

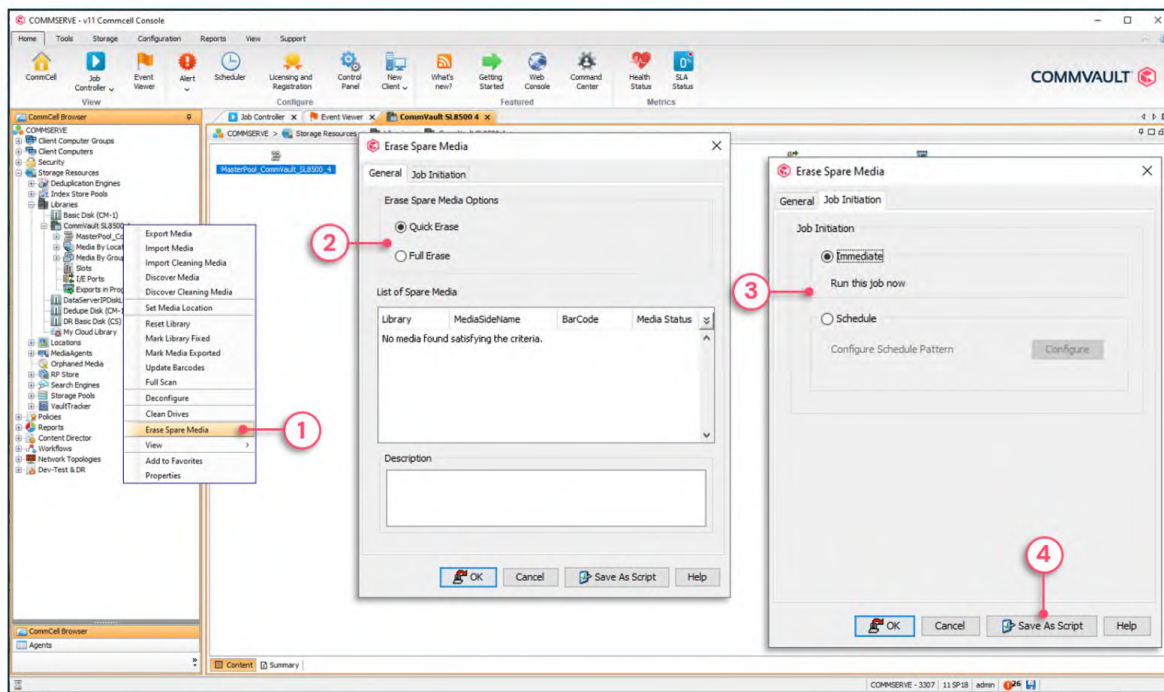
1. Manually erasing a tape in the Spare Media Group.
2. Automatically erasing tapes that are returned to the Spare Media Group (configured through the storage policy copy).

Manually Erase a Tape



1. Right-click **tape** > **Options** > **Erase Spare Media**
2. Select **Quick Erase** or **Full Erase**
3. Click **Yes** to confirm the operation
4. Erase media operations will appear in the Job Controller

Set Automatic Erase Operations

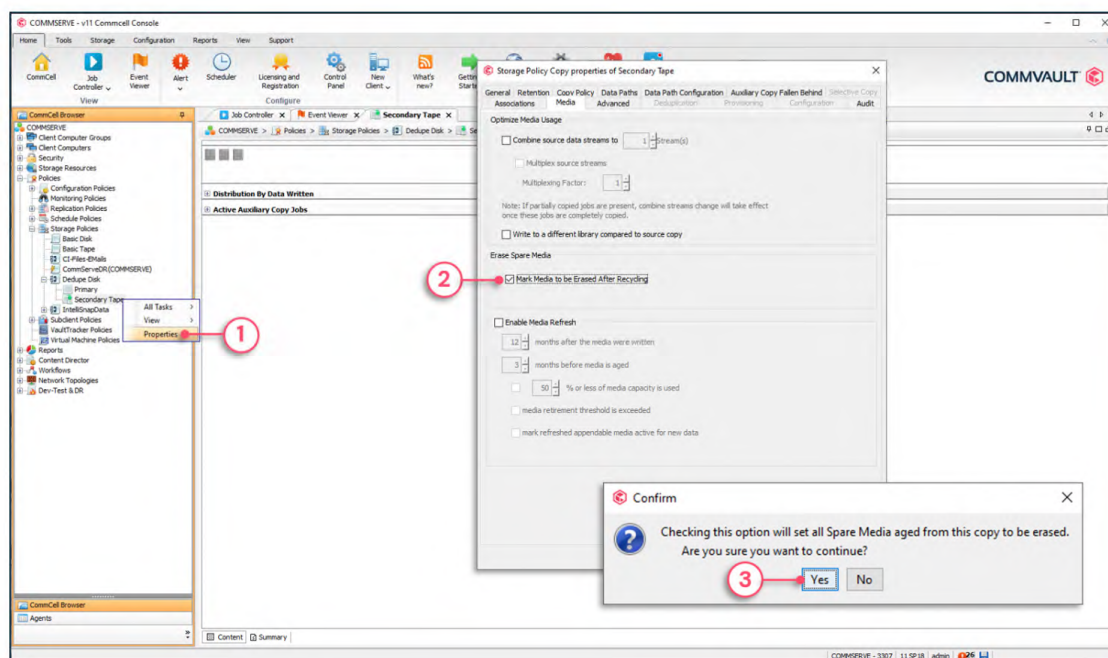


When tapes are recycled, they can automatically be marked to be erased. This is done by selecting the 'Erase Media' check box in the Media tab of a storage policy copy. An erase media operation must be scheduled for the library, which physically loads each marked tape and overwrites the OML header.

When the 'Mark Media to be Erased' option is enabled for storage policy copies, erase media operations must be scheduled for any library where media is marked for erasing.

1. Right-click **Library > Erase Spare Media**
2. Set the erase operation to be quick (overwrite OML header) or full (overwrite entire tape)
3. Choose **to** start the erase operation immediately
4. Alternatively, click **Save As Script** to save as a script

Erase Media Storage Policy Settings



The option 'Mark Media to be Erased After Recycling' in the Media tab of the storage policy copy marks all tapes managed by the policy to be erased once all jobs have aged. A schedule must then be set up at the tape library to erase the media. The erase media operation mounts the tape and write a new OML header to the tape. This makes the data completely unrecoverable through the CommCell console, Media Explorer, or through a tape catalog operation. It is important to note that the data on the tape is not actually erased. As such, it is recommended to encrypt all tapes.

1. Right-click a **storage policy copy** > **Properties**
2. In the pop-up window, from the *Media* tab, check the box for **Mark Media to be Erased After Recycling**
3. Click **Yes** to confirm